SIEVE LIMITS FOR THE ERDŐS–STRAUS EXCEPTIONAL SET: WHY 3/4 IS SHARP FOR CONGRUENCE SIEVES

ANONYMOUS

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Abstract. Let $E(N)$ count the $n \le N$ for which $4/n$ is not a sum of three positive unit fractions. The best internally proved bound in this programme is $E(N) \ll N \exp\{-c(\log N)^{3/4}\}$. We show that the exponent $3/4$ is a hard ceiling for a precisely defined class of congruence sieves. The class consists of nonnegative CRT majorants that are at least one on the whole avoider set of a family of forced congruence classes (and selector classes $0 \bmod p$), whose final bound has the form $N \cdot \mathbb{E}\nu + \sum_i |a_i|$, and whose family primes are at most $N^{O(1)}$. The main theorem: for every finite family mixing the forced classes $\mathscr{R}(M)$, the (a, D)-classes, the Case-A classes and selector classes $0 \bmod p$, with arbitrary moduli (no size condition, no dominant prime, any number of prime factors), every such sieve saves at most $O_A((\log N)^{3/4})$. So no sieve of this class proves $E(N) \ll N \exp\{-(\log N)^{3/4}\omega(N)\}$ with $\omega(N) \to \infty$; the previous version of this note lost a factor $(\log\log N)^{3/4}$ here. The Case-A part uses published divisor-sum estimates of Elsholtz and Tao, whose uniformity in the multiplier we re-derive (noting two harmless slips). The key tools are a weighted k-ary comparison theorem for sequential conditioned laws, proved by Lagrange extrapolation in artificial replacement coins and a supermartingale bound; a base of unit squares, which avoids the small classes of all four types because no $\mathscr{R}(M)$-, (a, D)- or Case-A class contains a square modulo its modulus; and a lossless first moment over smooth moduli, obtained from a local moment Z_y of the smooth part with Shiu's theorem in progressions. The majorant of the $3/4$ note lies in the class, so $3/4$ is sharp for its method. The cap $O_A((\log N)^{3/4})$ extends to every CRT-admissible large sieve with frequencies of polynomial level (twisted, multiplicative, hybrid and fibrewise forms), to interval counts of majorants whose moduli are at most $N/2$, and to tuple-count input of bounded order; tuple input of prime order k saves at most $O_A((\log N)^{3/4} + k\log\log N)$. The large sieve for primes and majorants positive only at primes are capped at $O_A((\log N)^{3/4}(\log\log N)^{3/4})$ (and at $O_A((\log N)^{3/4})$ at pointer level only). Gallagher's larger sieve saves only $O(\log\log N)$. For hybrid methods that count large classes blindly we reduce the cap to a combinatorial hypothesis (weak SPW), with a requirement on its margins that is exact for this route, and show that its fixed-margin form is false. For super-polynomial large-sieve frequencies we show that any cap needs family-specific input, such as the sparsity of forced classes; using sparsity through a smooth–rough splitting, every CRT-admissible large sieve, at any frequency level, saves at most $O((\log N)^{3/4}(\log\log N)^3)$ for mixtures with at most one prime factor above $\exp\{(\log N)^{1/4}\}$ per modulus, and at most $O_\gamma((\log N)^{3/4}(\log\log N)^3)$ whenever the residues of the classes with several such primes are γ-sparse at each such prime ($\gamma > 0$ fixed), which includes the residue-concentrated families; the remaining mixtures are reduced to a sup-decay hypothesis. We also show that the literal tuple-count hypothesis meets forced zeros above $2/3$ and give the repaired alternating form. We list exactly what the theorems exclude. Finally we explain the cap and the pointwise ceiling $1/4$ for the least witness modulus of a prime as two readings of one sieve limit, and record the $3/4$ bound for primes in $\pi(N)$ form. Every statement carries its status label. All proofs were checked internally only; nothing here has been externally refereed. This is not a proof of the Erdős–Straus conjecture.

1. Introduction and status

Call $n \ge 1$ *representable* if $4/n = 1/x_1 + 1/x_2 + 1/x_3$ with positive integers x_i, and let $E(N)$ count the non-representable $n \le N$. The Erdős–Straus conjecture asserts that every $n \ge 2$ is representable. Vaughan [V] proved $E(N) \ll N \exp\{-c(\log N)^{2/3}\}$ with the large sieve (see the reconstruction of his argument in [PW, §4]; for the inequality see [M]). Two companion notes of this programme

improve this. The first [LL] gains a factor $(\log\log N)^{1/3}$ in the exponent. The second [TQ] proves

$$E(N) \ll N \exp\{-c(\log N)^{3/4}\},$$

with label *internally proved; not externally refereed*. The 3/4 note itself explains 3/4 heuristically: the forced classes have cubic logarithmic mass, and a mass-driven optimisation gives $\theta = b/(b+1)$ with mass exponent $b = 3$ [TQ, §10].

This note turns that heuristic into a theorem for a precise class of sieves, and records exactly where the theorem stops.

Status labels. Every statement carries one of the following labels.

- [PROVED]: proved here, with the proof given or precisely cited to an internal document; checked by internal hostile review only.
- [PROVED MOD X]: proved assuming the published external result X, which was checked against the source but not re-proved.
- [CONDITIONAL]: proved assuming a named open hypothesis.
- [EVIDENCE]: numerical or model evidence; never a proof.
- *Assessment*: a reasoned judgement about methods or heuristics; not a theorem.
- [CONJECTURE] / [OPEN]: not proved.

[PROVED] for a result resting on Theorem 10.12 (or Lemma 10.9) means: given the [KA]/[KA2] framework (square base, leak, weighted sequential sieve limit) as reviewed there [KAr, KAr2, KA2r, KA2r2]; [KA3] does not re-prove it. Apart from the classical facts of Section 2, each [PROVED] result below appears in one of twenty-seven internal working documents, [ET], [EB], [TW], [TW2], [TW3], [KA], [TW4], [NC], [KA2], [KA3], [LS], [LS2], [LS3], [LS4], [PL], [IF], [IF2], [SPW], [SPW2], [TU], [TU2], [CU], [O10], [O13], [O14], [O15], [O16], or is assembled here from such results with the short argument given (marked "observation of this note"). Each document was hostile-reviewed; the reviews are [ETr, EBr, TWr, TW2r, TW3r, KAr, KAr2, TW4r, NCr, KA2r, KA2r2, KA3r, LSr, LS2r, LS2r2, LS3r, LS4r, PLr, IFr, IF2r, SPWr, SPW2r, TUr, TU2r, CUr] (those of [O10, O13, O14, O15, O16] are listed with them in the bibliography). Version 5 adds Sections 14.5 and 14.6, the refutation of fixed-margin SPW and the weak-SPW requirement in Section 14.9, and Section 18; the cap of Theorem 10.14 is unchanged. A few statements were added to these documents after their review and were checked by the coordinator of the programme only; this is said where they are used.

Attribution (literature audit [NA]). The linear-programming duality between upper-bound sieves and measures matching the sieve data is standard [T, Thm. 5] [BGP, Prop. 4]. In the exchangeable single-level case (identical densities, one weight), Theorem 4.5 is a weaker form of a known result. The optimal mean of a degree-k majorant is the maximal probability $M(n,k,p)$ that k-wise independent bits are all 1; the Selberg (square) value bounds it [BGP, Thm. 23]. Peled, Yadin and Yehudayoff [PYY, Thm. 1.1] showed that it is attained up to a factor $e^{O(k)}$, building on Prékopa's discrete moment problem [P]. To our knowledge the following are new: the weighted lower-set form with arbitrary densities, the Rankin-functional cap, the sequential and k-ary comparison theorems, and the applications to Erdős–Straus below. The arithmetic inputs of Section 10 are standard: Mordell's Jacobi-symbol obstruction, Rankin's trick, Shiu's theorem [S] and a small-divisor domination of τ (cf. [La]); the Case-A first moment uses Prop. 1.4, Thm 7.1, Cor. 7.4 and display

(7.10) of Elsholtz and Tao [ElT], published and not re-proved here (we re-derive only the uniformity of (7.10) in the multiplier, Remark 10.7). The local moment Z_y of Lemma 10.4 is an elementary device; we do not know a reference for its use in this way. That the (a, D)- and Case-A classes contain no squares is in the spirit of the Mordell–Schinzel obstructions quoted in [ElT]; our proof (Lemma 10.2) is direct.

Results.

(1) *The main theorem* (Theorem 10.14, [PROVED]; the Case-A part [PROVED MOD] Elsholtz–Tao Prop. 1.4, Thm 7.1, Cor. 7.4 and (7.10)). Take any finite family $\mathfrak{G}$ mixing $\mathscr{R}(M)$-, (a, D)-, Case-A and selector classes 0 mod p, with arbitrary moduli: no B-hypothesis, no dominant prime, any number and multiplicity of prime factors. Every method that bounds $\#(\mathcal{A}(\mathfrak{G})\cap[1, N])$ by $N\mathbb{E}\nu+\sum_i |a_i| < N$, with ν a CRT majorant ≥ 0 on $\mathbb{Z}$ and ≥ 1 on the whole avoider set, and with family primes $\leq N^{O(1)}$, saves at most $O_A((\log N)^{3/4})$. So no method of this class gives $\theta > 3/4$, nor a gain $(\log N)^{3/4}\omega(N)$ with $\omega \to \infty$. The underlying majorant cap is Theorem 10.12, $\log(1/\mathbb{E}\nu) \leq C\lambda^{3/4}$ at level λ. Version 3 of this note had an extra factor $(\log\log N)^{3/4}$ without B; it is removed here following [KA3]. All constants are absolute but astronomically large, so these are asymptotic statements.

(2) *The tools.* The sieve-limit theorem for prime-slice systems (Theorem 4.5): every majorant of level λ saves at most $C\inf_\alpha[\alpha\lambda + \sum_\ell \bar{p}_\ell \ell^{-\alpha}]$ beyond an admissible-set term, plus an error term that is $O(\log^2 \lambda)$ in all applications here (Remark 4.6); its sequential version (Theorem 4.9); the cubic profile $\sum \bar{p}_\ell \ell^{-\alpha} \ll \alpha^{-3}$ of the forced classes (Section 5); the quadratic-residue base built from Mordell's obstruction and the capped sequential measure (Section 7); and a weighted k-ary comparison theorem (Theorem 8.5), which replaces the one-coordinate-per-window hypothesis of the sequential sieve limit. It couples the sequential conditioned law with a product draw, freezes the path, extrapolates in artificial Bernoulli replacement coins by Lagrange interpolation for binomial laws (Lemma 8.3), and pays for the thinning with a supermartingale (Lemma 8.4). Only the mean activated mass enters, because the sequential sieve limit accepts random step costs (Theorem 9.1). To mix the four class types and drop B (Section 10): a base of unit squares, which avoids every small class because no $\mathscr{R}(M)$-, (a, D)- or Case-A class contains a square (Lemmas 10.2 and 10.3); first moments over all moduli with no loss, from the local weight $Z_y(m) = \sum_{p^\nu | m,\, p^\nu \leq y} \Lambda(p^\nu)/\log y$, which is $\geq u/2$ on y-smooth $m \approx y^u$ and has bounded moments, so that its fourth moment reduces smooth sums to congruences modulo small numbers, where Shiu's theorem (for $\mathscr{R}(M)$) and Elsholtz–Tao §7 (for Case A, split by the size of r) apply (Lemmas 10.4–10.9); and second moments by Rankin's trick and Cauchy–Schwarz (Lemma 10.11).

(3) *The campaign sieves are in the class* (Section 11, [PROVED]). The majorant of the 3/4 note is in the class, so 3/4 is sharp for its architecture; the large sieve of the 2/3-loglog note is capped at its own exponent $\frac{2}{3}$ with the $(\log\log N)^{1/3}$ factor.

(4) *Selberg-type majorants* (Section 12). A Λ^2 sieve limit for arbitrary systems in terms of noise stability (Theorem 12.1, [PROVED]), and Λ^2 caps $\ll$

$L^{3/4}(\log L)^{O_r(1)}$ for $\mathscr{R}(M)$-families with at most r prime factors above $(\log X)^8$, r fixed, with (Theorem 12.11) or without (Theorem 12.12) the condition $M \le P(M)^{1+B}$ ([PROVED]). Their new arithmetic input is a Brun–Titchmarsh bound for rough partners from the arithmetic large sieve (Lemma 12.10). As caps they are superseded by Theorem 10.12, which gives $C\lambda^{3/4}$ for every r (including the middle range $r \asymp \log L$ that the Λ^2 method leaves open) and without B (Remark 12.8).

(5) *Beyond CRT evaluation* (Section 13, [PROVED]). For the finite prime-slice families, the cap extends to every rounding bound $\sum_{\theta\neq 0}|\hat{\nu}(\theta)|w(\theta)$ with weight $w \ge 1$, with no bound on the size of the slice primes (Theorem 13.3); to majorants positive only at primes (Theorem 13.4); and to moment methods whose moments are evaluated through CRT densities.

(6) *Large sieves, prime laws and interval counts* (Section 14). By exact duality every CRT-admissible large sieve is bounded below by $N\mathbb{E}|g^*|^2$ for a Selberg-square CRT majorant, so it is capped at $C_A(\log N)^{3/4}$ for frequencies of polynomial level (Theorem 14.2); likewise twisted, multiplicative and hybrid Bessel forms (Theorem 14.5). The large sieve for primes (Theorem 14.6) is capped at $C_A(\log N)^{3/4}(\log\log N)^{3/4}$, and at $C_A(\log N)^{3/4}$ at pointer level only. Gallagher's larger sieve with prime-power kernels saves at most $26\log\log N + C$ over any mixture, unconditionally (Theorem 14.7). For super-polynomial frequencies a dense abstract family shows that comparison measures plus large-sieve axioms cannot give a cap (Theorem 14.11); the forced-class case is Conjecture 14.10. A smooth–rough splitting (Theorem 14.12) prices the coordinates of primes $\le \exp\{(\log N)^{1/4}\}$ by a density alone, so every large sieve at any frequency level is capped at $C(\log N)^{3/4}(\log\log N)^3$ for mixtures with at most one larger prime factor per modulus (Theorem 14.13, [PROVED]), and at $C\gamma^{-3}(\log N)^{3/4}(\log\log N)^3$ for every mixture whose classes with several such primes satisfy the residue-sparseness condition (RS_γ), $\gamma > 0$ fixed (Theorem 14.18, [PROVED]); via damped collisions the rest reduces to a sup-decay hypothesis (A^*) (Theorem 14.17, [PROVED] implication), open for mixtures whose multi-rough part violates (RS_γ) for every fixed $\gamma > 0$. Prime-only majorants are capped for all mixtures, with the same $(\log\log N)^{3/4}$ proviso (Theorem 14.19). Majorants built from classes of modulus $\le N/2$ are capped however the interval sum is evaluated (Corollary 14.23); hybrid methods that count large classes blindly are capped conditionally on a flat-minorant hypothesis, which follows from the combinatorial weak SPW (Hypothesis 14.32, Lemma 14.31, which makes the requirement on the margins exact for this route); SPW with a fixed margin, the form stated in version 4, is false (Theorem 14.30, [PROVED]) (Section 14.9). The results that use Theorem 10.12 as a black box are [PROVED]; labels are given individually there.

(7) *Tuple counts* (Section 15). Witness correlations of order $K \asymp (\log N)^\theta$ with CRT accuracy would give $E(N) \ll N\exp\{-(2/\mathrm{e}^2)(\log N)^\theta\}$ (Theorem 15.3, a [PROVED] implication), but the literal hypothesis meets forced zeros above $\theta = 2/3$ (Theorem 15.17); the alternating form (Hypothesis 15.19) cancels them and still suffices (Theorem 15.20). Under CRT-main-term evaluation, input of prime order k saves at most $C_A[(\log N)^{3/4} + k\log\log N]$, by a cap with truncated weights (Theorem 15.11, Corollary 15.12).

(8) *Exclusions and open problems* (Sections 16 and 17). A precise list of what lies outside: the large sieve at super-polynomial frequency levels against mixtures whose classes with two or more prime factors above $\exp\{(\log N)^{1/4}\}$ violate the residue-sparseness condition (RS_γ) for every fixed $\gamma > 0$; larger-sieve kernels with an uncontrolled factor; direct counts and hybrid accounting above modulus $N/2$; majorants ≥ 1 only on $[1, N]$ or on the exceptional set; non-CRT tuple counts and other non-CRT input; other class types (and non-selector admissible sets); family primes beyond $N^{O(1)}$.

(9) *One sieve limit* (Section 18). The cap $\lambda^{3/4}$ and the pointwise ceiling $1/4$ for the least witness modulus are two readings of one order-k sieve limit (Theorems 18.1 and 18.2, [PROVED] within the stated scopes); the $3/4$ bound holds for primes relative to $\pi(N)$ (a reproof, not an improvement).

None of this bounds $E(N)$ from below or says anything about methods outside the class; in particular it is not evidence against the conjecture.

2. Forced classes

A residue class $b \pmod G$ is *forced* if every positive integer $n \equiv b \pmod G$ is representable. We use three families of forced classes. For an integer $D \geq 1$ put $g(D) = \prod_p p^{\lceil v_p(D)/2 \rceil}$, so that $g(D) \mid D \mid g(D)^2$ and, for every integer x, $D \mid x^2 \iff g(D) \mid x$.

Lemma 2.1 (multiplier identity; [LL, Lemma 2.1]; [PROVED])**.** *Let $M \equiv 3 \pmod 4$, $A = (M+1)/4$, and $A = uvw$ with positive integers u, v, w. If $n \geq 1$ and $nv \equiv -u \pmod M$, then $s = (nv+u)/M$ is a positive integer and $4/n = 1/(suw) + 1/(nsvw) + 1/(nuvw)$.*

Proof. On the common denominator $nsuvw$ the numerators sum to $nv + u + s = s(M+1) = 4suvw$. □

Lemma 2.2 (the classes $\mathscr{R}(M)$; [N, Lemma 18.1]; [PROVED])**.** *For $M \equiv 3 \pmod 4$ let $\mathscr{R}(M)$ be the set of classes $-uv^{-1} \pmod M$ over all factorisations $A = uvw$. Then*

$$\mathscr{R}(M) = \{-4D \pmod M : \ D \mid A^2\}, \qquad |\mathscr{R}(M)| \leq \tau(A^2),$$

and every class of $\mathscr{R}(M)$ is forced.

Proof. Since $4A \equiv 1 \pmod M$, a factorisation $A = uvw$ gives $-uv^{-1} \equiv -u^2wA^{-1} \equiv -4u^2w \pmod M$ with $u^2w \mid A^2$. Conversely, if $A = \prod p^{e_p}$ and $D = \prod p^{d_p}$ with $d_p \leq 2e_p$, put $v_p(u) = \lfloor d_p/2 \rfloor$, $v_p(w) = d_p - 2\lfloor d_p/2 \rfloor$ and $v_p(v) = e_p - v_p(u) - v_p(w) \geq 0$; then $uvw = A$ and $u^2w = D$. Forcedness is Lemma 2.1. □

Lemma 2.3 ((a, D)-classes; [ET, Lemma 3.2]; [PROVED])**.** *For all integers $a, D \geq 1$ the class $n \equiv -(4D + a) \pmod{4a\, g(D)}$ is forced.*

Proof. Write $n+4D+a = 4a\, g(D)\, j$ with $j \geq 1$. Then $M := (n+4D)/a = 4g(D)j - 1$ is $\equiv 3 \pmod 4$ and $(M+1)/4 = g(D)j$, so $D \mid g(D)^2 \mid ((M+1)/4)^2$. Since $n \equiv -4D \pmod M$, n lies in a class of $\mathscr{R}(M)$, and Lemma 2.2 applies. □

These classes encode exactly the Case-B witnesses of the classical parametrisation [N, Thm 3.1(B)] (see [ET, Lemma 3.2]).

Lemma 2.4 (Case-A classes; [ET, §3]; [PROVED])**.** *For $d \geq 1$ and $m \mid 4d+1$ (so m is odd and $(m, 4g(d)) = 1$), the class $n \equiv -m^{-1} \pmod{4g(d)}$ is forced.*

Proof. Put $z_0 = (nm+1)/4$, an integer. Then $g(d) \mid z_0$, so $d \mid z_0^2$; and $m \mid 4d+1$ gives $m \mid 4d + 4z_0$, so $m \mid d + z_0$. With $x = (z_0 + d)/m$ and $y = (z_0 + z_0^2/d)/m = z_0(d + z_0)/(dm)$, both integers since $(m, d) = 1$, one has $1/x + 1/y = m/z_0$, hence $1/x + 1/y + 1/(nz_0) = (nm+1)/(nz_0) = 4/n$. □

We write *Case-B classes* for the classes of $\mathscr{R}(M)$ and the (a, D)-classes, and *Case-A classes* for those of Lemma 2.4. Two facts are used repeatedly.

Lemma 2.5 ($n = 1$ is never covered; [PROVED])**.** *No forced class contains $n = 1$.*

Proof. $4/1 > 3 \geq 1/x_1 + 1/x_2 + 1/x_3$. □

Lemma 2.6 (Mordell's obstruction; classical; [TW, Lemma 1.1]; [PROVED])**.** *Let $M \equiv 3 \pmod 4$, $A = (M+1)/4$ and $D \mid A^2$. Then $(D, M) = 1$ and the Jacobi symbol satisfies $\left(\frac{-4D}{M}\right) = -1$. Consequently, if n is a nonzero quadratic residue modulo every prime factor of M, then n lies in no class of $\mathscr{R}(M)$.*

Proof. $(A, M) = (A, 4A - 1) = 1$. Write $D = sr^2$ with s squarefree; then $D \mid A^2$ gives $sr \mid A$, so $M \equiv -1 \pmod{4s}$. Now $\left(\frac{-4D}{M}\right) = \left(\frac{-1}{M}\right)\left(\frac{s}{M}\right) = -\left(\frac{s}{M}\right)$. If s is odd, reciprocity and $\left(\frac{M}{s}\right) = \left(\frac{-1}{s}\right)$ give $\left(\frac{s}{M}\right) = 1$, since $(M-1)/2$ is odd. If $s = 2s'$, then $8 \mid M + 1$, so $\left(\frac{2}{M}\right) = 1$, and $\left(\frac{s'}{M}\right) = 1$ as before. For the consequence, $\left(\frac{n}{M}\right) = \prod_{p^e \| M} \left(\frac{n}{p}\right)^e = 1$. □

3. Congruence sieves: the architecture class

3.1. CRT systems and majorants.

Definition 3.1 (prime-slice CRT system [ET, §1])**.** The data are a modulus Q_0; a finite set $\mathcal{P}$ of primes not dividing Q_0 (the *slice primes*); a set $R \subseteq \mathbb{Z}/Q_0$ of admissible small residues; and, for each $c \in R$ and $\ell \in \mathcal{P}$, a set $F_\ell(c) \subseteq \mathbb{Z}/\ell$ of forbidden classes. The *avoider set* is

$$\mathcal{A} = \{n : \ n \bmod Q_0 \in R, \ n \bmod \ell \notin F_\ell(n \bmod Q_0) \text{ for all } \ell \in \mathcal{P}\}.$$

Put $p_\ell(c) = |F_\ell(c)|/\ell$ and $\bar{p}_\ell = |R|^{-1} \sum_{c \in R} p_\ell(c)$.

A family of conditions $n \equiv b \pmod{q_0 \ell}$ with $q_0 \mid Q_0$ and $\ell \in \mathcal{P}$ induces such a system with $F_\ell(c) = \{b \bmod \ell : \ b \equiv c \pmod{q_0}\}$. A typical R is a *selector* $R = \{c : (c, P) = 1\}$ with $P \mid Q_0$.

Definition 3.2 (majorant of level λ)**.** A *majorant* of $\mathcal{A}$ is a finite real combination

$$\nu(n) = \sum_i a_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$$

with $\nu \geq 0$ on all of $\mathbb{Z}$ and $\nu \geq 1$ on $\mathcal{A}$. It has *level* λ if $\sum_{\ell \in \mathcal{P},\ \ell \mid d_i} \log \ell \leq \lambda$ for every i. Its *mean* is $\mathbb{E}\nu = \sum_i a_i/d_i$, the average over a common period, and its *saving* is $\log(1/\mathbb{E}\nu)$.

The level charges only the slice primes; the small part of d_i is free. This is generous to the sieve. For every N,

$$\sum_{n \leq N} \nu(n) = N \cdot \mathbb{E}\nu + O\Big(\sum_i |a_i|\Big), \tag{1}$$

since each class meets $[1, N]$ in $N/d_i + O(1)$ points.

3.2. **The architecture class.**

Definition 3.3 (CRT-majorant architecture)**.** A *CRT-majorant architecture* for a family $\mathfrak{F}$ of forced classes bounds $\#(\mathcal{A}\cap[1,N])$, and hence the exceptional integers in $[1,N]$ that satisfy the admissibility condition $n \bmod Q_0 \in R$, as follows.

- (A1) ν is a majorant of the avoider set $\mathcal{A}$ of $\mathfrak{F}$ in the sense of Definition 3.2: $\nu \geq 0$ on $\mathbb{Z}$ and $\nu \geq 1$ on the *whole* avoider set, not only on exceptional primes.
- (A2) The final bound is $N\cdot\mathbb{E}\nu + T$ with $T = \sum_i |a_i|$, and it is non-trivial: $T < N$.
- (A3) The slice primes of the family are at most N^A for a fixed A.

Every sieve majorant in use satisfies (A1): Bonferroni truncations $Q_r(H) = \binom{H-1}{r}$ and Selberg squares. Both campaign notes satisfy (A1)–(A3) (Section 11). A selector admissible set $R = \{c : (c,P) = 1\}$ is the same as adding the classes $0 \bmod p$, $p \mid P$, to the family; in this form the main theorem (Theorem 10.14) needs no admissible set at all, provided all primes of the family, selector primes $p \mid P$ and the primes of the small parts q_0 included, are $\leq N^A$. Condition (A3) bounds only the slice primes, so this is an extra hypothesis.

Lemma 3.4 (coefficient budget implies level; [ET, Lemma 2.9], from [ETr, item 3]; [PROVED])**.** *Take a prime-slice system, or the sequential setting of Section 4.4. Let ν be a majorant of $\mathcal{A}$ with $T = \sum_i |a_i|$, and assume that every slice prime satisfies $\log \ell \leq \Lambda_0$. Then for every λ there is a majorant ν' of level at most λ with $\nu' \geq \nu$ pointwise and $\mathbb{E}\nu' \leq \mathbb{E}\nu + T\mathrm{e}^{\Lambda_0-\lambda}$. In particular $\lambda = \Lambda_0 + \log T + \log(1/\mathbb{E}\nu)$ gives $\mathbb{E}\nu' \leq 2\mathbb{E}\nu$.*

Proof. Keep every term of level at most λ. Drop each term of level $> \lambda$ with $a_i < 0$; this raises ν pointwise and raises the mean by $|a_i|/d_i \leq |a_i|\mathrm{e}^{-\lambda}$. For a term of level $> \lambda$ with $a_i > 0$, list its slice primes in increasing order and keep the longest prefix of level at most λ. The next prime has $\log \ell \leq \Lambda_0$, so the prefix has level $> \lambda - \Lambda_0$. Replace d_i by the divisor d_i' formed by the non-slice part of d_i and the kept prime powers. Then $\mathbf{1}[n \equiv b_i \ (d_i')] \geq \mathbf{1}[n \equiv b_i \ (d_i)]$, and the mean rises by at most $a_i/d_i' \leq a_i\mathrm{e}^{\Lambda_0-\lambda}$. So $\nu' \geq \nu$, ν' is a majorant, and summing gives the bound. □

Corollary 3.5 ([PROVED])**.** *Assume (A2)–(A3) and put $s = \log(1/\mathbb{E}\nu)$. Then there is a majorant $\nu' \geq \nu$ of level at most $(A+2)\log N$ with $\mathbb{E}\nu' \leq \mathbb{E}\nu + N^{-1}$. Hence, if every majorant of level at most $(A+2)\log N$ has saving at most $C(\log N)^{3/4} + C'$, then $s \leq \log 2 + C(\log N)^{3/4} + C'$ for all $N \geq N_0(C,C')$.*

Proof. Lemma 3.4 with $\Lambda_0 = A\log N$, $\lambda = (A+2)\log N$ and $T < N$ gives $\mathbb{E}\nu' \leq \mathbb{E}\nu + N\cdot N^A \cdot N^{-A-2} \leq \mathbb{E}\nu + N^{-1}$. So $\log(1/\mathbb{E}\nu') \geq \min(s, \log N) - \log 2$, and the hypothesis gives $\min(s,\log N) \leq \log 2 + C(\log N)^{3/4} + C'$. For $N \geq N_0$ the right side is $< \log N$, so the minimum is s. □

This form needs the cap only at level $(A+2)\log N$.

Remark 3.6 (positivity on all of $\mathbb{Z}$). Condition (A1) cannot be weakened to positivity on $[1,N]$. Restricted to $[1,N]$, the class $\mathbf{1}[n \equiv b \pmod N]$ is the point mass at b, so the span of level-$\log N$ class indicators contains every function on $[1,N]$. The relaxed linear programme is then the exact count, and evaluating it presupposes knowing the avoider set [ET, §2.4]. Such methods are not sieves in the present sense.

4. The sieve-limit theorem

All results of this section are from [ET, §2]; the review [ETr, items 1–2] found them sound and checked every intermediate inequality against exact linear programmes on small systems. Write $C_4 = 8\mathrm{e}^5$.

4.1. Two one-dimensional lemmas.

Lemma 4.1 (binomial mass; [PROVED]). *Let $\psi = \mathrm{Bin}(z,q)$ with $0 < q \le 1/4$ and $\mu' = zq$. For every integer $1 \le y \le z-1$, $\psi(y) \ge (8y)^{-1/2}\exp\{-\frac{4}{3}(y-\mu')^2/\mu'\}$.*

Proof. By [MS, Ch. 10, Lemma 7], $\binom{z}{y} \ge (8y(1-y/z))^{-1/2}\mathrm{e}^{zH(y/z)}$, so $\psi(y) \ge (8y)^{-1/2}\mathrm{e}^{-zD(y/z\|q)}$. The bound $D(a\|q) \le (a-q)^2/(q(1-q))$ gives $zD \le (y-\mu')^2/(\mu'(1-q)) \le \frac{4}{3}(y-\mu')^2/\mu'$. □

Lemma 4.2 (node lemma; [PROVED]). *Keep ψ, q, μ' as in Lemma 4.1. For a set Y of $k+1$ distinct integers in $[0,z]$ let $B(Y) = \max_{y\in Y}|\ell_y^Y(0)|/\psi(y)$, where ℓ_y^Y is the Lagrange basis polynomial of Y. Then every polynomial P of degree at most k with $P \ge 0$ on Y satisfies $|P(0)| \le B(Y)\sum_{y\in Y}\psi(y)P(y)$. Let $B(k) = \min_Y B(Y)$ and $C_1 = 16\mathrm{e}^6$. Then*

(a) $\log B(0) \le \frac{1}{2}\log(8\mu'+8) + 2$;
(b) $\log B(k) \le 4\log(4/3)\mu' \le 1.151\mu'$ *for every* $k \le z$;
(c) *if* $\mu' \ge 64$ *and* $1 \le k \le \mu'/16$, *then* $\log B(k) \le \frac{k}{2}\log(C_1\mu'/k) + \frac{1}{2}\log(16\mu')$.

Consequently, for all $\alpha, s > 0$ and $0 \le k \le z$,

$$\log B(k) \le 19\alpha k s + C_4\mu'\mathrm{e}^{-2\alpha s} + \tfrac{1}{2}\log(16\mu'+16) + 74. \tag{2}$$

Proof. The inequality for P is Lagrange interpolation $P(0) = \sum_y \ell_y(0)P(y)$ with $P \ge 0$ on Y. (a) If $\mu' \ge 1$ then $z \ge 4$; take the single node $y = \lceil\mu'\rceil \le z-1$ and apply Lemma 4.1 with $|y-\mu'| \le 1$. If $\mu' < 1$ take the node 0: $\psi(0) = (1-q)^z \ge \mathrm{e}^{-1.151\mu'}$. (b) Take $Y = \{0,\dots,k\}$; then $\ell_y(0) = \delta_{y0}$ and $B = 1/\psi(0) \le \mathrm{e}^{1.151\mu'}$. (c) Take nodes $y_i = a+ih$, $0 \le i \le k$, with $W = \lceil\sqrt{k\mu'}/2\rceil$, $h = \lceil 2W/k\rceil$, $a = \lceil\mu'\rceil - W$. Since $k \le \mu'/16$ and $\mu' \ge 64$, every node lies in $[\mu'/2, 2\mu'] \subseteq [1, z-1]$ and $|y_i - \mu'| \le \sqrt{k\mu'}$, so Lemma 4.1 gives $\psi(y_i) \ge (16\mu')^{-1/2}\mathrm{e}^{-4k/3}$. Next $|\ell_i(0)| = \prod_{j\ne i} y_j/(h^k i!(k-i)!)$, and $\prod_{j\ne i} y_j \le (2\mu')^k$, $k!/(i!(k-i)!) \le 2^k$, $k! \ge (k/\mathrm{e})^k$ and $h \ge \sqrt{\mu'/k}$ give $|\ell_i(0)| \le (4\mathrm{e}\sqrt{\mu'/k})^k$. Multiplying gives (c), since $16\mathrm{e}^{2+8/3} \le C_1$.

For (2): when (c) applies, use $\frac{k}{2}\log(C_1\mu'/k) \le \alpha k s + (C_1\mu'/2\mathrm{e})\mathrm{e}^{-2\alpha s}$ (the maximum over real k of the difference), and $C_1/(2\mathrm{e}) = C_4$. Otherwise use (a) or (b): if $\mu' < 64$ then $1.151\mu' \le 74$; if $k > \mu'/16$ and $\alpha s \ge 1$ then $1.151\mu' \le 18.5k \le 19\alpha k s$; if $\alpha s < 1$ then $1.151\mu' \le 8.6\mu'\mathrm{e}^{-2\alpha s}$. □

Lemma 4.3 (combination technique; standard; [PROVED]). *Let $\Lambda \subset \mathbb{Z}_{\ge 0}^G$ be a lower set and P_Λ the span of the monomials x^a, $a \in \Lambda$. For each coordinate g and degree k let $I_k^{(g)}$ be Lagrange interpolation at any $k+1$ distinct nodes, and put $c_j = \sum_{e\in\{0,1\}^G,\ j+e\in\Lambda}(-1)^{|e|}$. Then for $Q \in P_\Lambda$, $Q = \sum_{j\in\Lambda} c_j\big(\bigotimes_g I_{j_g}^{(g)}\big)Q$, and $|c_j| \le 2^G$.*

Proof. Put $\Delta_k = I_k - I_{k-1}$ with $I_{-1} = 0$; expanding $\sum_{j\in\Lambda}\bigotimes_g \Delta_{j_g}$ gives the coefficients c_j. For $a \in \Lambda$, $\Delta_{j_g}x_g^{a_g} = 0$ when $j_g > a_g$, and the box $\{j \le a\}$ lies in Λ, so the sum applied to x^a is $\prod_g\sum_{j_g\le a_g}\Delta_{j_g}x_g^{a_g} = x^a$. □

4.2. **The Boolean proposition.** Let $x_i \sim \mathrm{Bern}(p_i)$, $i \in I$, be independent, with weights $s_i \geq s_* > 0$. A function f on $\{0,1\}^I$ is *λ-level* if $f = \sum_T f_T$ where f_T depends only on x_T and $\sum_{i\in T} s_i \leq \lambda$.

Proposition 4.4 ([ET, Prop. 2.4]; [PROVED])**.** *Assume $p_i \leq 1/4$ whenever $s_i \leq \lambda$, and $p_i < 1$ for all i. Let f be λ-level with $f \geq 0$ and $f(0) \geq 1$, and let $\lambda \geq s_*$. Put $G = \lfloor \log_2(\lambda/s_*) \rfloor + 1$ and $\mu = \sum_{s_i \leq \lambda} p_i$. Then for every $\alpha > 0$,*

$$\log \frac{1}{\mathbb{E}f} \leq 19\alpha\lambda + C_4 \sum_{s_i \leq \lambda} p_i \mathrm{e}^{-\alpha s_i} + G\big(75 + \log(2 + \lambda/s_*)\big) + \frac{G}{2}\log(16\mu + 16). \quad (3)$$

Proof. Step 0. No f_T contains an i with $s_i > \lambda$, so f ignores those coordinates; discard them, coordinates with $p_i = 0$, and empty bands.

Step 1 (bands). Put $B_g = \{i : 2^g s_* \leq s_i < 2^{g+1} s_*\}$ for $0 \leq g < G$, $s_g = 2^g s_*$ and $q_g = \max_{i\in B_g} p_i \leq 1/4$.

Step 2 (thinning). Take independent $u_i \sim \mathrm{Bern}(q_g)$ and $w_i \sim \mathrm{Bern}(p_i/q_g)$; then $u_i w_i$ has the law of x_i. Fix w, let $Z_g = \{i \in B_g : w_i = 1\}$ and $z_g = |Z_g|$. Then $f_w(u) = f(u \circ w)$ depends only on u_Z, is λ-level and nonnegative, and $f_w(0) \geq 1$.

Step 3 (symmetrisation). Average f_w over $\prod_g \mathrm{Sym}(Z_g)$, which preserves the law of u. Each monomial u^S of a term f_T has multidegree $(|S \cap Z_g|)_g$ in the lower set $\Lambda = \{j : \sum_g j_g s_g \leq \lambda\}$, and symmetrises to $\prod_g \binom{K_g}{|S_g|}/\binom{z_g}{|S_g|}$ with $K_g = |u_{Z_g}|$. Hence $\mathbb{E}_u f_w = \mathbb{E}Q(K)$ where $Q \in P_\Lambda$, the $K_g \sim \mathrm{Bin}(z_g, q_g)$ are independent, $Q \geq 0$ on $\prod_g\{0,\dots,z_g\}$ and $Q(0) \geq 1$. Reducing modulo $\prod_{i=0}^{z_g}(K_g - i)$ we may take $Q \in P_{\Lambda'}$, $\Lambda' = \Lambda \cap \prod_g [0, z_g]$.

Step 4 (interpolation). Choose the node sets of Lemma 4.2 inside $\{0,\dots,z_g\}$ and apply Lemma 4.3 at 0. All grid points are support points, where $Q \geq 0$, so

$$1 \leq Q(0) \leq \Big(2^G |\Lambda'| \max_{j\in\Lambda'} \prod_g B_g(j_g)\Big)\mathbb{E}Q(K).$$

(If $z_g = 0$ then $j_g = 0$ on Λ' and $B_g(0) = 1$.)

Step 5 (bounds). Apply (2) with $s = s_g$ and $\mu' = q_g z_g$, using $\sum_g j_g s_g \leq \lambda$ and $|\Lambda| \leq (1 + \lambda/s_*)^G$: $\log(1/\mathbb{E}_u f_w) \leq \Phi(w)$, where $\Phi(w) = 19\alpha\lambda + \sum_g [C_4 q_g z_g \mathrm{e}^{-2\alpha s_g} + \frac{1}{2}\log(16 q_g z_g + 16) + 74] + G\log 2 + G\log(1 + \lambda/s_*)$. By Jensen, $\mathbb{E}f = \mathbb{E}_w \mathbb{E}_u f_w \geq \mathrm{e}^{-\mathbb{E}_w \Phi(w)}$. Finally $\mathbb{E}_w q_g z_g = \mu_g := \sum_{i\in B_g} p_i$, concavity of log, $\mathrm{e}^{-2\alpha s_g} \leq \mathrm{e}^{-\alpha s_i}$ on B_g, and $\sum_g \log(16\mu_g + 16) \leq G\log(16\mu + 16)$. □

4.3. **The CRT form.**

Theorem 4.5 (sieve limit for prime-slice systems; [ET, Thm 2.5]; [PROVED])**.** *Take a prime-slice system. Assume that for every $c \in R$, $|F_\ell(c)| \leq \ell/4$ for every $\ell \in \mathcal{P}$ with $\ell \leq \mathrm{e}^\lambda$, and $|F_\ell(c)| < \ell$ for every $\ell \in \mathcal{P}$. Let $\bar\mu = \sum_{\ell\in\mathcal{P},\ \ell\leq \mathrm{e}^\lambda} \bar p_\ell$ be the truncated mass, $s_* = \log\min\mathcal{P}$, and $G = \lfloor \log_2(\lambda/s_*)\rfloor + 1$. Then every majorant ν of $\mathcal{A}$ of level $\lambda \geq s_*$ satisfies, for every $\alpha > 0$,*

$$\begin{aligned}\log\frac{1}{\mathbb{E}\nu} \leq\ & \log\frac{Q_0}{|R|} + 19\alpha\lambda + C_4 \sum_{\ell\in\mathcal{P},\ \ell\leq \mathrm{e}^\lambda} \bar p_\ell\, \ell^{-\alpha} \\ & + G\big(75 + \log(2 + \lambda/s_*)\big) + \frac{G}{2}\log(16\bar\mu + 16).\end{aligned} \quad (4)$$

If $\lambda < s_*$, ν depends only on the small residue and the bound holds without the λ-terms.

Proof. Use the uniform measure on $\mathbb{Z}/Q_{\mathrm{tot}}$ for a common period. By CRT the coordinates $n \bmod Q_0^*$ (the Q_0-primary part of Q_{tot}), $n \bmod \ell^{e_\ell}$ ($\ell \in \mathcal{P}$) and the rest are independent and uniform. Since $\nu \geq 0$, $\mathbb{E}\nu \geq (|R|/Q_0)\operatorname{avg}_{c\in R}\mathbb{E}[\nu \mid n \bmod Q_0 = c]$. Fix $c \in R$. The hit indicators $x_\ell = \mathbf{1}[n \bmod \ell \in F_\ell(c)]$ are independent $\mathrm{Bern}(p_\ell(c))$, independent of the finer digits. Each term of ν, conditioned on (c, x), depends only on x_{T_i} with $T_i = \{\ell \in \mathcal{P} : \ell \mid d_i\}$, so $\nu_c = \mathbb{E}[\nu \mid c, x]$ is λ-level with weights $s_\ell = \log \ell$, and $\nu_c \geq 0$. The event $\{c\} \times \{x = 0\}$ has positive probability and lies in $\mathcal{A}$, so $\nu_c(0) \geq 1$. Apply Proposition 4.4 in each fibre, then Jensen over $c \in R$; the right side of (3) is linear in the p_ℓ except for the concave logarithmic term. □

Remark 4.6 (size of the error terms). The last two terms of (4) are $O(\log^2 \lambda)$ when $s_* \gg 1$ and $\log \bar{\mu} \ll \log \lambda$. In general $\bar{\mu}$ may be as large as $\pi(\mathrm{e}^\lambda)/4$, and then they are $\asymp \lambda \log \lambda$. In every application below $\bar{\mu} \ll \lambda^3$.

Remark 4.7 (the R-term). The term $\log(Q_0/|R|)$ is a saving that the theorem leaves uncontrolled: $\nu = \mathbf{1}_R$ is a majorant with no slice cost and saving exactly $\log(Q_0/|R|)$. For a selector $R = \{(c, P) = 1\}$ the term is $\log(P/\varphi(P)) \leq \log\log\log P + O(1)$. If R itself encodes forced classes with moduli dividing Q_0, the term is the void probability of a non-slice subsystem, about which Theorem 4.5 says nothing.

Define the *Rankin functional*

$$\Psi(\lambda) = \inf_{\alpha>0}\Big[\alpha\lambda + \sum_{\ell} \bar{p}_\ell\, \ell^{-\alpha}\Big]. \tag{5}$$

Theorem 4.5 says that no majorant of level λ saves more than $C\Psi(\lambda) + O(\log^2 \lambda)$ beyond the R-term, in the situation of Remark 4.6. No converse is claimed in general; attainability is discussed only for the two campaign families (Section 11).

Remark 4.8 (Montgomery's large sieve; [ET, Remark 2.6]; [PROVED]). Montgomery's large sieve [M] applied in a fibre c gives a bound $Y/S_c(Q)$ with $Q^2 \leq Y$, where $S_c(Q) = \sum_{q\leq Q} \mu^2(q) \prod_{\ell\mid q} g_\ell$ and $g_\ell = |F_\ell(c)|/(\ell - |F_\ell(c)|)$. Rankin's trick gives $S_c(Q) \leq Q^\alpha \prod_\ell (1 + g_\ell \ell^{-\alpha}) \leq \exp\{\alpha \log Q + 2\sum_\ell p_\ell(c)\ell^{-\alpha}\}$ when $p_\ell(c) \leq 1/2$. Summed over fibres, Jensen gives $\sum_c Y\mathrm{e}^{-\Phi_c} \geq |R|\, Y\mathrm{e}^{-\operatorname{avg}\Phi_c}$. So the large-sieve bound over a slice system obeys the same functional, with level $\lambda = \log Q$, directly and without duality. No sequential or shared-prime large-sieve statement is made here; for mixed and composite settings see Theorem 14.2. The condition $p_\ell(c) \leq 1/2$, which gives $g_\ell \leq 2p_\ell(c)$, is the reading of the Rankin step in [ETr, item 3.4].

4.4. **Sequential windows: shared large primes.** Theorem 4.5 needs the slice primes to be disjoint from the small modulus. The following extension allows a large prime to be the slice prime of one condition and a cofactor prime of another.

Setting [ET, §2.6]. Take Q_0, $R \subseteq \mathbb{Z}/Q_0$, and a set $\mathcal{P}$ of primes not dividing Q_0, split into ordered windows $\mathcal{P} = W_1 \sqcup \cdots \sqcup W_J$. A *condition* C is an event

$$E_C = \{n \equiv a_C \pmod{q_C},\ n \equiv b_{C,\ell} \pmod{\ell^{e_{C,\ell}}} \text{ for } \ell \in S(C)\}$$

with $q_C \mid Q_0$, $\emptyset \neq S(C) \subseteq \mathcal{P}$ and $e_{C,\ell} \geq 1$. Assume

(U) the last window meeting $S(C)$ meets it in exactly one prime $\ell(C)$, and $e_{C,\ell(C)} = 1$.

Fix $E_\ell \geq \max_C e_{C,\ell}$. The *history* before window j is $H_{<j} = (n \bmod Q_0, (n \bmod \ell^{E_\ell})_{\ell\in W_{<j}})$. For $\ell \in W_j$ and a history h let $F_\ell(h)$ be the set of $b_{C,\ell}$ over the conditions C with $\ell(C) = \ell$ whose other requirements are met by h, and $p_\ell(h) = |F_\ell(h)|/\ell$.

The avoider set is $\mathcal{A} = \{n : n \bmod Q_0 \in R,\ n \notin E_C\ \forall C\}$. Let Q_{seq} be the law of the history built window by window: $c = n \bmod Q_0$ uniform on R; given $H_{<j}$, the residues $n \bmod \ell$ ($\ell \in W_j$) independent and uniform on $\mathbb{Z}/\ell \setminus F_\ell(H_{<j})$, with uniform independent higher digits. Majorants of level λ charge all primes of $\mathcal{P}$.

Theorem 4.9 (sequential sieve limit; [ET, Thm 2.7]; [PROVED])**.** *Assume that on the support of Q_{seq}, $p_\ell(h) \le 1/4$ whenever $\log \ell \le \lambda$, and $p_\ell(h) < 1$ always. Then for all $\alpha_1, \dots, \alpha_J > 0$, every majorant ν of level λ satisfies*

$$\log \frac{1}{\mathbb{E}\nu} \le \log \frac{Q_0}{|R|} + \sum_j \mathbb{E}_{Q_{\mathrm{seq}}}\big[\Phi_j(H_{<j})\big],$$

where $\Phi_j(h)$ is the right side of (3) for the coordinates $\ell \in W_j$ with $\log \ell \le \lambda$, probabilities $p_\ell(h)$ and $\alpha = \alpha_j$. A window with $\log \min W_j > \lambda$ contributes 0.

Proof. Let $g_j(h) = \mathbb{E}[\nu \mid H_{<j} = h]$ under the uniform measure, and let $\mathcal{A}_{<j}$ be the histories in R satisfying no condition whose last window is before j. We show by downward induction that for $h \in \mathcal{A}_{<j}$, $-\log g_j(h) \le \mathbb{E}_{Q_{\mathrm{seq}}}[\sum_{i \ge j} \Phi_i(H_{<i}) \mid H_{<j} = h]$. For $j = J + 1$ the history decides membership in $\mathcal{A}$, and $\nu \ge 1$ on $\mathcal{A}$. For the step, fix $h \in \mathcal{A}_{<j}$. Given h, the residues $n \bmod \ell^{E_\ell}$, $\ell \in W_j$, are independent and uniform, so the indicators $x_\ell = \mathbf{1}[n \bmod \ell \in F_\ell(h)]$ are independent $\mathrm{Bern}(p_\ell(h))$. Put $f(x) = \mathbb{E}[g_{j+1}(H_{<j+1}) \mid h, x]$. Then $\mathbb{E}f = g_j(h)$, $f \ge 0$, and f is λ-level on W_j. The event $x = 0$ is exactly that no condition with last window j holds; given it, $H_{<j+1} \in \mathcal{A}_{<j+1}$ with the Q_{seq} transition law. The induction hypothesis and Jensen give $f(0) > 0$ and the bound for $f(0)$; Proposition 4.4 for $f/f(0)$ gives $\mathbb{E}f \ge f(0)\mathrm{e}^{-\Phi_j(h)}$. Finally $\mathbb{E}\nu \ge (|R|/Q_0)\,\mathrm{avg}_{c \in R} g_1(c)$ and Jensen over c. □

Lemma 4.10 (profile under Q_{seq}; [ET, Lemma 2.8]; [PROVED])**.** *For $\ell \in W_j$,*

$$\mathbb{E}_{Q_{\mathrm{seq}}} p_\ell(H_{<j}) \le \sum_{C:\ \ell(C) = \ell} \frac{1}{\ell}\, \mathbb{P}(c \equiv a_C\ (q_C) \mid c \in R) \prod_{\ell' \in S(C) \setminus \{\ell\}} \frac{1}{\ell'^{e_{C,\ell'}}(1 - p^*_{\ell'})},$$

where $p^*_{\ell'}$ is the supremum of $p_{\ell'}(h)$ over reachable histories.

Proof. Chain rule over windows: given the past, $n \bmod \ell'$ is uniform on at least $\ell'(1 - p^*_{\ell'})$ residues, and the higher digits are uniform. □

5. Supply profiles

We use Shiu's theorem [S, Theorem 1] in the form stated in [TQ, §3]: if $F \ge 0$ is multiplicative, $F(p^a) \le A_1^a$, and $F(n) \ll_\epsilon n^\epsilon$, then for fixed $0 < \alpha_s, \beta_s < 1/2$,

$$\sum_{\substack{V - Y < n \le V \\ n \equiv a\ (q)}} F(n) \ll \frac{Y}{\varphi(q) \log V} \exp\Big\{ \sum_{p \le V,\ p \nmid q} \frac{F(p)}{p} \Big\}$$

uniformly for $q \ge 2$, $(a, q) = 1$, $q < Y^{1-\alpha_s}$, $V^{\beta_s} < Y \le V$.

Lemma 5.1 (cubic Case-B profile, $\mathscr{R}(M)$; [ET, Lemma 3.1]; [PROVED])**.** *Put $S_B(x) = \sum_{M \le x,\ M \equiv 3\ (4)} \tau(A^2) M/\varphi(M)$ with $A = (M+1)/4$. Then $S_B(x) \ll x(\log 2x)^2$. Consequently, for $0 < \beta \le 1$, $\sum_{M \equiv 3\ (4)} \tau(A^2)(M/\varphi(M)) M^{-1-\beta} \le C_6 \beta^{-3}$.*

Proof. Write $M/\varphi(M)=\sum_{d\mid M}\mu^2(d)/\varphi(d)$ and swap sums; the inner sum runs over $A\le (x+1)/4$ with $4A\equiv 1 \pmod d$. For $2\le d\le x^{1/2}$, Shiu with $F(A)=\tau(A^2)$ ($F(p)=3$) in the reduced class $4^{-1} \pmod d$ gives $\ll (x/\varphi(d))(\log x)^2$; for $d=1$ the elementary bound $\sum_{A\le x}\tau(A^2)\ll x\log^2 x$ suffices; and $\sum_d \varphi(d)^{-2}<\infty$. For $d>x^{1/2}$ bound trivially by $(x/d+1)x^{1/10}$, for a total $\ll x^{0.61}$. The β-bound follows by partial summation, $\int(\log 2x)^2 x^{-1-\beta}dx\ll\beta^{-3}$. □

Numerically $S_B(x)/(x\log^2 x)=0.1067,\ 0.0990,\ 0.0963$ at $x=10^2,10^4,10^6$ [EVIDENCE].

Lemma 5.2 ((a,D) profile; [ET, Lemma 3.2]; [PROVED]). *With $G=4a\,g(D)$, $\sum_{a,D}(G/\varphi(G))G^{-1-\beta}\ll\beta^{-3}$ for $0<\beta\le 1$.*

Proof. Using $\#\{D: g(D)=g\}\le 2^{\omega(g)}$, the sum is at most

$$\ll\Big(\sum_a a^{-1-\beta}\frac{a}{\varphi(a)}\Big)\Big(\sum_g 2^{\omega(g)}\frac{g}{\varphi(g)}g^{-1-\beta}\Big)\ll\beta^{-1}\beta^{-2}.$$

□

Lemma 5.3 (weighted Case-A divisor sum; [ET, Lemma 3.7]; [PROVED MOD] Elsholtz–Tao Prop. 1.4). *For $x\ge 2$,*

$$\sum_{r,h\ge 1,\ rh\le x}\tau(4rh^2+1)\,\frac{rh}{\varphi(rh)}\ll x(\log x)^2. \tag{6}$$

The same holds with the extra weight $\gamma(rh)$ of Corollary 6.2 below. Consequently the Case-A classes $-m^{-1} \pmod{4g(d)}$, with the selector weight, have profile $\ll\beta^{-3}$.

Proof. The input is [ElT, Prop. 1.4]: for $A,B>1$ and every positive integer $k\ll(AB)^{O(1)}$, $\sum_{a\le A}\sum_{b\le B}\tau(kab^2+1)\ll AB\log(A+B)\log(1+k)$. Use $n/\varphi(n)\le(\pi^2/6)\sum_{d\mid n}1/d$ and write $r=sr'$, $h=th'$, so that $4rh^2=(4st^2)r'h'^2$. The left side of (6) is at most $C\sum_{s,t}(st)^{-1}\sum_{r'h'\le Y}\tau(4st^2r'h'^2+1)$ with $Y=x/(st)$. If $st\le\sqrt{x}$, cover $r'h'\le Y$ by $O(\log Y)$ rectangles with $AB\asymp Y\ge\sqrt{x}/2$, $A,B\ge 2$; then $k=4st^2\ll(AB)^3$, and Prop. 1.4 gives $\ll Y(\log x)^2\log(1+4st^2)$; finally $\sum_{s,t}\log(1+4st^2)/(st)^2<\infty$. If $st>\sqrt{x}$ then $Y<\sqrt{x}$ and $\tau\ll x^{\epsilon}$ give $\ll x^{1/2+2\epsilon}$. For the γ-weight replace $1/(st)$ by $\eta(s)\eta(t)$, where $\gamma(n)n/\varphi(n)=\sum_{d\mid n}\eta(d)$ and $\eta(p)=O(p^{-1}+p^{-\delta})$; the needed sums converge because $\eta(p)/p\ll p^{-1-\delta}$ and $\sum_n\eta(n)n^{-1+\delta/2}<\infty$. The profile statement follows by partial summation, since the Case-A class mass up to modulus G is at most $\sum_{4rh\le G}\tau(4rh^2+1)/(4rh)$. □

The review [ETr, item 4] checked Prop. 1.4 and its k-range against the source; it was not re-proved.

6. Dominant-prime families

This section treats families whose moduli have a dominant prime factor. It covers the (a,D)- and Case-A classes as well as the classes $\mathscr{R}(M)$; for the latter, Theorem 9.3 below removes the dominance hypothesis, and Theorem 10.12 removes it for all four class types at once. For CRT-majorant architectures with a selector admissible set, $\nu\ge 1$ on the whole avoider set, and all family primes $\le N^A$, the conclusions of this section are special cases of the bounded-B clause of Theorem 10.14 (a dominant prime $P(M)\ge M^{1/(1+C)}$ is the case $B=C$). Theorem 9.3 is implied

only for majorants ≥ 1 on the whole avoider set; its form for majorants ≥ 1 only on the avoiders in R_W is not implied by Theorem 10.14 and remains a separate statement.

6.1. **Prime-slice families.**

Corollary 6.1 ([ET, Cor. 3.4]; [PROVED]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4)**.** *Fix $0 < C < 1$ and $A \geq 1$. Take a prime-slice system in which every condition is a slice condition (no pure small-modulus conditions), and*

- *every condition is a forced class of $\mathscr{R}(M)$ (any splitting $M = q_0\ell$), an (a, D)-class, or a Case-A class;*
- *each condition modulus $q_0\ell$ satisfies $q_0 \leq \ell^C$;*
- *$R = \{c : (c, P) = 1\}$ for some $P \mid Q_0$;*
- *$\ell \geq \ell_0(C)$ for every slice prime.*

Then every majorant of level $\lambda \leq A \log N$ satisfies

$$\log \frac{1}{\mathbb{E}\nu} \leq C_7(A, C)(\log N)^{3/4} + \log \frac{P}{\varphi(P)},$$

and so does the large-sieve bound of Remark 4.8, summed over fibres.

Proof. Since $q_0 \leq \ell^C$ and the number of family classes modulo M is $M^{o(1)}$, $|F_\ell(c)| \leq \ell^{C+o(1)} \leq \ell/4$. For a selector, $\mathbb{P}(c \equiv b \ (q_0) \mid R) \leq 1/\varphi(q_0)$, so $\bar{p}_\ell \leq \sum_{M=q_0\ell} |\mathcal{C}(M)|(M/\varphi(M))/M$, where $|\mathcal{C}(M)|$ is the number of family classes modulo M. Since $\ell \geq M^{1/(1+C)}$, $\ell^{-\alpha} \leq M^{-\alpha/2}$. Lemmas 5.1, 5.2 and 5.3 give $\sum_\ell \bar{p}_\ell \ell^{-\alpha} \ll \alpha^{-3}$. Take $\alpha = \lambda^{-1/4}$ in (4); here $G = O(\log \lambda)$ and $\bar{\mu} \ll \lambda^3$. □

6.2. **Dominant-prime families.**

Corollary 6.2 ([ET, Cor. 3.6] and §3.7; [PROVED]; for Case-A classes [PROVED MOD] Elsholtz–Tao Prop. 1.4)**.** *Fix $0 < C < 1$ and $A \geq 1$, and put $\delta = (1-C)/2$. Consider any family of Case-B forced classes (classes of $\mathscr{R}(M)$, or (a, D)-classes modulo $G = 4a\,g(D)$), and Case-A classes (modulo $G = 4rh$), such that every modulus M has a prime factor $\ell(M) \geq M^{1/(1+C)}$. The other prime factors are unrestricted: they may be large, occur to higher powers, or be the dominant prime of other moduli. Let $w_0 = w_0(C)$ be large, $Q_0 = \operatorname{lcm}(P_{w_0},$ w_0-smooth parts of all moduli$)$ and $R = \{c : (c, P_{w_0}) = 1,$ c satisfies no condition with all primes $\leq w_0\}$. Then every majorant of level $\lambda \leq A \log N$, charging all primes $> w_0$, satisfies*

$$\log \frac{1}{\mathbb{E}\nu} \leq C_8(A, C)(\log N)^{3/4} + \log \frac{Q_0}{|R|}, \qquad \log \frac{Q_0}{|R|} = O_C(1).$$

Proof. The R-term. A w_0-smooth modulus is at most w_0^{1+C}, so there are finitely many. Let $L' = \operatorname{lcm}(P_{w_0}, w_0$-smooth moduli$)$. By Lemma 2.5, $c \equiv 1 \pmod{L'}$ lies in R, so $|R|/Q_0 \geq 1/L' = \mathrm{e}^{-O_C(1)}$.

Windows. With $s_0 = \log w_0$ take $W_j = (\mathrm{e}^{s_j}, \mathrm{e}^{s_j/C}]$, $s_j = s_0 C^{-j+1}$, and attach each condition to $\ell(M)$. Since $\ell(M) > \sqrt{M}$ it occurs to exponent one, and every other prime $\ell' \mid M$ has $\ell' \leq M/\ell \leq \ell^C$, so it lies in an earlier window or divides Q_0. Thus (U) holds.

Hypotheses of Theorem 4.9. The number of conditions with dominant prime ℓ is at most $\sum_{q \leq \ell^C} \ell^{o(1)} = \ell^{C+o(1)}$, so $p_\ell^* \leq \ell^{-\delta} \leq 1/4$ for $\ell > w_0$.

Profile. By Lemma 2.5, $\mathbb{P}(c \equiv a \ (q) \mid R) \leq L'/\varphi(q)$. By Lemma 4.10 each condition contributes at most $L'(M/\varphi(M))\gamma(M)/M$ with $\gamma(M) = \prod_{\ell' \mid M,\ \ell' > w_0}(1 -$

$\ell'^{-\delta})^{-1}$. Write $(M/\varphi(M))\gamma(M) = \sum_{d|M} h(d)$, h multiplicative on squarefree d with $h(p) \le 1/(p-1) + 3p^{-\delta}$. The proof of Lemma 5.1 goes through: in the Shiu range it uses $\sum_d h(d)/\varphi(d) < \infty$; for $d > x^{1/2}$, $\sum_d h(d)d^{-1+\delta/2} < \infty$ gives a tail $\ll x^{1-\delta/4+\epsilon} = o(x)$. For (a, D)-classes use $(G/\varphi(G))\gamma(G) \le 2[(a/\varphi(a))\gamma(a)][(g/\varphi(g))\gamma(g)]$, and for Case A use the γ-weighted form of Lemma 5.3. So the profile is $\ll_C \beta^{-3}$.

Summing windows. For $\ell \in W_j$, $M \le \ell^{1+C} \le \mathrm{e}^{2s_j/C}$, so the window mass is $\ll s_j^3$ and its profile is $\ll \mathrm{e}^{-\alpha s_j} s_j^3$. Optimising α_j per window, $\mathbb{E}\Phi_j \ll \min\{s_j^3, (\lambda/s_j)(1 + \log^+(s_j^4/\lambda))\} + O(\log^2 \lambda)$. Only $J \ll_C \log\lambda$ windows meet $\ell \le \mathrm{e}^{\lambda}$. Below the crossover $s \asymp \lambda^{1/4}$ the cubic branch is a geometric sum; above it, with $s = \lambda^{1/4}C^{-k}$, the window contributes $O_C(\lambda^{3/4}C^k(1+k))$; the remainders total $O_C(\log^3\lambda)$. Hence $\sum_j \mathbb{E}\Phi_j \ll_C \lambda^{3/4}$. □

The constants deteriorate as $C \to 1$. For unweighted integers, Dickman's function gives logarithmic density $\log(1 + C)$ to the moduli with $P(M) \ge M^{1/(1+C)}$; the share of the *weighted* cubic supply is not proved [CONJECTURE].

7. The quadratic-residue base and the capped measure

Corollary 6.2 needs $P(M) \ge M^{1/(1+C)}$ with $C < 1$. This section gives the two tools that remove this restriction for the classes $\mathscr{R}(M)$: a product base measure on the avoiders of all small-modulus classes (Lemma 7.2), and a capped sequential measure whose heavy coordinates enter only through a leak (Theorem 7.4). With them it proves the cap for gapped and window-resolved families (Theorems 7.6 and 7.7). These two theorems are now special cases of Theorem 9.3; they are kept because their proofs need only the Boolean step of Proposition 4.4. The results are from [TW, §§1–2, 4], building on the reductions of [EB, §2]; the review [TWr] (two rounds) found them sound.

7.1. **Gapped and twin moduli.** Write $P_2(M) = P(M/P(M))$, with $P_2(1) = 1$. Fix $w_0 \ge 3$, $0 < \eta < 1$ and $B \ge 0$. Following [EB, §2], M is *η-gapped* if $P(M) > w_0$ and $P(M) \ge P_2(M)^{1+\eta}$, and *η-twin* otherwise; it is *(η, B)-gapped* if moreover $M \le P(M)^{1+B}$. Put $s_0 = \log w_0$, $s_j = s_0(1+\eta)^{j-1}$ and use the windows $W_j = \{\ell \text{ prime} : s_j < \log\ell \le s_{j+1}\}$.

Lemma 7.1 (gapped implies (U); [EB, Lemma 2.1]; [PROVED])**.** *If M is η-gapped and $\ell = P(M)$, then $\ell \,\|\, M$, and every prime $\ell' \mid M/\ell$ with $\ell' > w_0$ lies in a window strictly before that of ℓ. So a family of η-gapped classes satisfies (U) with $\ell(C) = P(M_C)$.*

Proof. If $\ell^2 \mid M$ then $P_2(M) = \ell$, contradicting $\ell \ge \ell^{1+\eta}$. If $\ell' \in W_j$ divides M/ℓ, then $\log\ell \ge (1+\eta)\log\ell' > (1+\eta)s_j = s_{j+1}$. □

Dominant moduli with $P(M) > w_0$ are η-gapped for every $\eta < 1$ with $\eta \le 1/C - 1$. Balanced moduli $M = k\ell_1\ell_2$ with $P(k) < \ell_1$, $\ell_2 \ge \ell_1^{1+\eta}$ and $k\ell_1 \ge \ell_2$ are η-gapped. The same-scale pairs of Lemma 16.1 below are η-twin. For a gapped balanced modulus the cofactor $M/P(M)$ exceeds $P(M)$, so the counting bound $p_\ell \le \ell^{C-1+o(1)}$ of Corollary 6.2 is lost; the hit probabilities can be large for some histories (*heavy coordinates*).

7.2. **The quadratic-residue base.** Fix $W \geq 3$, let Q_0 be a common multiple of the W-smooth parts of all moduli with $P_W \mid Q_0$, and put $R_W = \{c \bmod Q_0 : (\frac{c}{p}) = 1 \text{ for every odd prime } p \leq W\}$.

Lemma 7.2 (QR base; [TW, Lemma 1.3]; [PROVED])**.** *(1) Every $c \in R_W$ avoids every class of $\mathscr{R}(M)$ with W-smooth M. (2) $\log(Q_0/|R_W|) = \sum_{3\leq p\leq W} \log(2p/(p-1))$. (3) Under the uniform law on R_W, residues at distinct primes are independent and $\mathbb{P}(c \equiv a \ (p^e)) \leq \gamma(p)/p^e$ for $p^e \mid Q_0$, with $\gamma(p) = 2p/(p-1)$ for odd p and $\gamma(2) = 1$.*

Proof. (1) is Lemma 2.6. (2) and (3): by CRT R_W is a product over the prime powers $p^e \| Q_0$; for odd $p \leq W$ the factor has $p^{e-1}(p-1)/2$ elements out of p^e. □

So Mordell's obstruction, which classically shows that polynomial identities never cover squares, is used constructively: it gives a *product* base measure on avoiders of all small-modulus classes, with polylogarithmic inflation.

7.3. **The capped sequential measure.** Work in the sequential setting of Section 4.4, with R satisfying (R1) every $c \in R$ avoids every condition with $S(C) = \emptyset$, and (R2) under the uniform law on R the residues at distinct primes are independent and $\mathbb{P}(c \equiv a \ (p^e)) \leq \gamma(p)/p^e$. Fix caps $\delta_\ell \in (0, 1/4]$. The *capped measure* Q' draws c uniformly from R; given $H_{<j} = h$, the residues $n \bmod \ell$ ($\ell \in W_j$) are independent, uniform on $\mathbb{Z}/\ell \setminus F_\ell(h)$ if $p_\ell(h) \leq \delta_\ell$ (*light*), and uniform on $\mathbb{Z}/\ell$ otherwise (*heavy*); higher digits are uniform. The *leak* is $\mathfrak{L} = Q'(\text{final history} \notin \mathcal{A})$. This is a variant of Hough's distortion method [H], in the form of Balister–Bollobás–Morris–Sahasrabudhe–Tiba [BBMST], which bounds the density of the uncovered set (the majorant $\mathbf{1}_\mathcal{A}$ only), adapted here to general level-λ majorants.

Lemma 7.3 (leak and inflation; [TW, Lemmas 2.1, 2.2]; [PROVED])**.** *(a) $\mathfrak{L} \leq \sum_{\ell\in\mathcal{P}} \mathbb{E}_{Q'}[p_\ell(H_{<j(\ell)})\mathbf{1}\{p_\ell(H_{<j(\ell)}) > \delta_\ell\}]$. (b) For m a product of prime powers p^e with $e \leq E_p$ ($p \in \mathcal{P}$) or $p^e \mid Q_0$, and every a, $Q'(n \equiv a \ (m)) \leq \prod_{p^e\|m} \gamma'(p)/p^e$, where $\gamma'(p) = \gamma(p)$ for $p \mid Q_0$ and $\gamma'(p) = (1-\delta_p)^{-1}$ for $p \in \mathcal{P}$.*

Proof. (a) A final history outside $\mathcal{A}$ satisfies some condition C with $S(C) \neq \emptyset$, by (R1). At the transition into the window of $\ell = \ell(C)$, $b_{C,\ell} \in F_\ell(h)$, and light coordinates never take such values, so ℓ was heavy and the hit has probability $p_\ell(h)$; take a union bound. (b) Chain rule: (R2) at the base; a light coordinate is uniform on at least $p(1-\delta_p)$ residues, a heavy one on all p. □

Theorem 7.4 (capped sequential sieve limit; [TW, Thm 2.3]; [PROVED])**.** *If $\mathfrak{L} \leq 1/2$, every majorant ν of level λ of $\mathcal{A}$ satisfies*

$$\log \frac{1}{\mathbb{E}\nu} \leq \log \frac{Q_0}{|R|} + \log 2 + 2 \sum_j \mathbb{E}_{Q'}\big[\Phi_j^{\mathrm{light}}(H_{<j})\big],$$

where $\Phi_j^{\mathrm{light}}(h)$ is the right side of (3) for the coordinates of W_j that are light at h and have $\log \ell \leq \lambda$.

Proof. Let $g_j(h) = \mathbb{E}_U[\nu \mid H_{<j} = h]$ under the uniform measure. By downward induction, for every history h at window j,

$$g_j(h) \geq \mathbb{E}_{Q'}\Big[\mathbf{1}_\mathcal{A}(H_{<J+1}) \exp\Big(-\sum_{i\geq j} \Phi_i^{\mathrm{light}}(H_{<i})\Big) \;\Big|\; H_{<j} = h\Big].$$

The base case is $g_{J+1} \geq \mathbf{1}_{\mathcal{A}}$. For the step, the hit indicators under U given h are independent $\mathrm{Bern}(p_\ell(h))$; split $x = (x_L, x_H)$ into light and heavy and put $\tilde{f}(x_L) = \mathbb{E}_U[g_{j+1}(H_{<j+1}) \mid h, x_L]$. Then $\mathbb{E}\tilde{f} = g_j(h)$, $\tilde{f} \geq 0$, $\tilde{f}$ is λ-level in x_L (averaging over x_H preserves this), light coordinates have $p_\ell(h) \leq 1/4$, and given $x_L = 0$ the law of $H_{<j+1}$ is the Q' transition. If $\tilde{f}(0) = 0$ the claim is trivial; otherwise Proposition 4.4 for $\tilde{f}/\tilde{f}(0)$ gives $\mathbb{E}\tilde{f} \geq \tilde{f}(0)\mathrm{e}^{-\Phi_j^{\mathrm{light}}(h)}$, and the induction hypothesis bounds $\tilde{f}(0)$. Finally, with $S = \sum_j \Phi_j^{\mathrm{light}} \geq 0$, $\mathbb{E}\nu \geq (|R|/Q_0)\mathbb{E}_{Q'}[\mathbf{1}_{\mathcal{A}}\mathrm{e}^{-S}]$, and Jensen on $Q'(\cdot \mid \mathcal{A})$ with $Q'(\mathcal{A}) \geq 1/2$ gives $\mathbb{E}_{Q'}[\mathbf{1}_{\mathcal{A}}\mathrm{e}^{-S}] \geq \frac{1}{2}\mathrm{e}^{-2\mathbb{E}_{Q'}S}$. □

Heavy coordinates thus cost nothing in the bound; they enter only through $\mathfrak{L}$, which is controlled by an average, never by a supremum.

7.4. **The gapped cap.** Take the windows above with $w_0 = W$, the base $R = R_W$, and caps $\delta_\ell = \ell^{-1/2}$. The family $\mathfrak{G}$ consists of $\mathscr{R}(M)$-classes with M (η, B)-gapped and $P(M) > W$, and any $\mathscr{R}(M)$-classes with W-smooth M. The latter are avoided by R_W (Lemma 7.2), so (R1) holds; (U) holds by Lemma 7.1. Then $\Gamma(m) := \prod_{p\mid m} \gamma'(p) \leq 3^{\omega(m)}$.

Lemma 7.5 (second moment; [TW, Lemma 2.4]; [PROVED])**.** *For every $\epsilon > 0$ there is $C(\epsilon, B)$, independent of W, η and the family, such that $\mathbb{E}_{Q'}[p_\ell(H)^2] \leq C(\epsilon, B)\ell^{-2+\epsilon}$ for every prime $\ell > W$.*

Proof. $|F_\ell(h)|$ is at most the number $N_\ell(n)$ of pairs (q, D) with $q \leq \ell^B$, $D \mid A_{q\ell}^2$ and $n \equiv -4D \pmod q$, a function of the full history. Expanding N_ℓ^2, a compatible pair of pairs contributes $Q'(n \equiv a\ (m)) \leq 3^{\omega(m)}/m$ with $m = \mathrm{lcm}(q, q')$, by Lemma 7.3(b). Each q carries $\tau(A^2) \ll_\epsilon \ell^{\epsilon/4}$ divisors D, and there are $\tau(m^2)$ pairs with $\mathrm{lcm} = m$. So $\mathbb{E}N_\ell^2 \ll_\epsilon \ell^{\epsilon/2}\sum_{m\leq \ell^{2B}} 3^{\omega(m)}\tau(m^2)/m \ll_{\epsilon,B} \ell^\epsilon$. Divide by ℓ^2. □

By Lemma 7.3(a) and Markov, $\mathfrak{L} \leq \sum_{\ell>W} \ell^{1/2}C(1/4, B)\ell^{-7/4} \ll_B W^{-1/4}$, so there is $W_0(B)$ with $\mathfrak{L} \leq 1/2$ for $W \geq W_0(B)$, uniformly in η and the family [TW, Cor. 2.5]. The same weighted Shiu argument as in Corollary 6.2 (now with $h(p) \leq 2$ for $p \leq W$ and $h(\ell) \leq 2\ell^{-1/2}$ for $\ell > W$) gives, for every window with $s_j \leq \lambda$,

$$\mathbb{E}_{Q'} \sum_{\ell\in W_j} p_\ell(H_{<j}) \leq K\,((1+B)s_j)^3, \qquad K = K(W, B) \tag{7}$$

[TW, Lemma 2.6], using $M \leq P(M)^{1+B} \leq \mathrm{e}^{(1+B)s_{j+1}}$ [EB, Lemma 2.4].

Theorem 7.6 (the gapped cap; [TW, Thm 2.7]; [PROVED])**.** *Fix $B \geq 0$ and take $W = W_0(B)$. For every $0 < \eta < 1$, every family $\mathfrak{G}$ as above, and every majorant ν of level $\lambda \geq \log W$,*

$$\log \frac{1}{\mathbb{E}\nu} \leq C\,K^{1/4}(1+B)^{3/4}\eta^{-1}\lambda^{3/4} + O_B(\eta^{-1}\log^2\lambda) + O_B(1),$$

with C absolute and $K = K(W_0(B), B)$.

Proof. Theorem 7.4 applies since $\mathfrak{L} \leq 1/2$; the R-term is $O_B(1)$ by Lemma 7.2(2). Within a window the costs lie in $(s_j, (1+\eta)s_j]$ and $1 + \eta < 2$, so after Step 0 of Proposition 4.4 there is one nonempty band, and $G = 1$ with s_* the least cost present [EB, Lemma 2.5a]. The light profile is at most the full profile, so (7) and concavity bound the logarithmic terms by $O(\log\lambda + \log K + \log(1+B))$ per window, and there are at most $1 + 2\log\lambda/\eta$ windows with $s_j \leq \lambda$. The main term of window

j is at most $\inf_\alpha[19\alpha\lambda + X_j e^{-\alpha s_j}] \le \min\{X_j, (19\lambda/s_j)(1+\log^+(X_j s_j/19\lambda))\}$ with $X_j = C_4 K(1+B)^3 s_j^3$. Put $s^* = (19\lambda/(C_4K(1+B)^3))^{1/4}$. Windows with $s_j \le s^*$ contribute at most $2X(s^*)/\eta = 38\lambda/(\eta s^*)$; the k-th window above s^* contributes at most $(19\lambda/s^*)(1+\eta)^{-k}(1+4(k+1)\log(1+\eta))$, summing to at most $(19\lambda/s^*)\cdot 18/\eta$ [EB, proof of Thm 2.5]. The total is $\le 380\lambda/(\eta s^*)$. Multiply by 2. □

For dominant moduli this recovers Corollary 6.2 for $\mathscr{R}(M)$-classes, with a different R-term. It needs no hypothesis on heavy coordinates: the extremal statement (E_δ) and hypothesis H_{light} of [EB, §§2.5, 4.4] are not used.

Scope of Theorem 7.6. It covers only $\mathscr{R}(M)$-classes: Lemma 2.6 is proved for $\mathscr{R}(M)$ only, and the analysis of [EB] is restricted to $\mathscr{R}(M)$. It needs B fixed: there is no summation over B. It does not cover η-twin moduli, for which (U) fails; Theorem 9.3 does.

Theorem 7.7 (window-resolved families; [TW, Thm 4.4]; [PROVED])**.** *Fix B, $W = W_0(B)$ and $\lambda \ge (2\log W)^4$. Let $\mathfrak{G}$ be any family of $\mathscr{R}(M)$-classes with $M \le P(M)^{1+B}$, plus any W-smooth classes, such that every modulus with top prime in $(e^{\lambda^{1/4}}, e^{\lambda/2}]$ is window-resolved: the last η-window meeting its primes $> W$ contains exactly one of them, to exponent one. Then every majorant of level λ satisfies $\log(1/\mathbb{E}\nu) \ll_B \eta^{-1}\lambda^{3/4}$.*

The proof [TW, §4] uses an abstract form of Theorem 7.4 with blocks: singletons below $e^{\lambda^{1/4}}$, where even twin and prime-power moduli cost $\ll \lambda^{3/4}$ in total [TW, Prop. 4.1]; the η-windows in $(e^{\lambda^{1/4}}, e^{\lambda/2}]$; one block above $e^{\lambda/2}$, where each term of ν sees at most one prime and a linear-window inequality gives cost $O(e^{-\lambda/4})$ [TW, Lemma 4.2, Cor. 4.3]; and a second-moment bound with prime-power top primes [TW, Lemma 4.0] for the leak. So all twin moduli with top prime outside $(e^{\lambda^{1/4}}, e^{\lambda/2}]$ are already covered. The *residual* was the set of unresolved moduli with top prime in that range. Theorem 9.3 closes it by replacing the η-windows with dyadic sequential blocks and the Boolean step with Theorem 8.5; the other three ingredients are reused.

8. A weighted k-ary comparison theorem

The sequential sieve limits of Sections 4.4 and 7 need, in each window, a single coordinate that decides each condition (hypothesis (U)). For twin moduli, prime-power tops, or several primes of one modulus at the same scale, (U) fails. This section proves an arithmetic-free comparison theorem of any arity that replaces the per-window Boolean step. Everything here is from [KA, §§1–3]; the reviews [KAr, KAr2] found it sound (the second re-derived all inputs; its presentational repairs E1, E2 are incorporated).

8.1. **The coupled sequential process.** Let V be a finite ordered set of coordinates, $y_\ell \in \Omega_\ell$ finite, and $\varpi = \bigotimes_\ell \varpi_\ell$ a product law. A *pattern* is a pair $C = (T_C, a_C)$ with $T_C \subseteq V$ and $a_C \in \Omega_{T_C}$; put $\operatorname{top}(C) = \max T_C$. Arities are arbitrary and unary patterns are allowed. A function f on $\prod_\ell \Omega_\ell$ is *d-local* if $f = \sum_T f_T$ with f_T a function of y_T and $|T| \le d$; a λ-level function on coordinates of cost $> s$ is d-local with $d = \lfloor \lambda/s \rfloor$. Fix thresholds $\delta_\ell \in (0, 1/4]$.

The coordinates are processed in order. Before ℓ the state is $(c_i, y_i)_{i<\ell}$, and the *plain activation rule* gives the set

$$F_\ell = F_\ell(y_{<\ell}) = \{a : \exists C,\ \mathrm{top}(C) = \ell,\ (y_{T_C\setminus\{\ell\}}, a) = a_C\},$$

known before ℓ is drawn; put $p_\ell = \varpi_\ell(F_\ell)$, and call ℓ *light* if $p_\ell \le \delta_\ell$. Draw $c_\ell \sim \varpi_\ell$ independently of the past. If ℓ is light and $c_\ell \in F_\ell$ (ℓ is *replaced*), draw y_ℓ from $\varpi_\ell(\cdot \mid \Omega_\ell \setminus F_\ell)$, independently of c_ℓ; otherwise $y_\ell = c_\ell$. Write $\omega = (c, y)$ for the path, $\mathcal{J}(\omega)$ for the replaced set, $z(\omega) = |\mathcal{J}(\omega)|$, and $m(\omega) = \sum_{\ell \text{ light}} p_\ell$ for the light activated mass. The law σ of y is the sequential capped law of [TW, §6]; the c's are an explicit coupling of σ with ϖ.

Lemma 8.1 (avoidance and inflation; [KA, Lemma 2.1]; [PROVED])**.** *(1) If* $\mathrm{top}(C)$ *is light then* $y_{T_C} \ne a_C$*; hence* $\mathbb{P}(y \text{ completes some } C) \le \sum_\ell \mathbb{E}[p_\ell \mathbf{1}\{p_\ell > \delta_\ell\}]$*. (2) Given the past,* $y_\ell \sim \varpi_\ell(\cdot \mid \Omega_\ell \setminus F_\ell)$ *if* ℓ *is light and* $y_\ell \sim \varpi_\ell$ *otherwise. Hence* $\sigma(y_S = b) \le \prod_{\ell\in S}(1-\delta_\ell)^{-1}\varpi(b)$ *for all* $S \subseteq V$ *and* $b \in \Omega_S$*.*

Proof. (1) At a light ℓ, $y_\ell \notin F_\ell$, and every completion of a pattern with top ℓ lies in F_ℓ. At a heavy ℓ, $y_\ell = c_\ell$ is independent of the past. (2) At a light ℓ, $\mathbb{P}(y_\ell = a \mid \text{past}) = \varpi_\ell(a)\mathbf{1}_{a\notin F_\ell}(1 + p_\ell/(1-p_\ell))$; the bound for y_S follows by the chain rule from the last element of S downwards. □

8.2. **Extrapolation in replacement coins.** Freeze a whole path ω. For $\rho \in \{0,1\}^V$ put $y^\rho_\ell = y_\ell$ if $\ell \in \mathcal{J}(\omega)$ and $\rho_\ell = 1$, and $y^\rho_\ell = c_\ell$ otherwise. So $y^0 = c$ and $y^{\mathbf{1}} = y$. The intermediate points y^ρ need not be paths of the process, and nothing below requires them to be.

Lemma 8.2 (locality; [KA, Lemma 2.2]; [PROVED])**.** *For fixed* ω *and a* d*-local* $f \ge 0$*,* $g_\omega(\rho) = f(y^\rho)$ *is nonnegative, multilinear of degree at most* d *in* $\rho_{\mathcal{J}(\omega)}$*, and independent of* $\rho_{V\setminus\mathcal{J}(\omega)}$*.*

Proof. y^ρ_ℓ depends on ρ_ℓ only, so $f_T(y^\rho_T)$ is a function of $\rho_{T\cap\mathcal{J}}$, and $|T| \le d$. □

The one-dimensional input is Lagrange *extrapolation* for binomial laws: the node lemma (Lemma 4.2) interpolates at the end point 0, and here we extrapolate to the other end point z.

Lemma 8.3 (binomial Lagrange extrapolation; [KA, Lemma 2.3 and §3]; [PROVED])**.** *Let* $z \ge 0$*,* $0 < t \le 1/4$*,* $d \ge 0$*, and let* $g \ge 0$ *on* $\{0,1\}^z$ *be multilinear of degree at most* d*. Then* $g(1,\dots,1) \le B(z,t,d)\,\mathbb{E}_{\rho\sim\mathrm{Bern}(t)^z} g(\rho)$*, where* $B(z,t,0) = 1$ *and, for* $d \ge 1$*,*

$$B(z,t,d) = \min_Y \max_{y\in Y} \frac{|\ell^Y_y(z)|}{\psi(y)}, \qquad \psi = \mathrm{Bin}(z,t),$$

over node sets $Y \subseteq \{0,\dots,z\}$ *with* $|Y| = \min(d,z)+1$*,* ℓ^Y_y *the Lagrange basis of* Y*. One has* $B \ge 1$*, and*

$$\log B(z,t,d) \le d\log\Big(C_1' \max\Big(\frac{1}{t}, \sqrt{\frac{z}{td}}\Big)\Big) + \tfrac{1}{2}\log(22zt+22) + 3, \qquad C_1' = 2\mathrm{e}^{4.31}. \tag{8}$$

Proof. Averaging g over $\mathrm{Sym}(z)$ changes neither side and gives $Q(K)$, $K = \sum \rho_i$, with Q a polynomial of degree at most d (the average of ρ^S is $\binom{K}{|S|}/\binom{z}{|S|}$) and $Q \ge 0$ on $\{0,\dots,z\}$. Lagrange interpolation at a minimising Y gives $Q(z) = \sum_y \ell_y(z)Q(y) \le B\sum_y \psi(y)Q(y) \le B\,\mathbb{E}Q(K)$. For $d \ge 1$, interpolating the constant 1 gives $\sum_y |\ell_y(z)| \ge 1 \ge \sum_y \psi(y)$, so $B \ge 1$.

For (8) let $d \ge 1$ and $m_0 = zt \le z/4$; we use three node sets [KA, §3], and $-\log(1-t) \le 1.151t$ for $t \le 1/4$.

(i) $z \le d$. Take $Y = \{0, \dots, z\}$. Then $\ell_y(z) = \mathbf{1}_{y=z}$ and $B \le 1/\psi(z) = t^{-z} \le t^{-d}$.

(ii) $z > d$ and $m_0 \le 2d$. Take $Y = \{0, \dots, d\}$. Then $|\ell_i(z)| \le z^d/(i!(d-i)!)$, $\binom{z}{i} \ge (z/i)^i \ge z^i/(i!\mathrm{e}^i)$ and $(1-t)^{z-i} \ge \mathrm{e}^{-1.151 m_0} \ge \mathrm{e}^{-2.31d}$, so $|\ell_i(z)|/\psi(i) \le z^{d-i}\mathrm{e}^i\mathrm{e}^{2.31d}/((d-i)!\,t^i)$. Since $z \le 2d/t$ and $(2d)^k/k! \le (2\mathrm{e}d/k)^k \le (2\mathrm{e})^d$ for $k \le d$, this is at most $(2\mathrm{e}^{4.31}/t)^d$.

(iii) $m_0 > 2d$. Put $h = \lfloor \sqrt{m_0/d} \rfloor \ge \frac{1}{2}\sqrt{m_0/d}$ and $y_i = \lceil m_0 \rceil - \lfloor dh/2 \rfloor + ih$, $0 \le i \le d$. Then $|y_i - m_0| \le dh/2 + 3/2 \le \frac{1}{2}\sqrt{dm_0} + \frac{3}{2}$, and $\sqrt{dm_0} < m_0/\sqrt{2}$, so every node lies in $[0.64m_0, 1.36m_0 + 1.5] \subseteq [1, z-1]$ (as $z \ge 4m_0 > 8$), and $8y_i \le 22m_0$. Since $m_0 > 2d \ge 2$, $\frac{4}{3}(y_i - m_0)^2/m_0 \le \frac{4}{3}(d/4 + 1.5\sqrt{d/m_0} + 2.25/m_0) \le d/3 + 2.92$, so Lemma 4.1 gives $\psi(y_i) \ge (22m_0)^{-1/2}\mathrm{e}^{-d/3-2.92}$. Next, $|\ell_i(z)| = \prod_{j \ne i} |z - y_j|/(h^d i!(d-i)!) \le (2z)^d/(h^d d!) \le (2\mathrm{e}z/(dh))^d \le (4\mathrm{e}\sqrt{z/(td)})^d$. Hence $\log B \le d\log(4\mathrm{e}^{4/3}\sqrt{z/(td)}) + 2.92 + \frac{1}{2}\log(22zt)$.

In each case the bound is at most the right side of (8), since $t^{-1} \le C_1'/t$ and $4\mathrm{e}^{4/3} \le C_1'$. □

The comparison with the product law rests on a supermartingale.

Lemma 8.4 (thinned law; [KA, Lemma 2.4]; [PROVED])**.** *Fix $t \in (0, 1/4]$ and put $\mathcal{W}(\omega) = \exp(-\frac{4}{3}t\,m(\omega))$. If ω is drawn from the process and, given ω, $\rho \sim \mathrm{Bern}(t)^V$, then for every x, $\mathbb{E}_\omega[\mathcal{W}(\omega)\mathbb{P}_\rho(y^\rho = x \mid \omega)] \le \varpi(x)$.*

Proof. At a replaced ℓ, $c_\ell \in F_\ell \not\ni y_\ell$, so $c_\ell = x_\ell$ and $y_\ell = x_\ell$ cannot both hold. Hence $\mathbb{P}_\rho(y^\rho = x \mid \omega) \le \prod_\ell \phi_\ell$ with $\phi_\ell = \mathbf{1}\{c_\ell = x_\ell\} + t\mathbf{1}\{\ell \in \mathcal{J},\ y_\ell = x_\ell\}$. Given the past, $\mathbb{E}[\phi_\ell \mid \text{past}] = \varpi_\ell(x_\ell) + \mathbf{1}\{\ell \text{ light}\}\, tp_\ell \varpi_\ell(x_\ell)\mathbf{1}_{x_\ell \notin F_\ell}/(1-p_\ell) \le \varpi_\ell(x_\ell) D_\ell$, $D_\ell = 1 + \frac{4}{3}tp_\ell\mathbf{1}\{\ell \text{ light}\}$, and D_ℓ is known before ℓ. So the partial products of $\phi_\ell/(\varpi_\ell(x_\ell)D_\ell)$ form a nonnegative supermartingale (if $\varpi(x) = 0$ both sides of the claim vanish, so assume $\varpi(x) > 0$) and $Z = \prod_\ell \phi_\ell/(\varpi_\ell(x_\ell)D_\ell)$ has $\mathbb{E}Z \le 1$. Pathwise $\prod_\ell D_\ell \le \mathrm{e}^{\frac{4}{3}tm} = 1/\mathcal{W}$, so $\mathbb{E}[\mathcal{W}\prod \phi_\ell] = \varpi(x)\mathbb{E}[\mathcal{W}\prod D_\ell \cdot Z] \le \varpi(x)$. □

Theorem 8.5 (weighted k-ary comparison; [KA, Thm 2.5]; [PROVED])**.** *In this setting, for every $t \in (0, 1/4]$ and every d-local $f \ge 0$,*

$$\mathbb{E}_\varpi f \ge \mathbb{E}_\omega\big[\mathrm{e}^{-\Phi(\omega)} f(y)\big], \qquad \Phi(\omega) = \log B(z(\omega), t, d) + \tfrac{4}{3}t\,m(\omega) \ge 0.$$

No incident-weight, codegree or sparsity hypothesis is needed, and the arity of the patterns is arbitrary.

Proof. By Lemmas 8.2 and 8.3, $f(y) = g_\omega(\mathbf{1}_\mathcal{J}) \le B\,\mathbb{E}_\rho g_\omega(\rho)$. Multiply by $\mathcal{W}/B$ and take $\mathbb{E}_\omega$: $\mathbb{E}_\omega[\mathrm{e}^{-\Phi} f(y)] \le \sum_x f(x)\mathbb{E}_\omega[\mathcal{W}\mathbb{P}_\rho(y^\rho = x \mid \omega)] \le \mathbb{E}_\varpi f$ by Lemma 8.4 and $f \ge 0$. □

Corollary 8.6 (mean cost; [KA, Cor. 2.6]; [PROVED])**.** *Let $d \ge 1$, $\bar m \ge \mathbb{E}m$ and $t = d/(\bar m + 4d)$. Then*

$$\mathbb{E}_\omega \Phi \le d\log\frac{C_0(\bar m + 4d)}{d} + \frac{4}{3}d + \frac{1}{2}\log(22d + 22) + 3, \qquad C_0 = 4\mathrm{e}^{4.31}.$$

Proof. Each light ℓ is replaced with conditional probability p_ℓ, so $\mathbb{E}z = \mathbb{E}m$. Bound the maximum in (8) by the sum and use concavity of log and $\sqrt{\cdot}$: with this t, $1/t = (\bar m + 4d)/d$, $\sqrt{\mathbb{E}z/(td)} \le (\bar m + 4d)/d$, $t\mathbb{E}z \le d$ and $\frac{4}{3}t\mathbb{E}m \le \frac{4}{3}d$. □

So the step cost is $d(O(1)+\log^+(\mathbb{E}m/d))$, the form of Proposition 4.4 in one band, for patterns of any arity, and with only the *mean* activated mass.

Remark 8.7 (weighted versus unweighted; [KA, Remark 2.8, §5]). Theorem 8.5 bounds $\mathbb{E}[\mathrm{e}^{-\Phi}f(y)]$, not $\log(\mathbb{E}_\sigma f/\mathbb{E}_\varpi f)$. No bound of the latter by the mean mass holds in general [PROVED]: take a trigger X with $\mathbb{P}(X=1)=\epsilon$, coordinates $Z_1,\dots,Z_q$ with $\mathbb{P}(Z_i=1)=1/4$, patterns $(X,Z_i)=(1,1)$, $\mu=q/4$ and $\epsilon=1/\mu$; every activated set is light, $\mathbb{E}m(\omega)=1$, but the 3-local $f=X(\sum Z_i-\mu)^2$ has $\mathbb{E}_\sigma f/\mathbb{E}_\varpi f=\mu/(1-1/4)\to\infty$. The incident weight of $(X,1)$ is about 1 here, so this does not contradict the unweighted comparison inequality conjectured in [TW, Conj. 6.4] (Conjecture 17.2 below), which assumes small incident weights. That unweighted form remains [OPEN]; Theorem 8.5 is its weighted form, for the conjecture's own law σ. The sieve application below needs only the weighted form, because the sequential sieve limit accepts random step costs (Theorem 9.1). An exact check of Theorem 8.5 on 100 random small systems (exhaustive path enumeration and floating-point linear programmes) found no violation [EVIDENCE].

9. The 3/4 cap for every fixed-B family of classes $\mathscr{R}(M)$

This section combines Theorem 8.5 with the quadratic-residue base and the capped sequential measure of Section 7. The results are from [KA, §4]; the second review [KAr2] re-derived every input taken from [TW] and found the argument sound.

9.1. **Sequential steps with random costs.** Take the sequential setting of Section 7.3: a base Q_0, an admissible set R with (R1)–(R2), coordinates $y_\ell=n \bmod \ell^{E_\ell}$ for $\ell\in\mathcal{P}$, now grouped into ordered *blocks* $V_1,\dots,V_J$ (in increasing order of primes) instead of windows. Each class is *decided* at its top prime $P(M)$: its other prime factors are smaller (earlier coordinates, or the base), and its requirement at $P(M)$ is one class modulo $P(M)^v$. Hypothesis (U) is *not* assumed.

Theorem 9.1 (weighted sequential sieve limit; [KA, Thm 4.1]; [PROVED])**.** *Suppose that for each block j and history h there are a probability law $\Pi_j(h)$ on auxiliary paths ω, a map $\omega\mapsto Y_j(h,\omega)\in\prod_{\ell\in V_j}\mathbb{Z}/\ell^{E_\ell}$, and a cost $\Phi_j(h,\omega)\ge 0$, such that*

(S_w) *every λ-level $f\ge 0$ on the block coordinates satisfies*

$$\mathbb{E}_U f \ge \mathbb{E}_{\omega\sim\Pi_j(h)}\big[\mathrm{e}^{-\Phi_j(h,\omega)}f(Y_j(h,\omega))\big],$$

U the uniform law. Let Q' be the law of the history built from the uniform law on R by drawing, at each block, $\omega_j\sim\Pi_j(H_{<j})$ and appending Y_j. If the leak $\mathfrak{L}=Q'(\text{final history}\notin\mathcal{A})$ is at most $1/2$, then every majorant ν of level λ satisfies

$$\log\frac{1}{\mathbb{E}\nu}\le\log\frac{Q_0}{|R|}+\log 2+2\sum_j\mathbb{E}_{Q'}\Phi_j(H_{<j},\omega_j).$$

Proof. Let $g_j(h)=\mathbb{E}_U[\nu\mid H_{<j}=h]$. By downward induction, $g_j(h)\ge\mathbb{E}_{Q'}[\mathbf{1}_{\mathcal{A}}\mathrm{e}^{-\sum_{i\ge j}\Phi_i}\mid H_{<j}=h]$; for $j=J+1$ this is $g_{J+1}\ge\mathbf{1}_{\mathcal{A}}$. Given h, $f(y)=g_{j+1}(h,y)$ is λ-level and nonnegative, as in the proof of Theorem 7.4. Apply (S_w) and the induction hypothesis at (h,Y_j); under Q' the later blocks depend on ω_j only through Y_j. Conclude with the Jensen step of Theorem 7.4, using $\sum_j\Phi_j\ge 0$. □

A step of Theorem 7.4 is an (S_w) with deterministic cost. The new instance is the *sequential step*: for a block V of primes with $\log \ell > s$, λ-level functions are d-local with $d = \lfloor \lambda/s \rfloor$. Given h, the patterns are the classes with top prime in V whose requirements outside V are met by h (a class modulo ℓ^v is a union of points of $\mathbb{Z}/\ell^{E_\ell}$). Take for $\Pi_j(h)$ the coupled process of Section 8 in increasing order with $\varpi = U$ and caps $\delta_\ell = \ell^{-1/2}$, $Y_j = y$, and for Φ_j the cost of Theorem 8.5 with $t = t_j(h) = d/(\mathbb{E}[m \mid h] + 4d)$, fixed before the block is drawn. Then (S_w) is Theorem 8.5, and $\Phi_j \geq 0$.

9.2. **Inflation, moments and leak.** Fix $B \geq 0$ and work with the base R_W of Lemma 7.2, caps $\delta_\ell = \ell^{-1/2}$, and $\Gamma(m) = \prod_{p|m} \gamma'(p)$ with $\gamma'(\ell) = (1-\ell^{-1/2})^{-1}$ for $\ell > W$ (Lemma 7.3(b)); assume $W \geq 16$, so that $\delta_\ell \leq 1/4$. The family $\mathfrak{G}$ consists of finitely many $\mathscr{R}(M)$-classes with $M \leq P(M)^{1+B}$, of any shape, plus any $\mathscr{R}(M)$-classes of W-smooth modulus. The latter are avoided by every element of R_W (Lemma 7.2(1)), so they do not change the avoider set in R_W, and Q_0 need only be a common multiple of P_W and the W-smooth parts of the finitely many other moduli.

Lemma 9.2 ([KA, Lemmas 4.2, 4.2′, 4.3]; [PROVED])**.** *Let Q' be built with sequential steps in some blocks and singleton steps in the others.*

(1) *(inflation) Lemma 7.3(b) holds for Q'.*
(2) *(first moment) There is $K = K(W,B)$ such that for every $s \geq \log W$ and every block $V \subseteq \{\ell : s < \log \ell \leq 2s\}$,*

$$\mathbb{E}_{Q'} m_V \leq \mathbb{E}_{Q'} \sum_{\ell \in V} p_\ell \leq \sum_{M:\, P(M) \in V} \frac{\tau(A_M^2)\Gamma(M)}{M} \leq K\,(2(1+B)s)^3,$$

the sum over the moduli of $\mathfrak{G}$, $A_M = (M+1)/4$.
(3) *(second moment) $\mathbb{E}_{Q'} p_\ell^2 \leq C(\epsilon, B)\ell^{-2+\epsilon}$ for every prime $\ell > W$.*
(4) *(leak) $\mathfrak{L} \leq \sum_\ell \mathbb{E}_{Q'}[p_\ell \mathbf{1}\{p_\ell > \delta_\ell\}] \ll_B W^{-1/4}$, so $\mathfrak{L} \leq 1/2$ for $W \geq W_0(B)$.*

Proof. (1) Chain rule, with Lemma 8.1(2): given the past, y_ℓ has density at most $(1-\delta_\ell)^{-1}$ with respect to U. (2) A class $-4D \pmod{q\ell^v}$ with top $\ell \in V$ adds at most ℓ^{-v} to p_ℓ, and only if $n \equiv -4D \pmod q$ holds for the history before ℓ, which has Q'-probability at most $\Gamma(q)/q$ by (1). There are at most $\tau(A_M^2)$ values of D. Write $\Gamma(M) = \sum_{e|M} h(e)$ with h multiplicative on squarefree e, $h(p) \leq 2$ for $p \leq W$, $h(\ell) \leq 2\ell^{-1/2}$ for $\ell > W$. The weighted Shiu argument of Lemma 5.1 and Corollary 6.2 gives $\sum_{M \leq x,\ M \equiv 3 \ (4)} \tau(A_M^2)\Gamma(M) \leq K_0(W) x \log^2(x+2)$, with no windows and no B. Every M in the sum has $M \leq P(M)^{1+B} \leq \mathrm{e}^{2(1+B)s}$; drop the condition $P(M) \in V$ and sum by parts. (3) is [TW, Lemma 4.0], the prime-power form of Lemma 7.5; it uses only (1). (4) A final history outside $\mathcal{A}$ satisfies a class that is not W-smooth, by (R1). When its top prime ℓ is processed its other requirements are met, so y_ℓ lies in the completing set; this is impossible if ℓ is light (Lemma 8.1(1)) and has conditional probability at most p_ℓ if ℓ is heavy. Markov and (3) give $\sum_{\ell > W} \ell^{1/2} O_B(\ell^{-7/4})$. □

The cubic growth s^3 of the block mass, against the depth $d \asymp \lambda/s$, is what makes the sum over blocks below converge.

9.3. **The cap.**

Theorem 9.3 (the 3/4 cap for fixed B; [KA, Thm 4.5]; [PROVED])**.** *Fix $B \geq 0$ and $W = W_0(B)$. There are $\lambda_0(B)$ and $C(B)$ such that for every finite family $\mathfrak{G}$ of $\mathscr{R}(M)$-classes with $M \leq P(M)^{1+B}$ (of any shape: dominant, gapped, η-twin, prime-power top, any number of prime factors at comparable scale), together with any $\mathscr{R}(M)$-classes of W-smooth modulus, every majorant ν of the avoider set in R_W of level $\lambda \geq \lambda_0(B)$, charging all primes $> W$, satisfies*

$$\log \frac{1}{\mathbb{E}\nu} \leq C(B)\, \lambda^{3/4}.$$

Proof. Put $s_1 = \lambda^{1/4}$, with λ_0 such that $s_1 > 2 \log W$. The blocks, in increasing order of primes, are: singletons $\{\ell\}$ for $W < \ell \leq \mathrm{e}^{s_1}$; sequential blocks $V_i = \{\ell : 2^i s_1 < \log \ell \leq 2^{i+1} s_1\} \cap (1, \mathrm{e}^{\lambda/2}]$ for $0 \leq i \leq I$, I the largest index with $2^I s_1 < \lambda/2$ (these cover $(\mathrm{e}^{s_1}, \mathrm{e}^{\lambda/2}]$ exactly); one sequential block $(\mathrm{e}^{\lambda/2}, \mathrm{e}^{\lambda}]$; and singletons above e^{λ}, of cost 0. Every class is decided at its top prime, and $\mathfrak{L} \leq 1/2$ by Lemma 9.2(4), so Theorem 9.1 applies. The base costs $O_B(1)$ (Lemma 7.2(2)). The singletons below e^{s_1} cost $\leq \frac{8}{3} K'(1+B)^3 \lambda^{3/4}$ in total [TW, Prop. 4.1], where $K' = K'(W, B)$ is the constant in $\sum_{W < \ell \leq \mathrm{e}^{s_1}} \mathbb{E}_{Q'} p_\ell \leq K'((1+B)s_1)^3$ (the bound of Lemma 9.2(2), applied to $(W, \mathrm{e}^{s_1}]$), and the top block costs $2\log(1 + 3\mathrm{e}^{-\lambda/4})$ [TW, Lemma 4.2, Cor. 4.3]: there each term of ν sees at most one prime. For V_i put $s = 2^i s_1$ and $d_i = \lfloor \lambda/s \rfloor \geq \lambda/(2s)$. Corollary 8.6 with $\bar{m} = \mathbb{E}[m \mid h]$ for each h, then Jensen over h (the map $m \mapsto \log(C_0(m + 4d)/d)$ is concave), give

$$\mathbb{E}_{Q'} \Phi_i \leq d_i \log \frac{C_0(\mathbb{E} m_{V_i} + 4d_i)}{d_i} + \frac{4}{3} d_i + \frac{1}{2} \log(22 d_i + 22) + 3.$$

By Lemma 9.2(2), $\mathbb{E} m_{V_i} \leq K_1 s^3$ with $K_1 = 8(1+B)^3 K$, so $(\mathbb{E} m_{V_i} + 4d_i)/d_i \leq 2K_1 s^4/\lambda + 4 = 2K_1 \cdot 16^i + 4$. Hence $\mathbb{E}\Phi_i \leq 2^{-i} \lambda^{3/4} (c_1(B) + 4i \log 2) + O(\log \lambda)$, and $2 \sum_i \mathbb{E}\Phi_i \leq 2\lambda^{3/4} \sum_i 2^{-i}(c_1(B) + 2.78\, i) + O(\log^2 \lambda) \ll_B \lambda^{3/4}$. □

Theorem 9.3 has no family hypothesis beyond B: no gap, no window resolution, no bound on the number of prime factors, and no hypothesis on heavy coordinates. It removes the residual of Theorem 7.7, and it makes Conjecture 17.2, the hypothesis K2 and the extra $\log \lambda$ of Proposition 17.3 unnecessary for the 3/4 question [KA, §4.3]. The constants $W_0(B)$, $\lambda_0(B)$ and $C(B)$ are not tracked.

10. The main theorem: every mixture of forced classes, no B

This section removes the hypothesis $M \leq P(M)^{1+B}$ from Theorem 9.3 and admits the (a, D)-classes, the Case-A classes and selector classes into the same family, with no condition on the moduli. The framework is from [KA2]. Two independent hostile reviews [KA2r, KA2r2] found it sound; the second re-derived the exponent chain and checked the mixing of types, and its second round verified all repairs. The first moments are from [KA3], which removes the factor $(\log\log N)^{3/4}$ that [KA2], and the previous version of this note, lost without B; its review [KA3r] found all claims sound. The labels of Theorems 10.12 and 10.14 rest on the [KA2]/[KA] framework as reviewed there (square base, leak, weighted sequential sieve limit), which [KA3] does not re-prove. All constants (W, λ_0, C) are absolute but astronomically large (the second review estimates $\log W \approx 10^{10}$–10^{11}, through the Case-A exponents), so every statement of this section is asymptotic only.

10.1. **Where B entered.** In the proof of Theorem 9.3 the hypothesis $M \le P(M)^{1+B}$ enters at exactly three places [KA2, §1]: the block first moment (Lemma 9.2(2), whose partial summation runs over $M \le \mathrm{e}^{2(1+B)s}$); the singleton first moment below e^{s_1}; and the second moment (Lemma 9.2(3)), through a pointwise divisor bound for the cofactor $q \le \ell^B$ of the top prime power, the range of lcm's and the range $v \le 1+B$ of exponents. Everything else is B-free: Theorems 8.5 and 9.1 are arithmetic-free and have no arity bound, the base enters only through its R-term, and every class is decided at its top prime. The number of prime factors of a modulus enters nowhere: the level gives d-locality with $d = \lfloor \lambda/s \rfloor$ block primes per term of ν, the first moments are sums over *all* classes with top prime in a range, and the second moment is a sum over all cofactors [KA2, Remark 5.3]. So it suffices to bound these three sums over all moduli, and to find a base that avoids the small classes of every type.

10.2. **The four class types and the square base.**

Definition 10.1 (forced-class families [KA2, Def. 2.0])**.** A *class* is a residue class $b \pmod G$. The four *types* are:

- $\mathscr{R}(M)$-classes $-4D \pmod M$, $M \equiv 3 \pmod 4$, $D \mid A_M^2$, $A_M = (M+1)/4$ (Lemma 2.2);
- (a,D)-classes $-(4D+a) \pmod{4a\,g(D)}$ (Lemma 2.3);
- Case-A classes $-m^{-1} \pmod{4rh}$, where $d = rh^2$ with r squarefree, so that $g(d) = rh$, and $m \mid 4d+1$ (Lemma 2.4);
- *selector classes* $0 \pmod p$, p prime.

The first three types are forced; selector classes are not. They encode rough-number restrictions: removing $0 \bmod p$ for all $p \le y$ turns the avoider set $\mathcal{A}$ into $\mathcal{A} \cap \{(n, P_y) = 1\}$, the selector of Definition 3.1 and of [TQ]. A *family* $\mathfrak{G}$ is any finite set of classes of these types, mixed arbitrarily, with no condition on the moduli: no B, no dominant prime, any number and multiplicity of prime factors at any scale. Its *avoider set* $\mathcal{A}(\mathfrak{G})$ is the set of **all** integers in none of its classes.

Lemma 10.2 (no squares; [KA2, Lemmas 2.1, 2.2]; [PROVED])**.** *The following hold.*

1. *No residue of a class of $\mathscr{R}(M)$ is a square modulo M.*
2. *No (a,D)-class contains a perfect square $x^2 \ge 1$. Hence no residue of an (a,D)-class is a square modulo its modulus.*
3. *No residue of a Case-A class $-m^{-1} \pmod{4rh}$ is a square modulo $4rh$.*

Proof. (1) If $-4D \equiv x^2 \pmod M$, then $(x,M) = 1$ since $(D,M) = 1$, and $(\frac{-4D}{M}) = (\frac{x^2}{M}) = 1$, against Lemma 2.6. (2) If $n = x^2 \ge 1$ lies in the class, the proof of Lemma 2.3 puts n in a class of $\mathscr{R}(M)$ with $M = (n+4D)/a$; by (1) n is not a square modulo M, a contradiction. If a residue c of the class were a square modulo the modulus, $c \equiv x^2$ with $1 \le x \le 4ag(D)$, then the perfect square x^2 would lie in the class. (3) Here $(m, 2rh) = 1$ since $m \mid 4rh^2+1$. Suppose $-m^{-1} \equiv y^2 \pmod{4rh}$. Then y is odd, so $y^2 \equiv 1 \pmod 8$ and $m \equiv -1 \pmod 4$; in particular $(\frac{-1}{m}) = -1$, and if r is even then $8 \mid 4rh$, $m \equiv -1 \pmod 8$ and $(\frac{2}{m}) = 1$. For each odd prime $p \mid r$, $y^2 m \equiv -1 \pmod p$ gives $(\frac{-m}{p}) = 1$; so the Jacobi symbol $(\frac{-m}{r_o}) = 1$, where r_o is the odd part of r. Modulo m, $4rh^2 \equiv -1$ gives $(\frac{-r}{m}) = 1$. On the other hand $(\frac{-r}{m}) = -(\frac{r_o}{m})$, and reciprocity with $(m-1)/2$ odd gives $(\frac{r_o}{m}) = (\frac{m}{r_o})(-1)^{(r_o-1)/2} = (\frac{-1}{r_o})(\frac{-m}{r_o})(-1)^{(r_o-1)/2} = 1$. So $(\frac{-r}{m}) = -1$, a contradiction ($r_o = 1$ included). □

Lemma 10.2(2)–(3) proves an observation that [TW] had made only numerically; [KA2, §7] re-checks it by brute force [EVIDENCE]. It is consistent with Schinzel's theorem, as quoted in [ElT], that no polynomial identity covers a class $b \bmod a$ with b a quadratic residue; we have not used or checked that theorem.

Lemma 10.3 (square base; [KA2, Lemma 2.3]; [PROVED])**.** *Fix $W \ge 3$, let P_W be the product of the primes $\le W$, and let Q_0 be the least common multiple of $8P_W$ and of the W-smooth parts of all moduli of $\mathfrak{G}$, so that Q_0 is W-smooth. Put*

$$R_W^{\square} = \{c \bmod Q_0 : \ c \bmod p^e \text{ is a unit square for every } p^e \,\|\, Q_0\}.$$

(1) *Every $c \in R_W^{\square}$ avoids every class $b \pmod G$ with G W-smooth that contains no unit square modulo G; in particular every W-smooth class of $\mathfrak{G}$, of all four types.*
(2) $\log(Q_0/|R_W^{\square}|) = 3\log 2 + \sum_{3\le p\le W} \log \frac{2p}{p-1} \le 2W$.
(3) *Under the uniform law on $R_W^{\square}$, residues at distinct primes are independent, and $\mathbb{P}(c \equiv a \pmod{p^e}) \le \gamma(p)/p^e$ for $p^e \mid Q_0$, with $\gamma(p) = 2p/(p-1)$ for odd p and $\gamma(2) = 8$.*

Proof. (1) Every $p^e \,\|\, G$ divides Q_0, so $c \equiv b \pmod G$ would make b a unit square modulo G by CRT. For the first three types this contradicts Lemma 10.2, which excludes even non-unit squares; a selector class $0 \pmod p$ contains no unit. (2) and (3): by CRT $R_W^{\square}$ is a product; the unit squares are a fraction $(p-1)/(2p)$ of the residues modulo p^e for odd p, and $1/8$ modulo 2^e, $e \ge 3$. □

This replaces the quadratic-residue base R_W of Lemma 7.2, which rests on Lemma 2.6 and so serves $\mathscr{R}(M)$ only. The inflation weights become $\Gamma(m) = \prod_{p\mid m} \gamma'(p)$ with $\gamma' = \gamma$ for $p \le W$ and $\gamma'(\ell) = (1-\ell^{-1/2})^{-1}$ for $\ell > W$, and $Q'(n \equiv b \pmod m) \le \Gamma(m)/m$ for all m, b, as in Lemma 9.2(1).

10.3. **Moments without B.** For a class $C = (b \bmod G)$ write $P(C) = P(G)$, and put

$$\mathfrak{M}(y) = \sum_{C\in\mathfrak{U},\ W<P(C)\le y} \frac{\Gamma(G)}{G},$$

where $\mathfrak{U}$ is the universe of all classes of the four types (each counted once per parametrisation). As in Lemma 9.2(2), $\mathbb{E}_{Q'} \sum_{\ell\in V} p_\ell \le \mathfrak{M}(\max V)$ for every block V and every family $\mathfrak{G} \subseteq \mathfrak{U}$.

The bounds in this subsection are from [KA3]; its hostile review [KA3r] found every claim sound after five minor repairs, which are incorporated here. They replace the first-moment bound of the previous version of this note [KA2, Lemmas 3.1–3.6, Cor. 3.7], which lost a factor $(\log\log y)^3$ (Remark 10.10). Write $\Gamma = 1 * h$ with h multiplicative and supported on squarefree numbers, $h(p) = \gamma'(p) - 1$; so $h(p) \le 7$ for $p \le W$ and $h(\ell) \le 2\ell^{-1/2}$ for $\ell > W \ge 16$. Let $\mathcal{P}_y$ be the set of prime powers $\le y$, and for $m \ge 1$ put

$$Z_y(m) = \frac{1}{\log y} \sum_{d\in\mathcal{P}_y,\ d\mid m} \Lambda(d), \qquad S_y(m) = \prod_{p\mid m,\ p^{v_p(m)}>y} p^{v_p(m)}.$$

The *local weight* Z_y has bounded mean, since $\sum_{p\le y} \log p/(p\log y) \to 1$, unlike $\Omega(m)$ or Rankin's m^η; and it is large on smooth numbers.

Lemma 10.4 (smooth numbers have a large local weight; [KA3, Lemma 2.1, (2.1′)]; [PROVED])**.** *Let $y \ge 2$ and let m be y-smooth. Then $S_y(m)$ is squarefull and $\log m \le \log S_y(m) + Z_y(m)\log y$. Hence, if $m \ge Y$ and $S_y(m) \le T$, then $Z_y(m)\log y \ge \log(Y/T)$. In particular, for $K \ge y$, $u = \log K/\log y$, $K < m \le 2K$ and every $k \ge 1$,*

$$\mathbf{1}[P(m) \le y] \le \mathbf{1}[P(m) \le y,\ S_y(m) > K^{1/2}] + (2Z_y(m)/u)^k.$$

Proof. If $p \mid S_y(m)$ then $p \le y < p^{v_p(m)}$, so $v_p(m) \ge 2$. If $p^{v_p(m)} \le y$, all p^ν with $\nu \le v_p(m)$ lie in $\mathcal{P}_y$, and $v_p(m)\log p = \sum_{\nu \le v_p(m)} \Lambda(p^\nu)$. Summing over the primes of m gives the inequality. For the last claim take $Y = K$, $T = K^{1/2}$: if $S_y(m) \le K^{1/2}$ then $Z_y(m) \ge u/2$. □

Lemma 10.5 (moments of the local weight; [KA3, Lemma 2.2]; [PROVED])**.** *For each $k \ge 1$ there is $C_k(W)$ such that for all $y \ge 2$,*

$$\sum_{d_1,\dots,d_k \in \mathcal{P}_y} \Lambda(d_1)\cdots\Lambda(d_k)\,\frac{\Gamma(D)}{D} \le C_k(W)(\log y)^k, \qquad D = \operatorname{lcm}(d_1,\dots,d_k).$$

Proof. Group the indices by the prime of d_i: a set partition π of $\{1,\dots,k\}$ into blocks b, distinct primes $p_b \le y$ and exponents $\nu_i \ge 1$. Then $\Gamma(D)/D = \prod_b \gamma'(p_b) p_b^{-\max_{i\in b}\nu_i}$. For a block of size m, $\sum_{\nu\in\mathbb{N}^m} p^{-\max\nu} \le \sum_{t\ge1} m t^{m-1} p^{-t} \le c_m/p$, as $p^{-t} \le p^{-1}2^{1-t}$. Dropping distinctness, π contributes at most $\prod_b c_{|b|}(\log y)^{|b|-1}\sum_{p\le y}\gamma'(p)\log p/p$, and by Mertens and $\gamma'(p) \le 1 + 2p^{-1/2}$ for $p > W$ the last sum is $\le c'(W)\log y$. Sum over the Bell(k) partitions. □

Lemma 10.6 (smooth $\mathscr{R}(M)$ blocks decay; [KA3, Lemma 2.3]; [PROVED])**.** *Fix $k \ge 1$. There are $C_k(W)$ and $y_0(k,W)$ such that for $y \ge y_0$, $K \ge y^{16k}$ and $u = \log K/\log y$,*

$$\sum_{K<M\le 2K,\ M\equiv 3\ (4),\ P(M)\le y} \tau(A_M^2)\Gamma(M) \le C_k(W)\,K(\log K)^2 u^{-k}.$$

Proof. By Lemma 10.4 the sum is $\le \Sigma_1 + (2/u)^k\Sigma_2$, where Σ_1 runs over the y-smooth $M \in (K, 2K]$ with $S_y(M) > K^{1/2}$ (smoothness is kept here, since otherwise $S_y(M)$ need not be squarefull), and $\Sigma_2 = \sum_{K<M\le2K,\,M\equiv3\,(4)} \tau(A_M^2)\Gamma(M)Z_y(M)^k$ drops smoothness.

Σ_1. Pointwise $\tau(A_M^2)\Gamma(M) \le 8\tau(A_M)^2 3^{\omega(M)} \le C_\epsilon K^\epsilon$. There are $\ll t^{1/2}$ squarefull numbers $\le t$, so $\sum_{s\ \text{squarefull}>Z} 1/s \ll Z^{-1/2}$, and the number of y-smooth $M \le 2K$ with $S_y(M) > K^{1/2}$ is $\le \sum_{s\ \text{squarefull}>K^{1/2}} 2K/s \ll K^{3/4}$. So $\Sigma_1 \ll K^{4/5}$.

Σ_2. Expanding Z_y^k and $\Gamma = 1 * h$,

$$\Sigma_2 = \frac{1}{(\log y)^k} \sum_{d_1,\dots,d_k\in\mathcal{P}_y} \prod_i \Lambda(d_i) \sum_e h(e)\,T(\operatorname{lcm}(D,e)),$$

where $T(q) = \sum_{K<M\le2K,\ M\equiv3\,(4),\ q\mid M} \tau(A_M^2)$. $T(q) = 0$ for even q. For odd q, $q \mid M$ if and only if $A_M \equiv 4^{-1} \pmod q$, a reduced class, with A in $((K+1)/4, (2K+1)/4]$. For $2 \le q \le K^{1/2}$ Shiu's theorem (as stated at the start of Section 5, with $F(A) = \tau(A^2)$, $F(p) = 3$, $F(p^a) = 2a+1 \le 3^a$, $V = (2K+1)/4$, $Y = K/4$,

$\alpha_s = \beta_s = 1/4$) gives

$$T(q) \ll \frac{K}{\varphi(q)\log K}\exp\Big\{\sum_{p\le K,\ p\nmid q}\frac{3}{p}\Big\} \ll \frac{K}{q}(\log K)^2\frac{q}{\varphi(q)}\prod_{p\mid q}\mathrm{e}^{-3/p} \le C_S\frac{K}{q}(\log K)^2,$$

since $(1-1/p)^{-1}\mathrm{e}^{-3/p} \le 1$. For $q=1$, $\sum_{A\le x}\tau(A^2) \ll x(\log x)^2$. For every q, $T(q) \le 3C_\epsilon K^{1+\epsilon}/q$. Now $D \le y^k \le K^{1/16}$, so $\operatorname{lcm}(D,e) \le K^{1/2}$ when $e \le K^{1/4}$, and

$$\sum_{e\le K^{1/4}}\frac{h(e)}{\operatorname{lcm}(D,e)} \le \frac{1}{D}\prod_{p\mid D}(1+h(p))\prod_{p\nmid D}\Big(1+\frac{h(p)}{p}\Big) \le c_1(W)\frac{\Gamma(D)}{D},$$

while, by Rankin with exponent $1/4$ and $\omega(D)\le k$,

$$\sum_{e>Z}\frac{h(e)\gcd(D,e)}{e} \le Z^{-1/4}\prod_{p\mid D}(1+h(p)p^{1/4})\prod_{p\nmid D}(1+h(p)p^{-3/4}) \le c_2(W)2^kZ^{-1/4}.$$

So $\sum_e h(e)T(\operatorname{lcm}(D,e)) \le c_3(W)[\Gamma(D)K(\log K)^2 + 2^kK^{1-1/16+\epsilon}]/D$, and Lemma 10.5 (with $\Gamma \ge 1$ for the second term) gives $\Sigma_2 \le C_k(W)[K(\log K)^2 + K^{1-1/20}]$. Finally $K^{4/5}+K^{1-1/20} \le K(\log K)^2u^{-k}$ for $K \ge K_0(k)$, which holds for $y \ge y_0(k,W)$ since $u \le \log K$. □

For Case A we use three results of [ElT, §7], the section from which its Prop. 1.4 (used in Lemma 5.3) is derived. They were checked against the source but not re-proved.

- [ElT, Cor. 7.4]: for natural numbers a,b,N with $a,b \le N^l$, $\sum_{n\le N}\tau(an+b) \ll_l \tau(\gcd(a,b))N\log N$.
- [ElT, Thm 7.1]: if P has degree D, nonnegative integer coefficients $\le N^l$, and $\rho_P(p^j) \le C$ for all prime powers, then $\sum_{n\le N}\tau(P(n)) \ll_{D,l,C} N\sum_{m\le N}\rho_P(m)/m$.
- [ElT, (7.10)]: for $A,B\ge 2$ and *every* positive integer k,

$$\sum_{a\le A}\sum_{m\le B}\frac{\rho_{ka}(m)}{m} \ll A\log B\cdot\log(1+k), \tag{9}$$

  where $\rho_{ka}(m) = \#\{b \bmod m :\ kab^2+1 \equiv 0\ (m)\}$.

Remark 10.7 (uniformity of (9) in k, and two slips; [KA3, §3], [KA3r, uniformity check, steps 3, 5]). In [ElT] the condition $k \ll (AB)^{O(1)}$ of Prop. 1.4 enters only through Thm 7.1. The proof of (9) [ElT, pp. 30–32], which splits into the ranges $q<A$, $A\le q\le kA$ and $q>kA$ after quadratic reciprocity, uses no size condition on k; so (9) holds uniformly in k, which is how we use it. Two slips in that proof were found and are harmless. (i) On p. 31 the character $a\mapsto(\frac{-ka}{q})$ is said to have mean zero for every q in the small range; this fails for square q (including $q=1$). Square q contribute at most $A\log B\sum_{q\text{ square}}1/q = O(A\log B)$, so (9) holds as stated. (ii) In the range $q>kA$ the factor $c(q)$ omits the reciprocity sign $(-1)^{\frac{q-1}{2}\cdot\frac{k'a-1}{2}}$. For fixed a the corrected factor is still 8-periodic in q, and $q\mapsto(\frac{-ka}{q})$ is the Kronecker character of the negative number $-ka$, hence never principal (even when $k'a$ is a square); so the mean-zero and partial summation step holds with an absolute constant, uniformly in k.

Lemma 10.8 (smooth Case-A blocks decay; [KA3, Lemma 3.1]; [PROVED MOD] Elsholtz–Tao Thm 7.1, Cor. 7.4 and (7.10))**.** *Fix $k \ge 1$. There are $C_k(W)$ and $y_0(k,W)$ such that for $y \ge y_0$, $K \ge y^{64k}$ and $u = \log K/\log y$,*

$$\sum_{r,h\ge 1,\ K<rh\le 2K,\ P(rh)\le y} \tau(4rh^2+1)\Gamma(4rh) \le C_k(W)\, K(\log K)^2 u^{-k}.$$

Proof. $\Gamma(4rh) \le 8\Gamma(r)\Gamma(h)$, and pointwise $\tau(4rh^2+1)\Gamma(4rh) \le C_\epsilon K^\epsilon$. We split the pairs by the size of r.

(i) $r \ge K^{1/4}$. Then $h \le 2K^{3/4}$ and r is y-smooth, so by Lemma 10.4 with $(Y,T) = (K^{1/4}, K^{1/8})$, $1 \le \mathbf{1}[S_y(r) > K^{1/8}] + (8Z_y(r)/u)^k$. The pairs with r y-smooth and $S_y(r) > K^{1/8}$ number $\le \sum_{h\le 2K} \sum_{s\ \text{squarefull}>K^{1/8}} 2K/(hs) \ll K^{1-1/16}\log K$, and contribute $\ll K^{1-1/20}$. The moment part, with smoothness and the lower bound $rh > K$ dropped, is at most

$$8\Big(\frac{8}{u}\Big)^k \frac{1}{(\log y)^k} \sum_{d_1,\dots,d_k\in\mathcal{P}_y} \prod_i \Lambda(d_i) \sum_{s,t} h(s)h(t) \times \sum_{\substack{h\le 2K^{3/4}\\ t\mid h}} \sum_{\substack{r\le 2K/h\\ L\mid r}} \tau(4h^2r+1), \qquad L = \operatorname{lcm}(D,s).$$

For $s \le K^{1/16}$ we have $D \le K^{1/64}$, $L \le K^{1/8}$; write $r = Lr'$ with $r' \le R' = 2K/(hL)$, so $R' \ge K^{1/8}$. The linear form $ar'+1$, with $a = 4h^2L \le 16K^{13/8} \le R'^{14}$, is coprime, and Cor. 7.4 ($l = 14$) gives $\ll R'\log R' \le 2K\log K/(hL)$. Summing over $h = th'$ gives a factor $\le 2\log K/t$. These terms total $\ll K(\log K)^2(h(t)/t)(h(s)/\operatorname{lcm}(D,s))$, and $\sum_t h(t)/t \le c(W)$, $\sum_s h(s)/\operatorname{lcm}(D,s) \le c_1(W)\Gamma(D)/D$ as in Lemma 10.6. For $s > K^{1/16}$ the inner sum is bounded trivially by $(2K/(hL))C_\epsilon K^\epsilon$, and the Rankin tail of Lemma 10.6 gives $\ll 2^k K^{1+\epsilon-1/64}\log K/D$. Lemma 10.5 bounds part (i) by $C_k(W)[K(\log K)^2u^{-k} + K^{1-1/80}]$.

(ii) $r < K^{1/4}$. Then $h > K^{3/4}$ is y-smooth and, with $(Y,T) = (K^{3/4}, K^{1/4})$, $1 \le \mathbf{1}[S_y(h) > K^{1/4}] + (2Z_y(h)/u)^k$. The first term contributes $\ll K^{1-1/20}$ as in (i). The moment part is at most

$$8\Big(\frac{2}{u}\Big)^k \frac{1}{(\log y)^k} \sum_{d_1,\dots,d_k\in\mathcal{P}_y} \prod_i \Lambda(d_i) \sum_{s,t} h(s)h(t) \times \sum_{\substack{r<K^{1/4}\\ s\mid r}} \sum_{h'\le 2K/(rL)} \tau(4rL^2h'^2+1), \qquad L = \operatorname{lcm}(D,t).$$

Take $t \le K^{1/16}$, so $L \le K^{1/8}$. Fix r, and put $P(n) = 4rL^2n^2+1$, $N = \lfloor 2K/(rL)\rfloor \ge K^{1/2}$; its coefficients are $\le 4K^{1/2} \le N^2$. For $p \mid 2rL$, $P \equiv 1 \pmod p$, so $\rho_P(p^j) = 0$. For odd $p \nmid rL$, $n \mapsto Ln$ is a bijection modulo p^j carrying the roots of P to those of $4rb^2+1$, so $\rho_P(p^j) = \rho_{4r}(p^j) \le 2$ by Hensel ($P'(n) = 8rL^2n$ is a unit at a root). By CRT $\rho_P(m) \le \rho_{4r}(m)$ for all m. Thm 7.1 (degree 2, $l = 2$, $C = 2$) gives $\sum_{n\le N}\tau(P(n)) \ll (2K/(rL))\sum_{m\le 2K}\rho_{4r}(m)/m$. Write $r = sa$, so $\rho_{4r} = \rho_{ka}$ with $k = 4s$, and split $a \in [R, 2R)$ dyadically ($\le \log K$ ranges). By (9), uniformly in $k = 4s$, each range gives $\ll (2K/(sRL))\cdot 2R\log(2K)\log(1+4s)$. These terms total $\ll K(\log K)^2(h(s)\log(1+4s)/s)(h(t)/\operatorname{lcm}(D,t))$, and $\sum_s h(s)\log(1+4s)/s \le c(W)$

since $\sum_s h(s)s^{-3/4} < \infty$. The terms $t > K^{1/16}$ are bounded trivially as in (i), and Lemma 10.5 bounds part (ii) by $C_k(W)[K(\log K)^2u^{-k} + K^{1-1/80}]$.

Finally $K^{1-1/80} \le K(\log K)^2u^{-k}$ for $K \ge K_0(k)$. □

In part (ii) the multiplier $k = 4s$ of (9) is a divisor of r, of size up to $K^{1/4}$; this is why uniformity in k matters.

Lemma 10.9 (first moments; [KA3, Cors. 2.4, 3.2, 3.3], [KA2, Lemmas 3.3, 3.3′]; [PROVED]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4, Thm 7.1, Cor. 7.4, (7.10))**.** *For $y \ge y_0(W)$,*

$$\mathfrak{M}(y) \le K_3'(W)(\log y)^3.$$

More precisely, the $\mathscr{R}(M)$-part is $\le K_R(W)(\log y)^3$, the Case-A part is $\le K_A(W)(\log y)^3$, the (a,D)-part is $\ll_W (\log y)^3$ and the selector part is $\le \log\log y + O(1)$. If every modulus satisfies $G \le P(G)^{1+B}$, the sum over the classes of $\mathfrak{G}$ with $P(G) \le y$ is $\le C(W,B)(\log y)^3$, and for Case A only Prop. 1.4 is used.

Proof. $\mathscr{R}(M)$. Take $k = 4$ and $X = y^{64}$. On the body $M \le 2X$ drop the smoothness condition; the weighted Shiu argument of Lemma 9.2(2), which uses no B, gives $\le c(W)(\log 2X)^3 \le c'(W)(\log y)^3$. The blocks $(K, 2K]$, $K = 2^t \ge X$, cover $M > 2X$; by Lemma 10.6 the block contributes $\le K^{-1}C_4K(\log 2K)^2u_t^{-4} \le 4C_4(\log y)^2u_t^{-2}$, $u_t = t\log 2/\log y$. Summing over $t \ge t_0 = \lceil \log X/\log 2\rceil$ gives $\le 4C_4(\log y)^4(\log 2)^{-2}\sum_{t\ge t_0} t^{-2} \le (C_4/(8\log 2))(\log y)^3$. *Case A.* Since the Case-A classes with $g(d) = rh$ are counted by $\tau(4rh^2+1)$, the Case-A part is at most $\sum_{P(rh)\le y}\tau(4rh^2+1)\Gamma(4rh)/(4rh)$. Take $k = 4$ and $X = y^{256}$. The body $rh \le 2X$ is Lemma 5.3 with its γ-weight clause and partial summation, $\le C(W)(\log 2X)^3$; this uses Prop. 1.4. The tail is summed by Lemma 10.8 exactly as for $\mathscr{R}(M)$. *(a,D) and selectors.* The classes are counted by $(a, g(D))$ with multiplicity at most $2^{\omega(g)}$, so the (a,D)-sum is at most $2\sum_{P(a)\le y}\Gamma(a)/a\cdot\sum_{P(g)\le y}2^{\omega(g)}\Gamma(g)/g$, an Euler product of size $\ll (\log y)^{1+2}$; the selector sum is $\sum_{W<p\le y}\gamma'(p)/p$ [KA2, Lemmas 3.3, 3.3′]. *Bounded B.* Drop smoothness and sum over $G \le y^{1+B}$, using the body bounds only [KA2, §3]. □

Remark 10.10 (where the previous version lost $(\log\log y)^3$; [KA3, §1]; *Assessment*). The previous version of this note (following [KA2, §3]) dropped smoothness on $M \le y^{u_0}$ and used Rankin's trick, with Cauchy–Schwarz against a shifted-divisor moment, beyond. Rankin bounds a dyadic block of y-smooth moduli by e^{-u} times the *whole* smooth sum, which is $\asymp \log y$, so each block loses a full power of $\log y$; this forces $u_0 \asymp \log\log y$ and a mass $(\log y)^3(\log\log y)^3$. Hölder in place of Cauchy–Schwarz loses in the same way. The local weight avoids this: expanding Z_y^4 only imposes a divisibility $D \mid M$ with $D \le y^4$, a congruence to a *small* modulus, to which Shiu's theorem (resp. Elsholtz–Tao §7) applies with no loss; the decay u^{-4} is crude (the truth is like $\rho(u)$) but beats the u^2 growth of the block masses. Numerically the smooth first moments are $\asymp (\log y)^3$ up to 10^{12}, with block profile $\max_u u^4b(u) = 0.385$ [KA3, §8] [EVIDENCE].

Lemma 10.11 (second moment and leak; [KA2, Lemmas 4.1–4.3]; [PROVED], no external input)**.** *For every prime $\ell > W$ and every family $\mathfrak{G} \subseteq \mathfrak{U}$, $\mathbb{E}_{Q'}p_\ell^2 \le C(W)\ell^{-7/4}(\log\ell)^c$, with $C(W) \le C(\log W)^c$ and c absolute. Hence there is an*

absolute W_0 such that for $W \ge W_0$ and the block structure of Theorem 10.12 below, the leak satisfies $\mathfrak{L} \le 1/2$.

Proof sketch. For classes with top prime power ℓ^v and cofactor q, so $G = q\ell^v$ with $P(q) < \ell$, one has $p_\ell \le \sum_v \ell^{-v} N_{\ell,v}$, where $N_{\ell,v}$ counts the active classes. Expanding $\mathbb{E}N_{\ell,v}^2$ and replacing the pointwise divisor bound by $\mu(q)\mu(q') \le (\mu(q)^2 + \mu(q')^2)/2$ gives $\mathbb{E}N_{\ell,v}^2 \ll_W \log \ell \sum_{P(q)<\ell} \mu_{\ell,v}(q)^2 \tau(q)\Gamma(q)/q$, with no condition on the size of q. For each type this cofactor sum is $\ll_\kappa (v+1)^c[(\log \ell)^c + \ell^{\kappa v}]$: long dyadic blocks of q by Cauchy–Schwarz, Rankin and Shiu's theorem (resp. the Case-A box moments), short blocks by pointwise divisor bounds; selector classes contribute 1. With $\kappa = 1/4$, Minkowski over v and types gives the bound, since $\sum_v \ell^{-7v/8}$ converges. The leak is then $\le \sum_{\ell > W} \ell^{1/2} \mathbb{E}p_\ell^2 \ll W^{-1/4}(\log W)^{3c+2}$, as in Lemma 9.2(4). This lemma uses no external input, Case A included. □

10.4. **The cap.**

Theorem 10.12 (the 3/4 cap without B; [KA3, Thm 4.1], improving [KA2, Thm 5.1]; [PROVED]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4, Thm 7.1, Cor. 7.4, (7.10))**.** *There are absolute constants W, λ_0, C such that the following holds. Let $\mathfrak{G}$ be any finite family of $\mathscr{R}(M)$-, (a,D)-, Case-A and selector classes, mixed arbitrarily, with arbitrary moduli, and let ν be a majorant of level $\lambda \ge \lambda_0$ of the whole avoider set $\mathcal{A}(\mathfrak{G}) \subseteq \mathbb{Z}$: $\nu = \sum_i a_i \mathbf{1}[n \equiv b_i \pmod{d_i}] \ge 0$ on $\mathbb{Z}$, $\nu \ge 1$ on $\mathcal{A}(\mathfrak{G})$, and $\sum_{\ell \mid d_i,\, \ell > W} \log \ell \le \lambda$ for every i. Then*

$$\log \frac{1}{\mathbb{E}\nu} \le C\lambda^{3/4}.$$

If $\mathfrak{G}$ contains no Case-A classes, the only external input is Shiu's theorem.

Proof. Theorem 9.1 with the base $R_W^{\square}$ of Lemma 10.3, whose R-term is $\le 2W$ and which avoids every W-smooth class of $\mathfrak{G}$; selector classes $0 \bmod p$ with $p > W$ are ordinary classes decided at p. The blocks are those of Theorem 9.3, with singletons $W < \ell \le \mathrm{e}^{s_1}$, $s_1 = \lambda^{1/4}$; the leak is $\le 1/2$ by Lemma 10.11, and W is fixed by it, so $K_3' = K_3'(W)$ of Lemma 10.9 is absolute. The singletons cost $\le \frac{8}{3}\mathfrak{M}(\mathrm{e}^{s_1}) \le \frac{8}{3}K_3' s_1^3 = \frac{8}{3}K_3'\lambda^{3/4}$. For the block V_i, $s = 2^i s_1$, $d_i = \lfloor \lambda/s \rfloor$, Corollary 8.6 and Jensen give, as in Theorem 9.3, $\mathbb{E}\Phi_i \le d_i \log(C_0(\mathbb{E}m_{V_i} + 4d_i)/d_i) + \frac{4}{3}d_i + O(\log\lambda)$, and now $\mathbb{E}m_{V_i} \le \mathfrak{M}(\mathrm{e}^{2s}) \le 8K_3' s^3$, so $(\mathbb{E}m_{V_i} + 4d_i)/d_i \le 16K_3' s^4/\lambda + 4 = 16K_3' \cdot 16^i + 4$, and $\mathbb{E}\Phi_i \le (\lambda/(2^i s_1))(c_1 + i\log 16) + O(\log\lambda)$. Hence $2\sum_i \mathbb{E}\Phi_i \le C\lambda/s_1 + O(\log^2\lambda) = C\lambda^{3/4} + O(\log^2\lambda)$. The linear block $(\mathrm{e}^{\lambda/2}, \mathrm{e}^{\lambda}]$ costs $2\log(1 + 3\mathrm{e}^{-\lambda/4})$, and primes above e^{λ} cost nothing. The constant $2W + \log 2$ is absorbed into C for $\lambda \ge \lambda_0$. □

This is the proof of the previous version [KA2, Thm 5.1] verbatim, with the lossless first moment of Lemma 10.9 in place of $\mathfrak{M}(y) \ll (\log y)^3(\log\log y)^3$. The previous version had to shrink the singleton range to $s_1 = \lambda^{1/4}(\log\lambda)^{-3/4}$ to balance the singletons against the blocks, which gave $\lambda^{3/4}(\log\lambda)^{3/4}$; with the lossless moment the balance is at $s_1 = \lambda^{1/4}$ as in Theorem 9.3. The constants are absolute but astronomically large; the second review of [KA2] estimates $\log W \approx 10^{10}$–10^{11}, and [KA3] does not change W.

Theorem 10.13 (bounded B, all four types; [KA2, Thm 5.2]; [PROVED]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4)**.** *Fix $B \ge 0$. If every modulus of $\mathfrak{G}$ satisfies $G \le P(G)^{1+B}$ (automatic for selector classes), then in Theorem 10.12 $\log(1/\mathbb{E}\nu) \le C(B)\lambda^{3/4}$ for $\lambda \ge \lambda_0(B)$.*

Proof. As Theorem 9.3 ($s_1 = \lambda^{1/4}$), with Lemma 10.3 for the base, the bounded-B clause of Lemma 10.9 for the first moments and Lemma 10.11 for the leak. □

So Theorem 9.3 extends verbatim to (a, D)-, Case-A and selector classes. Theorem 10.12 now implies it with a constant independent of B; Theorem 10.13 is kept because its Case-A part needs only Prop. 1.4 of [ElT], not [ElT, §7]. Nowhere in this section does the number of prime factors of a modulus enter; this is why the k-ary route has no analogue of the middle window $r \asymp \log L$ of the Λ^2 route (Section 12).

10.5. **The main theorem.**

Theorem 10.14 (no $\theta > 3/4$ for coefficient-sum CRT sieves; [KA3, Cor. 4.2], improving [KA2, Cor. 6.1]; [PROVED]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4, Thm 7.1, Cor. 7.4, (7.10))**.** *Fix $A \geq 1$. Let a method bound $\#(\mathcal{A}(\mathfrak{G}) \cap [1, N])$ by $N \cdot \mathbb{E}\nu + \sum_i |a_i|$, where*

- *$\mathfrak{G}$ is any finite family of $\mathscr{R}(M)$-, (a, D)-, Case-A and selector classes (Definition 10.1; arbitrary moduli, no B, no dominant prime, any number and multiplicity of prime factors);*
- *$\nu = \sum_i a_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$ is a CRT majorant of any level, ≥ 0 on $\mathbb{Z}$ and ≥ 1 on all of $\mathcal{A}(\mathfrak{G}) \subseteq \mathbb{Z}$;*
- *every prime dividing a modulus of $\mathfrak{G}$ is at most N^A;*
- *$\sum_i |a_i| < N$.*

Then the saving $s = \log(N/\text{bound})$ satisfies

$$s \leq C_A (\log N)^{3/4}$$

for $N \geq N_0(A)$. In particular no method of this class proves $E(N) \ll N \exp\{-(\log N)^\theta\}$ with $\theta > 3/4$, nor $E(N) \leq N \exp\{-(\log N)^{3/4}\omega(N)\}$ with $\omega(N) \to \infty$.

Proof. Projection. Let Q be the least common multiple of the moduli of $\mathfrak{G}$; $\mathcal{A}(\mathfrak{G})$ is Q-periodic. Replace ν by its conditional average $\bar\nu(n) = \mathbb{E}[\nu(n') \mid n' \equiv n \pmod{Q}]$. A term $a\mathbf{1}[n \equiv b \pmod{d}]$ becomes $a(d'/d)\mathbf{1}[n \equiv b \pmod{d'}]$ with $d' = (d, Q)$, or vanishes. So $\bar\nu \geq 0$, $\bar\nu \geq 1$ on $\mathcal{A}(\mathfrak{G})$, $\mathbb{E}\bar\nu = \mathbb{E}\nu$, $\sum |\bar a_i| \leq \sum |a_i| < N$, and every prime of every modulus of $\bar\nu$ is a family prime, hence $\leq N^A$. *Coarsening.* Put $S = \log(1/\mathbb{E}\nu)$; since the bound is $\geq N\mathbb{E}\nu$, $s \leq S$. Lemma 3.4, whose proof uses only $\log \ell \leq \Lambda_0$ for every charged prime (here every prime $> W$), with $\Lambda_0 = A \log N$ and $\lambda = (A+2)\log N$, gives a majorant $\nu' \geq \bar\nu$ of $\mathcal{A}(\mathfrak{G})$ of level $\leq \lambda$ with $\mathbb{E}\nu' \leq \mathbb{E}\nu + N^{-1}$, as in Corollary 3.5. So $\log(1/\mathbb{E}\nu') \geq \min(S, \log N) - \log 2$, and Theorem 10.12 at level λ gives $\min(S, \log N) \leq \log 2 + C\lambda^{3/4} = \log 2 + C(A+2)^{3/4}(\log N)^{3/4}$. For $N \geq N_0$ the right side is $< \log N$, so the minimum is S, and $s \leq S$ is bounded as claimed. □

For whole-avoider architectures, Theorem 10.14 subsumes the previous form of the main theorem: case (i), families whose moduli have a dominant prime (Corollary 6.2, whose R is a selector together with family classes), and case (ii), $\mathscr{R}(M)$-families with $M \leq P(M)^{1+B}$ (Theorem 9.3). Case (ii) is subsumed only for majorants that are ≥ 1 on the whole avoider set. Theorem 9.3 also applies to majorants that are ≥ 1 only on the avoiders in the quadratic-residue set R_W; that form is not implied by Theorem 10.14 and remains a separate statement. The square base $R_W^{\square}$ is a proof device: Theorem 10.14 needs no admissible set. A selector $R = \{c : (c, P) = 1\}$ in the sense of Definition 3.3 is the same as adding the selector classes $0 \bmod p$, $p \mid P$,

to $\mathfrak{G}$. So every CRT-majorant architecture (Definition 3.3) for such a family, with a selector admissible set and with all primes of the family (selector primes and small parts of moduli included) $\le N^A$, saves at most $C_A(\log N)^{3/4}$. The previous version of this note proved $C_A(\log N)^{3/4}(\log\log N)^{3/4}$ in general and $C_{A,B}(\log N)^{3/4}$ only under bounded B; the gap, gains $(\log N)^{3/4}\omega(N)$ with $1 \ll \omega(N) \le (\log\log N)^{3/4}$ for unbounded-B families, is now closed [KA3, Cor. 4.2]. So, up to constants, the cap and the saving $c(\log N)^{3/4}$ of [TQ] agree for every family of the four types.

Remark 10.15 (the 3/4 note's majorant is literally covered; [KA2, Remark 5.4]; [PROVED]). The majorant of [TQ] is $\nu_X = S_y\, Q_r(H_X)$ with $S_y = \mathbf{1}[(n, P_y) = 1]$ and $y > W$. It is ≥ 1 only on y-rough avoiders, so it is *not* a majorant of $\mathcal{A}(\mathfrak{G}_0)$ for the atom family $\mathfrak{G}_0$ alone; it *is* a majorant of $\mathcal{A}(\mathfrak{G}_0 \cup \{0 \bmod p : p \le y\})$, since $Q_r(H) = \binom{H-1}{r} \ge 0$ for even r and $Q_r(0) = 1$. The atoms $-uv^{-1} \bmod k\ell$ are $\mathscr{R}(k\ell)$-classes with $k\ell \le \ell^{1+2\kappa}$, $\kappa < 1/240$ (checked in [KA2r2, check (3)] on 124,464 atoms), so this family lies in the class of Theorem 10.13 with $B < 1/120$. The note's coefficient budget $T_{\mathrm{abs}} \le N^{1/2} < N$ and its primes $\le X \le N$ give the remaining hypotheses of Theorem 10.14. Hence its saving $c(\log N)^{3/4}$ matches the cap $C(\log N)^{3/4}$ in order. The constants cannot be compared: the note's are ineffective (Bombieri–Vinogradov), and those here are astronomically large. The selector primes $p \le W$ are absorbed by the base, and those in $(W, y]$ cost $O(\log\log y)$ in the singleton steps, the same order as the selector term $\log(P_y/\varphi(P_y))$ of Lemma 11.1. The note's transfer from exceptional primes to $E(N)$ by the multiplicative semigroup is non-CRT post-processing that only loses, so it does not affect this reading. This complements Lemma 11.1, which places the same majorant in a prime-slice system.

What the class contains [KA2, §6], [KA2r2, (C)]. Nonnegative CRT majorants of the *whole* avoider set of any finite mixture of the four types, used with the coefficient-sum rounding bound. This includes Bonferroni and inclusion–exclusion truncations, Selberg's Λ^2 over any set system of these classes (its error $\sum |\lambda_T \lambda_{T'}|$ is at least the merged $\sum |a_i|$), the β- and Rosser sieves and any combinatorial upper-bound sieve on the class system, CRT-evaluated moment and variance methods (Section 13), the majorant of the 3/4 note (Remark 10.15), and the sequential, Λ^2 and Selberg-type majorants of Sections 4–12 when they are ≥ 1 on the whole avoider set (several of them, such as Theorems 7.6 and 9.3, are stated for the avoiders in R_W only; the Λ^2 majorants of [TW2] are whole-avoider majorants, see Remark 12.8). *Not covered by Theorem 10.14 itself*, but capped in Section 14: the large sieve, which is not a pointwise majorant with rounding $\sum |a_i|$ but is bounded below by one through duality (Theorem 14.2); and interval counts of majorants whose moduli are $\le N/2$ (Corollary 14.23). The per-frequency rounding of Theorem 13.3 is proved for finite prime-slice families only. Theorem 10.14 needs the rounding $\sum |a_i|$.

11. The campaign sieves lie in the class

11.1. The three-quarter note.

Lemma 11.1 ([ET, Lemma 4.1]; [PROVED]). *The atom family $\mathcal{A}_X$ of [TQ, §2] is a prime-slice system with $Q_0 = \mathrm{lcm}(L_K, P_y)$, $\mathcal{P}$ the primes in $(X^{1/2}, X]$, the selector $R = \{(c, P_y) = 1\}$, $|F_\ell(c)| \le \ell^{1/3}$, and $\sum_\ell p_\ell(c)\ell^{-\alpha} \le C_u t^3 X^{-\alpha/2}$ for all c (the upper half of [TQ, Cor. 4.3], "uniform fibre masses"), where $t = \log X$. Its majorant $\nu_X = S_y\, Q_r(H_X)$ is ≥ 0 on $\mathbb{Z}$, equals 1 on the whole avoider set $\mathcal{A}$ (there $S_y = 1$,*

$H_X = 0$ and $Q_r(0) = 1$; [ETr, item 3.1]; [TQ, Thm 8.2] itself states only $\nu_X \geq 1$ on exceptional primes), and has level $\lambda = rt$. Hence every majorant on this family has

$$\log \frac{1}{\mathbb{E}\nu} \leq \log\log y + \min\Big\{Ct^3,\ C\frac{\lambda}{t}\Big(1 + \log^+ \frac{Ct^4}{\lambda}\Big)\Big\} + O(\log^2 \lambda).$$

Proof. The structure is [TQ, Lemma 2.2] (deduplication and conditional independence): conditional on $c \bmod L_K$ the coordinates $n \bmod \ell$ are independent and uniform, and an atom (k, ℓ, u, v) forbids $-uv^{-1} \bmod \ell$ in the fibres with $u + cv \equiv 0 \pmod k$; at most $z_j^2 \leq \ell^{1/3}$ atoms live at each ℓ. The majorant is a product of the selector S_y and $Q_r(H) = \binom{H-1}{r}$, both nonnegative, and $Q_r(0) = 1$. A term of $Q_r(H_X)$ involves at most r atoms, hence at most r slice primes $\leq X$, so its level is at most rt. The bound is Theorem 4.5 with α optimised and the selector term $\log(P_y/\varphi(P_y)) = \log\log y + O(1)$. □

With $\lambda \leq A \log N$ the maximum over t is $\asymp (\log N)^{3/4}$, attained only for $t \asymp (\log N)^{1/4}$. What forces $\lambda \lesssim \log N$ is not the note's modulus bound, which it calls "not necessary", but its coefficient budget $T_{\mathrm{abs}} \leq N^{1/2}$ [TQ, proof of Thm 8.2]: the slice primes are $\leq X$, so $\Lambda_0 = t \leq \log N$, and Lemma 3.4 converts any majorant on this family with coefficient sum $< N$ into one of level $\leq t + \log N + s$ with at most twice the mean. So the note satisfies (A1)–(A3), and *its exponent 3/4 is sharp for its own architecture* [PROVED]. This was the review's one substantive correction [ETr, item 3, SC1]. Independently of the prime-slice reading, the same majorant is a majorant of the whole avoider set of the atom family together with the selector classes $0 \bmod p$, $p \leq y$, with $B < 1/120$, so Theorems 10.13 and 10.14 apply to it directly (Remark 10.15).

Lemma 11.2 (Bonferroni depth; [ET, Lemma 4.2]; [PROVED])**.** *For even r and any integer $H \geq 0$, $\mathbb{E}Q_r(H) \geq \mathbb{P}(H \geq r + 1)$. If H is a sum of independent Bernoulli variables with mean μ and $r + 1 \leq \mu - 2\sqrt{\mu}$, then $\mathbb{E}Q_r(H) \geq 3/4$.*

Proof. $Q_r(h) = \binom{h-1}{r} \geq 1$ for $h \geq r + 1$, and Chebyshev with $\operatorname{Var} H \leq \mu$. □

So the depth $r \gg t^3$ in [TQ] is forced. The binding constraint in the 3/4 proof is the pair (cubic supply $\mu_c \leq C_u t^3$, budget $\log T_{\mathrm{abs}} \leq \log N$). Bonferroni depth, the Bombieri–Vinogradov level, the selector and the pruning cannot improve the exponent: the cap of Theorem 4.5 holds for every majorant, and Lemmas 5.1–5.3 bound the *entire* forced-class supply independently of any level of distribution, so Elliott–Halberstam cannot raise the profile either [ET, Lemma 4.3 and remark] [PROVED]. In the exchangeable *binomial* model, Selberg's square majorant attains the optimum up to a factor $e^{O(k)}$; this is a theorem [PYY, Thm. 1.1, Cor. 1.3]. For the Poisson model the numerically computed linear-programming optimum agrees with the Selberg value to within 0.01 in $-\log$ in the tested cases [ET, §2.5]. Attainment for the multi-band families here is only [EVIDENCE].

11.2. **The two-thirds loglog note.** The note [LL] is not a majorant sieve but Montgomery's large sieve over progressions modulo $L_{\mathcal{K}}$, with multipliers $\mathcal{K} = \{k \leq \delta \log N\}$. It is covered through Remark 4.8.

Corollary 11.3 (restricted multiplier sets; [ET, Cor. 3.5]; [PROVED] given the notes' Brun–Titchmarsh upper bounds)**.** *Suppose a family at scale $X = \mathrm{e}^t$ uses*

slice primes in $(X^{1/2}, X]$ *and multipliers* $k \in \mathcal{K}$*, with* $\sum_\ell p_\ell(c) \le A_0 t^2 h(\mathcal{K})$ *for all* c*, where* $h(\mathcal{K}) = \sum_{k\in\mathcal{K}} \varphi(k)/k^2$*. Then*

$$\text{saving} \le \min\Big\{ Ct^2 h,\ \frac{38\lambda}{t}\Big(1 + \log^+ \frac{Ct^3 h}{\lambda}\Big)\Big\} + O(\log^2 \lambda),$$

and the supremum over t *is* $\asymp \lambda^{2/3} h^{1/3}$*. For* $\mathcal{K} = \{k \le \delta \log N\}$*,* $h \asymp \log\log N$*, giving* $(\log N)^{2/3}(\log\log N)^{1/3}$*; the same holds for any* $\mathcal{K}$ *with* $\operatorname{lcm}(\mathcal{K}) \le N$*, since then* $h(\mathcal{K}) \le \prod_{p\mid \operatorname{lcm}}(1 + 1/p) \ll \log\log N$*.*

Proof. Insert $\sum_\ell \bar{p}_\ell \ell^{-\alpha} \le A_0 t^2 h \mathrm{e}^{-\alpha t/2}$ into (4). With $t_0 = (\lambda/h)^{1/3}$, for $t \le t_0$ the first entry is $\le C\lambda^{2/3}h^{1/3}$; for $t = ut_0 \ge t_0$ the second is $\le C\lambda^{2/3}h^{1/3}(1 + 3\log u)/u$. □

The note's large-sieve level is $Q^2 \le N/L_{\mathcal{K}}$, so $\lambda \le \log N$, and its admissible set is the reduced residues modulo $L_{\mathcal{K}}$, with R-term $\log(L_{\mathcal{K}}/\varphi(L_{\mathcal{K}})) \ll \log\log\log N$. By Remark 4.8 and Corollary 11.3 its saving is at most $C(\log N)^{2/3}(\log\log N)^{1/3}$, i.e. its bound is never smaller than $N\exp\{-C(\log N)^{2/3}(\log\log N)^{1/3}\}$: *the* 2/3*-loglog note is sharp for its architecture* [ET, Lemma 4.4], [PROVED]. Vaughan's argument and that of Pomerance–Weingartner [PW] are the case $\mathcal{K} = \{1\}$, giving $(\log N)^{2/3}$. For the 3/4 note, $h \asymp \kappa t$ and the same computation gives $\lambda^{3/4}$.

12. Selberg-type majorants

12.1. **A sieve limit with no structural hypothesis.** Let $\Omega = \prod_i \Omega_i$ be a finite product probability space (CRT coordinates), with costs $s_i \ge 0$: $s_i = \log \ell$ for a slice prime and $s_i = 0$ for free coordinates. Let $\mathcal{A} \subseteq \Omega$ be *arbitrary*. Let V_κ be the span of functions of ω_T over sets T with $\sum_{i\in T} s_i \le \kappa$; it is the sum of the Efron–Stein components H_T over these T.

Theorem 12.1 (Selberg-type sieve limit; [ET, Thm 5.5]; [PROVED])**.** *Let* $g \in V_{\lambda/2}$ *with* $g \ge 1$ *on* $\mathcal{A}$*, so that* $\nu = g^2$ *is a majorant of level* λ*. Then for every* $\alpha > 0$*,*

$$\mathbb{E} g^2 \ge \mathrm{e}^{-\alpha\lambda/2}\,\frac{\mathbb{P}(\mathcal{A})^2}{\mathbb{P}(\omega \in \mathcal{A},\ \omega' \in \mathcal{A})},$$

where ω' *is the* ρ*-correlated copy: independently for each* i*,* $\omega'_i = \omega_i$ *with probability* $\rho_i = \mathrm{e}^{-\alpha s_i}$ *and otherwise a fresh sample. Equivalently,* $\log(1/\mathbb{E}\nu) \le \alpha\lambda/2 + \Xi_{\mathcal{A}}(\alpha)$ *with the* noise-stability excess $\Xi_{\mathcal{A}}(\alpha) = \log[\mathbb{P}(\mathcal{A} \cap \mathcal{A}'_\alpha)/\mathbb{P}(\mathcal{A})^2] \in [0, \log(1/\mathbb{P}(\mathcal{A}))]$.

Proof. $\mathbb{E}[g\mathbf{1}_{\mathcal{A}}] \ge \mathbb{P}(\mathcal{A})$ and, since $g \in V = V_{\lambda/2}$, $\mathbb{E}[g\mathbf{1}_{\mathcal{A}}] = \langle g, \Pi_V \mathbf{1}_{\mathcal{A}}\rangle \le \|g\|\,\|\Pi_V \mathbf{1}_{\mathcal{A}}\|$. Now

$$\begin{aligned}\|\Pi_V \mathbf{1}_{\mathcal{A}}\|^2 &= \sum_{c(T)\le\lambda/2} \|(\mathbf{1}_{\mathcal{A}})_T\|^2 \le \mathrm{e}^{\alpha\lambda/2}\sum_T \mathrm{e}^{-\alpha c(T)}\|(\mathbf{1}_{\mathcal{A}})_T\|^2 \\ &= \mathrm{e}^{\alpha\lambda/2}\langle \mathbf{1}_{\mathcal{A}}, T_\rho \mathbf{1}_{\mathcal{A}}\rangle = \mathrm{e}^{\alpha\lambda/2}\mathbb{P}(\mathcal{A}\cap\mathcal{A}'),\end{aligned}$$

where $c(T) = \sum_{i\in T} s_i$ and the noise operator T_ρ acts on H_T by $\prod_{i\in T}\rho_i$. □

This covers Selberg's Λ^2 sieve. For a prime-slice system the excess factorises fibrewise, $\Xi_{\mathcal{A}_c}(\alpha) = \sum_\ell \log(1 + \rho_\ell p_\ell(c)/(1 - p_\ell(c))) \le \frac{4}{3}\sum_\ell p_\ell(c)\ell^{-\alpha}$ when $p_\ell(c) \le 1/4$, so Theorem 12.1 reproduces the Rankin functional [ET, Cor. 5.6]. Its point is that balanced, twin and shared-prime moduli are allowed: for Λ^2-majorants, beating 3/4 with balanced moduli requires the following hypothesis to fail.

Hypothesis 12.2 ($H_{\mathrm{MS}}^{\mathrm{Sel}}$; [ET, §5.7]; [OPEN])**.** For the complete Case-B and Case-A forced-class system with moduli $\le \mathrm{e}^{\lambda}$, with primes $\le \lambda^4$ given $\rho = 1$, $\Xi_{\mathcal{A}}(\alpha) \ll \alpha^{-3}(\log\lambda)^{O(1)}$ uniformly for $\lambda^{-1} \le \alpha \le 1$.

Under Hypothesis 12.2, Theorem 12.1 caps Selberg-type majorants at $\lambda^{3/4}(\log\lambda)^{O(1)}$ for all moduli [CONDITIONAL]. A fibrewise version alone would not suffice: $\mathcal{A} = \{c = c_0\} \times \Omega'$ has $\Xi = 0$ in every nonempty fibre, yet $g = \mathbf{1}_{c=c_0}$ saves $\log Q_0$. The remedy is to tilt the fibre law.

Lemma 12.3 (fibre tilting; [TW, Lemma 6.6], [TW2, Lemma 2.1]; [PROVED])**.** *In the setting of Theorem 12.1, let F be a set of coordinates whose residues c (the fibre) range over $\mathbb{Z}/Q_F$; put $\mathcal{A}_c = \mathcal{A} \cap \{\text{fibre} = c\}$ and $\Xi_c(\alpha) = \log[\mathbb{P}(\mathcal{A}_c \cap \mathcal{A}'_c \mid c)/\mathbb{P}(\mathcal{A}_c \mid c)^2]$, where in $\mathcal{A}'_c$ the coordinates outside F are ρ-correlated and the fibre is shared. For any probability law P on fibres with $\mathcal{A}_c \ne \emptyset$ on its support, every $g \in V_{\lambda/2}$ with $g \ge 1$ on $\mathcal{A}$ satisfies*

$$\log\frac{1}{\mathbb{E}g^2} \le \frac{\alpha\lambda}{2} + \log\Big\|\frac{dP}{dU}\Big\|_\infty + \mathbb{E}_P\,\Xi_c(\alpha).$$

For P uniform on a set R the middle term is $\log(Q_F/|R|)$.

Proof for uniform P on R. Let $\tilde\sigma = \sum_{c\in R}\pi_c U(\cdot \mid c, \mathcal{A})$ and $h = d\tilde\sigma/dU$. Then $\mathbb{E}_U[gh] \ge 1$, so $\mathbb{E}g^2 \ge 1/\|\Pi_{V_{\lambda/2}}h\|^2$, and $\|\Pi_V h\|^2 \le \mathrm{e}^{\alpha\lambda/2}\langle h, T_\rho h\rangle$. Since $\langle h, T_\rho h\rangle$ is nondecreasing in each ρ_i, put $\rho_i = 1$ on F; then $\langle h, T_\rho h\rangle = Q_F\sum_{c\in R}\pi_c^2\mathrm{e}^{\Xi_c}$. Choosing $\pi_c \propto \mathrm{e}^{-\Xi_c}$ and Jensen give $\le (Q_F/|R|)\mathrm{e}^{\mathrm{avg}_R\Xi_c}$. The general case is [TW2, Lemma 2.1]. □

12.2. **Two large primes, twins included.** *Setting* [TW2, §3]. Fix $B \ge 0$, $A_0 \ge 1$; let X be large, $L = \log X$, $W_1 = L^{1/2}$, $w_2 = L^8$ and $\alpha = L^{-1/4}$. The family is any set of classes $-4D \pmod M$ of $\mathscr{R}(M)$ with $M \le X$, $M \le P(M)^{1+B}$, and *at most two distinct prime factors of M above w_2*. Twin moduli $k\ell_1\ell_2$ ($\ell_i > w_2$, k w_2-smooth), dominant moduli and prime powers are included. All primes are charged, and the level is $\lambda \le A_0 L$. The fibre is $c = n \bmod$ the w_2-smooth part, with a law P built from the QR base at primes $\le W_1$, conditioning on avoidance of the small classes, and a good-fibre event; then $\log\|dP/dU\|_\infty \le 4L^{1/2}$ [TW2, Lemmas 3.1–3.4]. In a fibre, a class $k\ell^v$ is unary at ℓ and a class $k\ell^v m^u$ is a binary *edge*.

The probabilistic core is a binary noise-stability bound. Take independent coordinates y_ℓ with laws ν_ℓ and a finite set of edges $e = \{(\ell, a), (m, c)\}$, $\ell \ne m$, hit when $y_\ell = a$ and $y_m = c$. Put $\pi_e = \nu_\ell(a)\nu_m(c)$, $\deg(\ell, a) = \sum \nu_m(c)$ over the edges at (ℓ, a), $w_\ell = \sum_{e\ni\ell}\pi_e$ and $q_\ell = \sum_a \nu_\ell(a)\deg(\ell, a)^2$, and assume $w_\ell \le \delta \le 1/16$ for all ℓ. Let Z_1 be the probability that no edge is hit, and $Z_2(\tilde\rho)$ the probability that no edge is hit in either copy of the coupled pair in which, independently for each ℓ, $y'_\ell = y_\ell$ with probability $\tilde\rho_\ell$ and is otherwise an independent copy.

Theorem 12.4 (binary noise stability; [TW2, Thm 1.4]; [PROVED])**.** *For every $\tilde\rho \in [0,1]^{\text{index set}}$,*

$$\log\frac{Z_2(\tilde\rho)}{Z_1^2} \le (1 + 25\delta)\Big[\sum_{e=\{\ell,m\}}\tilde\rho_\ell\tilde\rho_m\pi_e + \sum_j \tilde\rho_j q_j\Big].$$

The proof [TW2, §1] differentiates $\log Z_2$ along $\kappa = t\tilde\rho$, uses an identity for $\partial_{\kappa_j}\log Z_2$ under the unconditioned law of the other coordinates, and bounds the

result with a conditional form of the Lovász local lemma; no cluster expansion is needed. An exact check on 1200 small systems found no violation [EVIDENCE].

For a fibre c, a prime $j > w_2$ and a residue a, let $\deg_c(j,a)$ be the partner mass of the active binary classes through (j,a), and $S_j(c) = \sum_a \nu_j(a) \min(\deg_c(j,a),1)^2$.

Theorem 12.5 (two-prime Λ^2 cap, reduced; [TW2, Thm 5.1]; [PROVED])**.** *In this setting, for L large, every $g \in V_{\lambda/2}$ with $g \geq 1$ on the avoiders of the family satisfies*

$$\log \frac{1}{\mathbb{E} g^2} \leq \frac{A_0}{2} L^{3/4} + C_B L^{3/4} (\log L)^C + 11\, \mathbb{E}_P \sum_{j > w_2} \rho_j S_j(c), \qquad \rho_j = j^{-\alpha},$$

with an absolute exponent C.

The proof combines Lemma 12.3 with the fibre law P, Theorem 12.4 in each good fibre after a free quarantine of hub residues [TW2, Lemma 2.2], and a profile bound $\ll \alpha^{-3} (\log L)^{O(1)}$ for the unary and diagonal binary terms [TW2, Lemma 4.1].

The off-diagonal term was left open in [TW2]. It is now bounded unconditionally [TW3]; the review [TW3r] found the argument sound after repairs.

Theorem 12.6 (the off-diagonal term (H_O); [TW3, Thm 4.1]; [PROVED])**.** *In this setting, $\mathbb{E}_P \sum_{j > w_2} \rho_j S_j(c) \ll_B \alpha^{-3} (\log L)^{O(1)}$.*

Proof sketch. The prime-power classes are split off as in [TW2, Lemma 5.4], at cost $o(1)$. For the classes with moduli kjm ($j \neq m$ primes $> w_2$, k w_2-smooth), call a class through j *small* if $m \leq (kj)^{C_0}$ and *large* otherwise, for a fixed $C_0 \geq 6$, and write $\deg_c(j,a) = x(a) + z(a)$ accordingly. Since $\min(x+z,1) \leq \min(x,1) + z$ and $\min(x,1)^2 \leq x$,

$$S_j \leq 2 w_j^{\mathrm{sm}}(c) + 2 \sum_a \nu_j(a) z(a)^2,$$

where w_j^{sm} is the binary mass at j of the small active classes. *Small part* [TW3, Lemma 2.1]: by Shiu's theorem along the top prime, the divisor mass of a small class is $(\log kj)^3$ rather than L^3, and $\sum_j \rho_j (\log j)^3 / j \ll \alpha^{-3}$; so its first moment is $\ll \alpha^{-3} (\log L)^{O(1)}$. *Large part* [TW3, Lemmas 3.1–3.4]: parametrise $D \mid A^2$ by $A = uvt$, $(u,v) = 1$, $D = u^2 t$. The largest of u, v, t is at least $A^{1/3}$, and $m > (kj)^{C_0}$ makes the range of the prime m long enough for Brun–Titchmarsh in a class modulo four times the product of the two shorter variables. The residue modulo j depends only on those two variables, and their second moments give $\sum_{a \bmod j} V_{j,k}(a)^2 \ll (\log L)^6$ uniformly in j, k. Activity and the k-sum are handled as in [TW2, Lemma 5.4]. The large part contributes $\ll (\log L)^{O(1)}$. □

Theorem 12.7 (two-prime Λ^2 cap, twins included; [TW2, Cor. 5.2], [TW3, Cor. 4.2]; [PROVED])**.** *Fix $B \geq 0$ and $A_0 \geq 1$. Let the family consist of classes of $\mathscr{R}(M)$ with $M \leq X$, $M \leq P(M)^{1+B}$, and at most two distinct prime factors of M above $(\log X)^8$ (twin moduli $k\ell_1\ell_2$, dominant moduli and prime powers included). Then every $g \in V_{\lambda/2}$, $\lambda \leq A_0 \log X$, with $g \geq 1$ on the avoiders of the family satisfies*

$$\log \frac{1}{\mathbb{E} g^2} \ll_{A_0, B} L^{3/4} (\log L)^{O(1)}, \qquad L = \log X.$$

Proof. Theorems 12.5 and 12.6. □

Scope. This is a statement about Selberg-type majorants g^2 only. It needs at most two prime factors above $(\log X)^8$; the extension to any fixed number is

Theorem 12.11 below. The restriction $M \le P(M)^{1+B}$ with B fixed excludes, for fixed B, twins $k\ell_1\ell_2$ with a w_2-smooth cofactor $k > M^{(B-1)/(B+1)}$; for a fixed number of large primes it is removed by Theorem 12.12. The bound is stated in $L = \log X$, with moduli $\le X$ and level $\lambda \le A_0 L$. The level hypothesis $\lambda \le A_0 L$ is assumed directly: the coarsening of Lemma 3.4 does not preserve the square form g^2, so the reduction from a coefficient budget to a level (Corollary 3.5) is not available here as stated (observation of this note, not from [TW2, TW3]).

Remark 12.8 (relation to the main theorem; observation of this note; [PROVED]). A Λ^2-majorant g^2 with $g \in V_{\lambda/2}$ and $g \ge 1$ on the avoiders of a family is a nonnegative majorant of level λ (the setting of [TW2, §3] charges all primes). Under the B-hypothesis Theorem 9.3 already applies to g^2, as anticipated in [TW4, §11]. In general, a Λ^2-majorant g^2 with $g \ge 1$ on the *whole* avoider set of a family of the four types is a majorant in the sense of Theorem 10.12, which then gives $\log(1/\mathbb{E}g^2) \le C\lambda^{3/4}$ for any number of large prime factors and with no B. The settings of [TW2, §3] and [TW4] are of this kind [KA3, Lemma 7.1]: the admissible $g \in V_{\lambda/2}$ is required to be ≥ 1 on the avoiders of the family $\mathfrak{F}$ itself (the fibre law and the base are devices of the proof, not restrictions on g), the classes are $\mathscr{R}(M)$-classes, and all primes are charged, so g^2 is a CRT majorant of level $\le \lambda$ in the sense of Theorem 10.12; the savings agree, both being $\log(1/\mathbb{E}_U g^2)$ under the uniform law. Hence every Λ^2 cap of this section (Theorems 12.7, 12.11 and 12.12, and the uniformity statements after them) is superseded as a cap by $\log(1/\mathbb{E}g^2) \le CA_0^{3/4}L^{3/4}$ for every r, with or without B, including the middle range $r \asymp \log L$ left open below [PROVED]. [KA3, Lemma 7.1] was added after the review [KA3r] and checked by the coordinator of this programme only. What the Λ^2 route still gives is its mechanism (fibre tilting, star sums, per-prime efficiency) and the noise-stability tools.

Remark 12.9 (what was bypassed). The route of [TW2] split the off-diagonal term into a same-label part ($H_O^{=}$) (proved there, [TW2, Lemma 5.4]) and a cross-label part ($H_O^{\neq}$), both as *un-quarantined* pair sums, with a sufficient form H_{div} [TW2, §5.5]. Theorem 12.6 does not prove ($H_O^{\neq}$) or H_{div}: small partners are paid for through $\min(x,1)^2 \le x$, which says nothing about pair sums. Both remain [OPEN] as stated, but are no longer needed. The exponent budget α^{-3} is used only by the small part, so this route does not lower the exponent $3/4$.

12.3. **Any fixed number of large primes, and removing B.** The results of this subsection are from [TW4]; the review [TW4r] (two rounds) found them sound after nits and one minor repair (G1), all applied. Their new arithmetic input is an upper-bound sieve for the *whole* partner of a modulus, from the arithmetic large sieve, in place of Brun–Titchmarsh for a prime partner.

Lemma 12.10 (rough-partner Brun–Titchmarsh; [TW4, Lemma 2.3]; [PROVED])**.** *Let $s \ge 1$, $w \ge 2$, $q \ge 1$, $(b,q) = 1$ and $x \ge q\,2^{s+1}$. Let $\mathcal{R}_s(w)$ be the set of integers all of whose prime factors exceed w and with $\Omega(R) \le s$. Put $Y = x/q$, $Z = Y^{1/(s+1)}$ and $H = \sum_{w<p\le Z} 1/(p-1)$ ($H = 0$ if $Z \le w$). Then*

$$\sum_{\substack{R\in\mathcal{R}_s(w),\ x<R\le 2x\\ R\equiv b\ (q)}} \frac{1}{R} \le \frac{3(s+1)(1+H+\cdots+H^{s-1})}{\varphi(q)\log Y}.$$

Proof. Write $R = dn$, d the part of R at primes $\le Z$. If $d = R$ then $R \le Z^s < x$; so $\Omega(d) \le s-1$, $d \le Z^{s-1}$, $(d,q) = 1$, and $n > 1$ has all prime factors $> Z$,

with $n \equiv b\bar{d} \pmod q$ and $n \in (x/d, 2x/d]$. Fix d and write $n = n_0 + qk$. The admissible k lie in an interval of at most $x/(dq)+1$ consecutive integers and avoid one class modulo each prime $p \le Z$, $p \nmid q$. The arithmetic large sieve [MV] (in the form $|\mathcal{N}| \le (N+Q^2)/J$) with $Q = Z$, and $J \ge (\varphi(q)/q)\log Z$ by [vLR], give $\#\{n\} \le (x/(dq)+1+Z^2)/((\varphi(q)/q)\log Z) \le 3x/(d\varphi(q)\log Z)$, since $x/(dq) \ge Z^2$ and $x/(dq) \ge 1$. Each R has $1/R < 1/x$. Sum over d: $\sum_d 1/d \le \sum_{i<s} H^i$, and $\log Z = \log Y/(s+1)$. □

For $s = 1$ this is Brun–Titchmarsh up to the constant. Neither the primality of the partner nor its splitting is used. In the proof of Theorem 12.11 below it takes the place of the Brun–Titchmarsh step used in Theorem 12.6, and it closes the ternary residual left open in [TW3, §6.3]: there the partner $R = \ell_a\ell_b$ with $\ell_b < w_2 q$ is itself rough to level $(R/q)/w_2$, so only an *upper* bound for rough integers in one progression of length R/q is needed, not an asymptotic of Bombieri–Friedlander–Iwaniec type [TW4, §2].

Theorem 12.11 (r-prime Λ^2 cap; [TW4, Thm 7.1]; [PROVED])**.** *Fix $r \ge 2$, $B \ge 0$ and $A_0 \ge 1$. Let the family consist of classes of $\mathscr{R}(M)$ with $M \le X$, $M \le P(M)^{1+B}$, and at most r distinct prime factors of M above $w_2 = (\log X)^8$. Then every $g \in V_{\lambda/2}$, $\lambda \le A_0 \log X$, with $g \ge 1$ on the avoiders of the family satisfies*

$$\log \frac{1}{\mathbb{E} g^2} \ll_{A_0,B,r} L^{3/4} (\log L)^{3r+O(1)}.$$

Proof sketch. The noise-stability input is the any-arity bound with codegree terms and free codegree quarantine of [TW3, Lemmas 6.1, 6.2]; it reduces the saving to star sums [TW4, Prop. 3.1]. The fibre law of [TW2, §3] extends with threshold δ_r [TW4, Lemma 4.1]. Whole-event and unary terms are bounded by Shiu along the top prime ($\ll \alpha^{-3}(\log L)^{r+O(1)}$), prime-power classes cost $o(1)$, small partners are paid through $\min(x,1)^2 \le x$, and large partners through a largest-variable reduction with Lemma 12.10 and box counting modulo squarefree Q, $\ll (\log L)^{3r+O(1)}$ [TW4, §§5–6]. □

The r-dependence is explicit [TW4, Prop. 9.1] [PROVED]: every implied constant is $\le (C_B \log L)^{Cr}$, so for $r \le \epsilon \log L/\log\log L$, $\epsilon \le \epsilon_0$, the cap is $\ll_{A_0,B} L^{3/4+C\epsilon}$. At the other end, classes with at least $330 \log L$ prime factors above w_2, with no B-hypothesis, can be added to any such family at cost $o(1)$, provided the fibre law also conditions on a low-mass event [TW4, Lemma 9.2, Cor. 9.3] [PROVED].

Theorem 12.12 (no B-hypothesis for fixed r; [TW4, Thm 10.4]; [PROVED])**.** *Fix $r \ge 2$ and $A_0 \ge 1$. For families of $\mathscr{R}(M)$-classes with $M \le X$ and at most r distinct prime factors above $(\log X)^8$, with no condition relating M to $P(M)$, every $g \in V_{\lambda/2}$, $\lambda \le A_0 \log X$, with $g \ge 1$ on the avoiders satisfies*

$$\log \frac{1}{\mathbb{E} g^2} \ll_{A_0,r} L^{3/4} (\log L)^{O_r(1)}.$$

The same holds if the family also contains arbitrary $\mathscr{R}(M)$-classes with at least $330 \log L$ prime factors above $(\log X)^8$.

Proof sketch. Write $M = kM_L$ with k the w_2-smooth part. The classes split into *smooth-dominated* ($k > M_L^2$), *high-power* (not smooth-dominated, $\ell^{12} \mid M$ for some $\ell > w_2$), and the rest, which satisfy $M \le P(M)^{33r}$ and are handled by Theorem 12.11 with $B = 33r-1$. For the smooth-dominated classes, Cauchy–Schwarz, Rankin's

trick on the smooth k and Shiu's theorem along k (the modulus M_L is now short) give a polylogarithmic *unweighted* first moment [TW4, Lemmas 10.1, 10.2]; the high-power classes have first moment $\ll L^{-3}$. The fibre law survives without B [TW4, Lemma 10.3]. □

The middle range $\epsilon \log L/\log\log L < r < 330 \log L$ is not covered by the Λ^2 method. By a Poisson-type heuristic it carries most of the α^{-3}-weighted mass (*Assessment*, [TW4, §9.3]). There the method needs a hub count and an off-diagonal pair-sum bound that are efficient per prime [TW4, §§9.3, 12] [OPEN]. As a *cap* the middle range is no longer open: by Remark 12.8, every Λ^2 majorant of these settings saves at most $CA_0^{3/4}L^{3/4}$, for every r and without B [KA3, §§6–7]. The open question concerns only the Λ^2-internal mechanism.

13. Beyond CRT evaluation: rounding, primes and moments

The caps above concern bounds of the shape $N\mathbb{E}\nu + \sum_i |a_i|$ for majorants ≥ 0 on $\mathbb{Z}$ and ≥ 1 on the whole avoider set. This section treats three natural ways out: signed rounding, majorants that are positive only at primes, and moment methods. Everything here is from [NC], for the *finite prime-slice* families of Corollary 6.1 only (not the sequential or balanced settings of Sections 7–9); the review [NCr] (two rounds) found it sound after repairs, which narrowed the scope statements reproduced below, and led to the withdrawal of one conjecture.

13.1. **Per-frequency rounding.** For a majorant ν with common period Q_{tot} put $\hat\nu(\theta) = \mathbb{E}_n \nu(n)\mathrm{e}(-n\theta)$ for $\theta \in Q_{\mathrm{tot}}^{-1}\mathbb{Z}/\mathbb{Z}$, so $\hat\nu(0) = \mathbb{E}\nu$ and, exactly,

$$\sum_{n\leq N} \nu(n) - N\mathbb{E}\nu = \sum_{\theta\neq 0} \hat\nu(\theta) S_N(\theta), \qquad S_N(\theta) = \sum_{n\leq N} \mathrm{e}(n\theta). \tag{10}$$

A *per-frequency rounding bound* is $R_w(\nu) = \sum_{\theta\neq 0} |\hat\nu(\theta)| w(\theta)$ with a weight $w \geq 1$. Every class indicator has $\sum_\theta |\hat{\mathbf{1}}(\theta)| = 1$, so $R_1(\nu) \leq \sum_i |a_i|$, and R_1 can be much smaller. The weight $w(\theta) = \min(N, 1/(2\|\theta\|))$ makes R_w a valid bound for (10), and complete Gauss, Kloosterman, Ramanujan or divisor-exponential sums over the classes sharing one frequency are already inside $|\hat\nu(\theta)|$.

For independent $x_i \sim \mathrm{Bern}(p_i)$ write $y_i = x_i - p_i$; every f on $\{0,1\}^I$ has a unique biased Walsh expansion $f = \sum_S d_S y^S$, $d_\emptyset = \mathbb{E}f$.

Proposition 13.1 (Boolean sieve limit at arbitrary level; [NC, Prop. 2.1]; [PROVED])**.** *In the setting of Proposition 4.4, assume $p_i \leq 1/4$ for all i, and let $f \geq 0$ with $f(0) \geq 1$ be of* any *level. Put $r_0 = \sum_{s(S)>\lambda} |d_S| \prod_{i\in S} p_i$ and $r_1 = \sum_{s(S)>\lambda} |d_S| \prod_{i\in S} 2p_i(1-p_i)$, $s(S) = \sum_{i\in S} s_i$. Then $\mathbb{E}f \geq (1-r_0)\mathrm{e}^{-\Phi} - 2r_1$, with Φ the right side of (3).*

Proof sketch. Split $f = f_{\mathrm{lo}} + f_{\mathrm{hi}}$ at level λ. Then $\mathbb{E}f = \mathbb{E}f_{\mathrm{lo}}$, $f_{\mathrm{lo}}(0) \geq 1 - r_0$, $\mathbb{E}|f_{\mathrm{hi}}| \leq r_1$, and $f_{\mathrm{lo}} \geq -h$ with $h = \mathbb{E}[|f_{\mathrm{hi}}| \mid x_{\{s_i\leq\lambda\}}]$. Run the proof of Proposition 4.4 on f_{lo}, carrying h: at the interpolation step write $LQ \leq |L|(Q+H) + |L|H$ with $Q + H \geq 0$. □

Lemma 13.2 (Walsh coefficients versus Fourier mass; [NC, Lemma 2.2]; [PROVED])**.** *Take a prime-slice system with $\mathcal{P}$ finite and a majorant ν (any moduli). For $c \in R$ let $\nu_c = \sum_S d_S(c) y^S$ be the conditional expectation of ν given $n \equiv c \pmod{Q_0}$ and the hit indicators. For $S \neq \emptyset$ let Θ_S be the frequencies $a/Q_0 + \sum_{\ell\in S} h_\ell/\ell$ with every $h_\ell \not\equiv 0 \pmod\ell$, and $A_S(\nu) = \sum_{\theta\in\Theta_S} |\hat\nu(\theta)|$. Then $|d_S(c)| \prod_{\ell\in S} p_\ell(c)(1-p_\ell(c)) \leq A_S(\nu) \prod_{\ell\in S} p_\ell(c)$, and $\sum_{S\neq\emptyset} A_S(\nu) \leq R_1(\nu)$.*

Proof. $d_S(c)\prod_S p_\ell(1-p_\ell) = \mathbb{E}[\nu_c y^S] = Q_0\mathbb{E}_n[\nu(n)g(n)]$ with $g = \mathbf{1}[n \equiv c\ (Q_0)]\prod_{\ell\in S} y_\ell(n \bmod \ell)$. By CRT, $\hat g$ is supported on Θ_S with $|\hat g| \le Q_0^{-1}\prod_S p_\ell(c)$; apply Parseval. The sets Θ_S are pairwise disjoint and avoid 0. □

Theorem 13.3 (sieve limit with per-frequency rounding; [NC, Thm 2.3, Cors 2.4, 2.5]; [PROVED]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4)**.** *Take a finite prime-slice system with $p_\ell(c) \le 1/4$, weights $s_\ell = \log(1/(2p_\ell^+))$, $p_\ell^+ = \max_c p_\ell(c)$, and a majorant ν of arbitrary level. Let $\bar\Phi(\lambda,\alpha)$ be the right side of (4) minus $\log(Q_0/|R|)$, computed with these weights and $s_* = \log 2$. Then for all $\lambda \ge s_*$ and $\alpha > 0$,*

$$\mathbb{E}\nu \ge \frac{|R|}{Q_0}\big[(1-\varepsilon)\mathrm{e}^{-\bar\Phi(\lambda,\alpha)} - 2\varepsilon\big], \qquad \varepsilon = \sum_{s(S)>\lambda} A_S(\nu)\mathrm{e}^{-s(S)} \le \mathrm{e}^{-\lambda}R_1(\nu).$$

Consequently, for the families of Corollary 6.1, every bound $\#(\mathcal{A}\cap[1,N]) \le N\mathbb{E}\nu + R_w(\nu)$ with $w \ge 1$ has saving at most $C(\log N)^{3/4} + \log(P/\varphi(P))$, with no bound on the size of the slice primes.

Proof sketch. In each fibre apply Proposition 13.1; by Lemma 13.2 both tails are at most ε. Average over c as in Theorem 4.5. If the bound equals $N\mathrm{e}^{-s}$ then $R_1 \le N$, and for $N \ge N_0(C)$ the level $\lambda = 2\log(4N)$ gives $\varepsilon \le \mathrm{e}^{-\bar\Phi}/4$. For the families of Corollary 6.1, $s_\ell \ge \frac{1-C}{2}\log\ell$ for $\ell \ge \ell_0(C)$, so the profile is $\ll \alpha^{-3}$ as there. □

Only the Fourier mass above level λ enters. This removes, for finite prime-slice families, the proviso "slice primes $\le N^{O(1)}$" (item (A3) of Definition 3.3), and it covers the coefficient sum, the sawtooth bound in the form $\min(N, 1/(2\|\theta\|))$, and complete exponential sums over the classes of one frequency. It does *not* cover [NC, §§2.3–2.5]: Erdős–Turán/Vaaler rounding with the factor $2|\sin\pi N\theta|$, per-class floor/ceiling rounding, the exact weight $|S_N(\theta)|$, smooth windows, or Kloosterman/dispersion cancellation over the moduli; these have weight < 1 or cancel between frequencies. For smooth windows, $Q_0 = 1$ and hit-pattern majorants, the cap follows from an equidistribution statement for sums $\sum_{\ell\in S} h_\ell/\ell$ near 0, Conjecture (H_{eq}) of [NC, (2.6)] [CONDITIONAL]; the other cases are [OPEN].

13.2. **Prime-only majorants.** A *prime majorant* is ≥ 0 at all primes and ≥ 1 at the primes of $\mathcal{A}$; the bound is $\sum_{p\le N}\nu(p)$.

Theorem 13.4 ([NC, Lemma 3.1, Thms 3.2, 3.3]; [PROVED])**.** *(1) Assume $|F_\ell(c)\setminus\{0\}| \le (\ell-1)/4$ (so in particular $< \ell-1$). By Dirichlet's theorem, a prime majorant is ≥ 0 on every reduced class modulo a common period L and ≥ 1 on every reduced class contained in $\mathcal{A}$. Theorem 4.5 and Lemma 3.4 hold for the uniform law on $(\mathbb{Z}/L)^\times$, with $\varphi(Q_0)$, $R\cap(\mathbb{Z}/Q_0)^\times$ and $p_\ell^*(c) = |F_\ell(c)\setminus\{0\}|/(\ell-1)$ in place of $Q_0, R, p_\ell(c)$ (the budget lemma up to a factor $\ll \log\lambda$). For the families of Corollary 6.1 with a selector, the cap $C(\log N)^{3/4} + \log(P/\varphi(P))$ at level $A\log N$ is unchanged. (2) If an integer majorant is ≥ 1 only on $\mathcal{A}\cap\{(n,P(z))=1\}$ with $z \le \mathrm{e}^\lambda$, has level $\le\lambda$ in the system augmented by the class 0 at each $\ell \le z$ (the primes $\ell < 5$ go into the selector), and the augmented probabilities are $\le 1/4$, then the bound of Theorem 4.5 grows by at most $C_4\sum_{\ell\le z}\ell^{-1-\alpha} + O(\log^2\lambda)$, i.e. by $O((\log\log N)^2)$ at $\alpha = \lambda^{-1/4}$, $\lambda \asymp \log N$.*

So prime equidistribution at level $N^{O(1)}$ (Siegel–Walfisz, Bombieri–Vinogradov, Bombieri–Davenport–Halberstam, Elliott–Halberstam, GRH) cannot lift a prime

majorant above the cap. That such majorants live at level $N^{O(1)}$, and that unconditional prime error terms are weaker than the integer count at relative accuracy $\mathrm{e}^{-(\log N)^{3/4}}$, is an *Assessment* [NC, Remark 3.4], not a theorem.

13.3. **Moment methods.** For a finite witness family $\mathcal{W}$ of classes put $f(n) = \#\{W \in \mathcal{W} : n \in W\}$. If P is a real polynomial with $P \geq 0$ on $\mathbb{Z}_{\geq 0}$ and $P(0) \geq 1$, then $P \circ f$ is a CRT majorant of $\{f = 0\}$ [NC, Prop. 4.1] [PROVED]. Hence integer moment methods of any order, whose moments are evaluated as $N\cdot$(CRT density) plus a per-frequency error, are capped by Theorem 13.3; prime moments are capped at level $\leq A \log N$ by Theorem 13.4. Beyond that level they are capped only under the Assessment of [NC, Remark 3.4] (every high-level term costs $\geq |a_i|$) together with $\sum_{\text{high}} |a_i| < \pi(N)$ and slice primes $\leq N^{O(1)}$ [NC, Prop. 4.1]. For slice witness classes and $\deg P \leq k$, $\log(1/\mathbb{E}[P \circ f]) \leq 19k(2 + \log^{+}(C_4\bar{\mu}/(19k))) + O(\log(\bar{\mu}+k)) + \log(Q_0/|R|)$ [NC, Prop. 4.2] [PROVED]; with witness moduli $\leq N^{O(1)}$, $\bar{\mu} \ll (\log N)^3$ and a degree-k method saves $O(k \log\log N)$ in the CRT mean, so the second moment gives only a power of $\log N$.

13.4. **Direct interval counts.** What is left of signed rounding is cancellation *between* frequencies in (10), i.e. a direct count of $\sum_{n \leq N} \nu(n)$. Theorem 13.3 turns into a dichotomy [NC, Thm 8.1] [PROVED]: if $\sum_{n \leq N} \nu(n) \leq N\mathrm{e}^{-s}$ then, for every λ and α, either the Fourier mass above level λ satisfies $\varepsilon > \mathrm{e}^{-\bar{\Phi}(\lambda,\alpha)}/4$, or $N\mathbb{E}\nu - \sum_{n \leq N} \nu(n) \geq N[(|R|/(4Q_0))\mathrm{e}^{-\bar{\Phi}} - \mathrm{e}^{-s}]$. For the families of Corollary 6.1, $\bar{\Phi}(\lambda) \leq C\lambda^{3/4}$, so for $s = (\log N)^{\theta}$, $\theta > 3/4$, the second branch is non-vacuous at every level $\lambda \leq c(\log N)^{4\theta/3}$ (a range containing levels $\gg \log N$): there a direct count needs either Fourier mass $\gtrsim \mathrm{e}^{-C\lambda^{3/4}}$ above level λ, or an interval count a factor $\geq \mathrm{e}^{s - C\lambda^{3/4}}$ below the CRT mean [NC, §8.1]. Two facts about $[1, N]$ are proved. Case-B classes of modulus $> M_0 := 8\lfloor (N+1)/3 \rfloor^2$ contain no $n \leq N$, so for hit-pattern majorants they give neither sieving power nor cost [NC, Lemma 8.2, Cor. 8.3]; and the multiplier-one witnesses of all moduli have mean $\leq \frac{1}{4}(1+1/N)(1+\log N)^2$ on $[1, N]$ [NC, Prop. 8.4]. The moduli in $(N, M_0]$ still carry real hit mass $\asymp (\log N)^3$ per integer, a constant factor below the CRT value [EVIDENCE]; a conjecture of [NC] to the contrary was withdrawn after review. For hit-count polynomials of degree k evaluated exactly on $[1, N]$, beating $3/4$ needs $k \geq (\log N)^{3/4 + o(1)}$, [CONDITIONAL] on a local-limit hypothesis H_{node} for the witness count, which is untested, and under the further hypothesis that the mean witness count on $[1, N]$ is $m \geq 16(\log N)^{3/4+\delta}$ [NC, Prop. 8.6, Cor. 8.7]. For the full Case-B family this hypothesis is supported only by enumeration and a heuristic ($m \asymp (\log N)^3$, [EVIDENCE]). Numerically, for polynomials of degree ≤ 10 in the hit count of a prime-modulus subfamily, the exact interval optimum saves less than the CRT optimum [EVIDENCE] [NC, §8.3].

14. Large sieves, prime laws and interval counts

This section summarises further internal documents: [LS] (review [LSr]), [LS2] (reviews [LS2r] of §§1–7 and [LS2r2] of §§8–9), [LS3] (review [LS3r]), [LS4] (review [LS4r]), [PL] (review [PLr]), [IF] (review [IFr]), and, in Section 14.9, [IF2], [SPW] and [SPW2]. No review found a fatal defect. The IF review found one major scope overclaim (hybrid methods), repaired by restricting the claim ([IF, Rem. 2.6] and Section 14.9 here); the other repairs were minor, and all are applied in the sources. Throughout, $\mathfrak{G}$ is a finite mixture of the four types of Definition 10.1 with arbitrary

moduli, $\mathcal{A} = \mathcal{A}(\mathfrak{G}) \subseteq \mathbb{Z}$, and M_0 is the lcm of its moduli. The *level* of a modulus d, or of a rational frequency of denominator d, is $\lambda(d) = \sum_{\ell \mid d,\, \ell > W} \log \ell$, with W from Theorem 10.12. For a common period M' fixed in each statement, $\mathcal{M}(\mathcal{A})$ is the set of probability measures on $\mathbb{Z}/M'$ supported on $\mathcal{A} \bmod M'$, and $\hat{\rho}(\theta) = \sum_n \rho(n)\mathrm{e}(n\theta)$.

The sources cite [KA2, Thm 5.1], with cap $C\lambda^{3/4}(\log\lambda)^{3/4}$. The integer-measure results use it only as a black-box lower bound for $\mathbb{E}\nu$, so Theorem 10.12 may be substituted, giving $C\lambda^{3/4}$ and the reading $C_A(\log N)^{3/4}$ ([KA3, §4.3] for [LS, Thm 3.1] and [IF, Cor. 2.3]; the update note at the head of [LS2] and ledger item (D)24 for the results of [LS2]; for [IF, Thm 2.5] the substitution is checked in this note). Below it is a one-line step in each proof. The unit-measure results (Theorem 14.19 and its consequences) rerun the proof instead; for them the improvement is pointer-level only, and both forms are stated. Case-A classes enter only through the cap used (Theorem 10.12, Theorem 14.19 or Corollary 6.1) and carry its dependence on [ElT]; Theorem 14.7 and Theorem 14.8(3) have none.

14.1. **Duality and the cap.** A finite $\Theta \subset \mathbb{R}/\mathbb{Z}$ with weights $w_\theta > 0$ is an N-*large-sieve system* if $\sum_\theta w_\theta |\sum_{n\in I} a_n \mathrm{e}(n\theta)|^2 \le \sum_{n\in I} |a_n|^2$ for every interval I of N consecutive integers: Montgomery–Vaughan [MV], the Farey form [M] with any prime, prime-power or composite moduli, weighted forms. Always $w_\theta \le 1/N$ (take $a_n = \mathrm{e}(-n\theta)$) and $\sum_\theta w_\theta \le 1$ (random signs). Let $A \subseteq \mathcal{A} \cap I$, $Z = |A|$, $\Theta_{\mathbb{Q}}$ the rational points of Θ, M' a common multiple of M_0 and their denominators, and $F_w(\rho) = \sum_{\theta\in\Theta_{\mathbb{Q}}} w_\theta |\hat{\rho}(\theta)|^2$. With $a_n = \mathbf{1}_A$ the inequality gives $Z^2 F_w(\rho_A) \le Z$, ρ_A the law of $A \bmod M'$. A bound $Z \le 1/L$ is *CRT-admissible* if $L \le F_w(\rho)$ for every $\rho \in \mathcal{M}(\mathcal{A})$, as in every arithmetic large sieve in use. Irrational frequencies are excluded by definition.

Theorem 14.1 (duality; [LS, Thm 2.1, Cor. 2.2]; standard; [PROVED])**.** *Put* $F_w^* = \min_{\rho\in\mathcal{M}(\mathcal{A})} F_w(\rho)$ *and* $m_w = \inf\{\sum_\theta |\gamma_\theta|^2/w_\theta : g = \sum_{\theta\in\Theta_{\mathbb{Q}}} \gamma_\theta \mathrm{e}(-n\theta),\ \mathrm{Re}\, g \ge 1 \text{ on } \mathcal{A}\}$. *Then* $F_w^* = 1/m_w$, *attained when finite. For an* N*-large-sieve system every CRT-admissible bound has* $1/L \ge m_w \ge N \cdot \mathbb{E}\nu^*$, *where* $\nu^* = |g^*|^2$ *for an optimal* g^*. ν^* *is* ≥ 0 *on* $\mathbb{Z}$, ≥ 1 *on* $\mathcal{A}$, *and a real combination of classes with moduli dividing* $\mathrm{lcm}(\mathrm{den}\,\theta, \mathrm{den}\,\theta')$, $\theta, \theta' \in \Theta_{\mathbb{Q}}$.

Proof sketch. For feasible g, $1 \le \mathrm{Re}\sum_n \rho(n) g(n) = \mathrm{Re}\sum_\theta \gamma_\theta \overline{\hat{\rho}(\theta)}$, and Cauchy–Schwarz gives $F_w(\rho) \ge 1/m_w$. The reverse is the minimax theorem on $\mathcal{M}(\mathcal{A})$ times the unit ball. Finally $\sum |\gamma|^2/w \ge N \sum |\gamma|^2 = N\mathbb{E}|g^*|^2$ by $w \le 1/N$ and Parseval. □

This is the classical large-sieve/Selberg duality, not claimed as new. There is no rounding term, since the factor N sits in the large-sieve constant; only the *level* of ν^* matters. The multiplicative sieve is covered: by Gauss sums, $q\sum_{\chi \text{ prim}} |\sum_n \rho(n)\chi(n)|^2 \le \varphi(q) \sum^*_{a \bmod q} |\hat{\rho}(a/q)|^2$, so the character functional lies pointwise below the Farey one [LS, Rem. 2.5]. For prime product systems Montgomery's sieve is CRT-optimal, $F_1^* = S(Q)$ [LS, Prop. 2.6] [PROVED].

Fibrewise sieves. Fix $Q_0 \le N/2$. In the fibre $c \bmod Q_0$ write $n = c + Q_0 m$, with m in an interval of $N_c \ge \lfloor N/Q_0 \rfloor$ integers, avoiding $\mathcal{A}_c = \{m : c + Q_0 m \in \mathcal{A}\}$. Each fibre uses its own N_c-large-sieve system with a CRT-admissible L_c, or the trivial bound; $B = \sum_c 1/L_c$. Let λ_Θ be the largest level of a rational frequency used.

Theorem 14.2 (large-sieve cap; [LS, Thm 3.1, Cor. 3.2], [KA3, §4.3]; [PROVED] via Theorem 10.12; Case A as there)**.** *With* $\lambda = \max(\lambda_0, \lambda(Q_0) + 2\lambda_\Theta)$, *every*

fibrewise CRT-admissible large-sieve bound for $A \subseteq \mathcal{A} \cap I$, $|I| = N$, has $\log(N/B) \le \log 2 + C\lambda^{3/4}$. *If $Q_0 \le \min(N^A, N/2)$ and every frequency has denominator $\le N^A$, the saving is $\le C_A(\log N)^{3/4}$. So no CRT-admissible large sieve with polynomially bounded denominators (any moduli, forced classes used in any form, any weights, fibrewise or not) proves $E(N) \ll N\exp\{-(\log N)^\theta\}$ with $\theta > 3/4$.*

Proof. In fibre c, Theorem 14.1 for the M_0-periodic set $\mathcal{A}_c$ gives $1/L_c \ge N_c \mathbb{E}\nu_c$; put $\nu_c \equiv 1$ in trivial fibres, $\nu_c \equiv 0$ if $\mathcal{A}_c = \emptyset$. Let $\nu(n) = \nu_c((n-c)/Q_0)$ for $n \equiv c \pmod{Q_0}$. A term $a\mathbf{1}[m \equiv b \pmod d]$ becomes $a\mathbf{1}[n \equiv c + Q_0 b \pmod{Q_0 d}]$, so ν is a majorant of $\mathcal{A}$ of level $\le \lambda$, and $B \ge \lfloor N/Q_0 \rfloor Q_0 \mathbb{E}\nu \ge (N/2)\mathbb{E}\nu$. Apply Theorem 10.12, which charges only primes $> W$ (so smooth parts such as Farey $q = 2^k$ are free). For the reading, $\lambda \le 3A\log N$. □

Averaging g^* modulo M_0 kills every frequency whose denominator does not divide M_0, so λ_Θ may be computed on those only [LS, Rem. 3.3]. The large sieve of the 2/3 note [LL] is an instance ($Q_0 \le N^{2\delta}$, prime moduli, $Q = (N/Q_0)^{1/2}$, atoms $\mathscr{R}(k\ell)$ with $k\ell \le \ell^{1+o(1)}$), capped with Theorem 10.13 in place of Theorem 10.12. So are [V] and [PW, §4], on the 2/3 note's description of them; neither was re-read [LS, Rem. 3.4]. This supersedes Remark 4.8.

Theorem 14.3 (prime slices, any rational frequencies; [LS, Thm 4.1, Cor. 4.2, Rem. 4.4]; [PROVED]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4)**.** *Take a prime-slice system (Definition 3.1) with $f_\ell(c) = |F_\ell(c)| \le \min(\ell/2, \ell^\kappa)$, $0 \le \kappa < 1$, and $\bar p_\ell = \mathbb{E}_{c\in R} f_\ell(c)/\ell$. Every CRT-admissible bound B from any N-large-sieve system, with any rational frequencies, satisfies for $0 < \alpha \le 1-\kappa$*

$$\log(N/B) \le \frac{\alpha \log N}{2(1-\kappa)} + \log\frac{2Q_0}{|R|} + 8\sum_\ell \bar p_\ell \ell^{-\alpha}.$$

For the families of Corollary 6.1 with all slice primes $\ge \ell_0'(C)$ (an enlarged threshold) the saving is $\le C'(C)(\log N)^{3/4} + \log(P/\varphi(P)) + \log 2$, also fibrewise with $Q_0 \le N/2$.

Proof sketch. Let $\beta = \alpha/(2(1-\kappa))$, $p' = 2 + 2\beta$, and let ρ be the product measure (residues mod ℓ uniform outside $F_\ell(c)$) on the at least $|R|/2$ fibres of at most twice the mean supply. Hausdorff–Young on $\mathbb{Z}/Q_0$ gives $\sum_\theta |\hat\rho(\theta)|^{p'} \le (2Q_0/|R|)\exp\{8\sum \bar p_\ell \ell^{-\alpha}\}$; Hölder with $\sum w \le 1$, $w \le 1/N$ bounds $F_w(\rho)$. Then take $\alpha = (\log N)^{-1/4}$. □

For composite moduli, the Euler product of the prime-local system obtained by conditioning on small primes does *not* dominate the large sieve [LS, Ex. 5.2] [PROVED]; the top-prime sequential functional does, through Theorem 14.2.

14.2. **Twisted, hybrid and prime large sieves.**

Lemma 14.4 (comparison measure; [LS2, Lemmas 1.1, 1.2]; [PROVED] via Theorems 10.12, 14.19; Case A as there)**.** *Let $\mathcal{D} \ni 1$ be a finite set of moduli of level $\le \lambda$, $\lambda \ge \lambda_0$, M' a common multiple of M_0 and $\mathcal{D}$, $V_\mathcal{D}$ the real span of the classes with moduli in $\mathcal{D}$, and $\mathbb{E}_U$ the uniform mean on $\mathbb{Z}/M'$. Some $\rho \in \mathcal{M}(\mathcal{A})$ has $\mathbb{E}_\rho f \le \mathrm{e}^{C\lambda^{3/4}}\mathbb{E}_U f$ for every $f \in V_\mathcal{D}$ with $f \ge 0$. The same holds on units: ρ on $\mathcal{A} \cap (\mathbb{Z}/M')^\times$, $\mathbb{E}^*$ uniform on units, $f \ge 0$ on units, and $\mathrm{e}^{S^*(\lambda)}$ with S^* from Theorem 14.19.*

Proof. The linear programme $m^* = \min\{\mathbb{E}_U\nu : \nu \in V_{\mathcal{D}},\ \nu \ge \mathbf{1}_{\mathcal{A}}\}$ has $m^* \ge \mathrm{e}^{-C\lambda^{3/4}}$ by Theorem 10.12. Duality gives $\mu \ge 0$ with $\sum_x \mu(x)\nu(x) = \mathbb{E}_U\nu$ on $V_{\mathcal{D}}$ and $\mu(\mathcal{A}) = m^*$; take $\rho = \mu|_{\mathcal{A}}/m^*$. □

One measure tests every nonnegative low-level function, not only majorants, so Cauchy–Schwarz can strip twists and fibres.

A *periodic Bessel system* of length N consists of periodic rows ϕ_j (period d_j) and $\Delta > 0$ with $\sum_j |\sum_{n\in I} a_n\overline{\phi_j(n)}|^2 \le \Delta\sum_{n\in I}|a_n|^2$ for every interval I of length N; its level is $\max_j \lambda(d_j)$. Examples: $\sqrt{w_\theta}\,\mathrm{e}(n\theta)$ ($\Delta = 1$); $(q/\varphi(q))^{1/2}\chi$, χ primitive mod $q \le Q$ ($\Delta = N + Q^2$); hybrid rows $\chi(n)\mathrm{e}(n\theta)$; Gauss-sum rows. It is applied to $a_n = \psi(n)\mathbf{1}_A(n)$ with a periodic twist, $|\psi| \ge 1$ on $\mathcal{A}$; a bound $Z \le \Delta/L$ is CRT-admissible if $L \le \sum_j |\mathbb{E}_\rho[\psi\bar{\phi}_j]|^2/\mathbb{E}_\rho|\psi|^2$ for all $\rho \in \mathcal{M}(\mathcal{A})$. A *fibrewise hybrid* over $Q_0 \le N/2$ uses in each fibre (i) such a system, (ii) a majorant ν_c of $\mathcal{A}_c$ with bound $N_c\mathbb{E}\nu_c + \sum|a_i|$, or (iii) the trivial bound N_c; $B = \sum_c B_c$.

Theorem 14.5 (hybrid cap; [LS2, Thm 2.4, Cor. 2.5]; [PROVED] via Theorem 10.12; Case A as there)**.** *Let the type-(i) systems have level $\le \lambda_\Phi$, every prime of $\mathfrak{G}$ be $\le \mathrm{e}^{\Lambda_0}$, and $\lambda = \max(\lambda_0, \lambda(Q_0) + \max(2\lambda_\Phi, \Lambda_0 + \log N))$. Then $B \ge (N/2)\mathrm{e}^{-C\lambda^{3/4}}$. If $Q_0 \le \min(N^A, N/2)$ and all row periods and all primes of $\mathfrak{G}$ are $\le N^A$, the saving is $\le C_A(\log N)^{3/4}$. The twist's period is unrestricted.*

Proof sketch. Averaging the adjoint inequality over shifts of I shows that every c has $N\mathbb{E}_U|\sum_j c_j\phi_j|^2 \le \Delta\|c\|^2$. Take ρ from Lemma 14.4 with $\mathcal{D}$ containing 1, Q_0, $Q_0\,\mathrm{lcm}(d_j, d_k)$ and the coarsened fibre moduli, and its fibre laws ρ_c. Type (i): Cauchy–Schwarz removes ψ, and the lifted $|\sum_j c_j\bar{\phi}_j|^2 \ge 0$ lies in $V_{\mathcal{D}}$, so $B_c \ge N_cQ_0\rho(c)\mathrm{e}^{-S}$. Type (ii): after projection to the family period (as in the proof of Theorem 10.14), Lemma 3.4 coarsens ν_c to level $\le \Lambda_0 + \log N_c$ and mean $\le B_c/N_c$; then $1 \le \mathbb{E}_{\rho_c}\nu_c' \le \mathrm{e}^S\mathbb{E}\nu_c'/(Q_0\rho(c))$. Type (iii): $\rho(c) \le \mathrm{e}^S/Q_0$. Sum over c. □

So twisted, multiplicative, Gauss-sum and hybrid large sieves, and fibrewise mixtures with rounding fibres, give no $\theta > 3/4$. Not covered: non-periodic rows (n^{it}, smooth weights in n), which use the position of n in I.

Theorem 14.6 (the large sieve for primes; [LS2, Thm 3.1]; [PROVED] via Theorem 14.19; Case A as in Theorem 14.19)**.** *Let A consist of primes $p \le N$, $p \in \mathcal{A}$, $p \nmid M'$; let admissibility range over all probabilities on $\mathcal{A} \cap (\mathbb{Z}/M')^\times$, and let all fibres be of type (i) or (iii), $c \in (\mathbb{Z}/Q_0)^\times$. Then $B \ge (N/2)\mathrm{e}^{-S^*(\lambda)}\varphi(M')/M'$, $\lambda = \max(\lambda_0, \lambda(Q_0) + 2\lambda_\Phi)$. If all primes of M' (family, rows, twists, Q_0) are $\le N^A$ and $\lambda \le 3A\log N$, the saving against $\pi(N)$ is $\le C_A(\log N)^{3/4}(\log\log N)^{3/4}$, and $\le C_A(\log N)^{3/4}$ at pointer level.*

The proof is that of Theorem 14.5 with the unit measure, using $\mathbb{E}^*f \le (M'/\varphi(M'))\mathbb{E}_Uf$ for $f \ge 0$ and $M'/\varphi(M') \le 2A\log N$.

14.3. **The larger sieve.** For moduli $q \in \mathcal{S}$ with weights $w(q) \ge 0$ let $K(m) = \sum_q w(q)\mathbf{1}[q \mid m]$, $W_K = K(0)$, $h \ge \max_{0<|m|<N} K(m)$, $\mathrm{coll}_q(\rho) = \sum_b \rho(b \bmod q)^2$ and $D(\rho) = \sum_q w(q)\mathrm{coll}_q(\rho)$. Counting pairs gives $Z^2D(\rho_A) \le ZW_K + (Z^2 - Z)h$, so $Z \le (W_K - h)/(L - h)$ if $h < L \le D(\rho_A)$; the bound is CRT-admissible if $L \le D(\rho)$ for all $\rho \in \mathcal{M}(\mathcal{A})$. Gallagher's larger sieve [G] takes $w = \Lambda$ on prime powers $\le Q$, $h = \log N$ and $L = \sum\Lambda(q)/\nu(q)$, $\nu(q) \ge |\mathcal{A} \bmod q|$. With $\chi_q^2(\rho) = q\,\mathrm{coll}_q(\rho) - 1$ and $X(\rho) = \sum_q (w(q)/q)\chi_q^2(\rho)$, every CRT-admissible kernel

bound satisfies $\log(N/B) \le \log(1 + NX(\rho)/(W_K - h))$ for every $\rho \in \mathcal{M}(\mathcal{A})$ [LS, Thm 6.2] [PROVED], since $\sum_q w(q)/q \le h + (W_K - h)/N$. On the prime slices of Corollary 6.1 Gallagher's sieve saves $O(1)$ [LS, Cor. 6.3].

Theorem 14.7 (Gallagher's sieve over any mixture; [LS2, Thm 4.3]; [PROVED], unconditional)**.** *There are absolute C, N_0 such that for every $\mathfrak{G}$ (Case A included) and $Q \ge 2$ some $\rho \in \mathcal{M}(\mathcal{A})$ has $\sum_{q=\ell^v \le Q} (\Lambda(q)/q)\chi_q^2(\rho) \le 24 \log\log(3Q) + C$. Hence for $N \ge N_0$ Gallagher's larger sieve, in CRT-optimal or Cauchy–Schwarz form, with any Q, saves at most $26 \log\log N + C$.*

Proof sketch. Take $W = \max(W_0, (\log 3Q)^8)$, the sequential law σ of Section 8 on the base $R_W^{\square}$ (Lemma 10.3), singletons at the primes $\ell > W$, and $\rho = \sigma(\cdot \mid \mathcal{A})$. The leak is $\le C(\log W)^c W^{-1/4}$ (Lemma 10.11, uniformly in W). At $\ell > W$, $\chi_{\ell^v}^2(\sigma) \ll (\log W)^c \ell^{-7/8+o(1)}$; at odd $p \le W$, $\chi^2 = (p+1)/(p-1) \le 2$ (and $\chi^2 \le 7$ at $p = 2$); Mertens and $\mathfrak{L} \log W = o(1)$ give $3 \log W + C$. □

It uses Shiu's and Mertens' theorems only, not [ElT] or Theorem 10.12.

Theorem 14.8 (composite kernels; [LS2, Thms 4.2, 9.1, Prop. 9.2]; [PROVED] via Theorem 10.12, except (3))**.** *(1) If every kernel modulus has level $\le \lambda_{\mathcal{S}}$ and $\lambda = \max(\lambda_0, 2\lambda_{\mathcal{S}})$, every CRT-admissible kernel bound has $B \ge (N/2)\mathrm{e}^{-C\lambda^{3/4}}/(1 + Nh/(W_K - h))$. (2) The same holds with no level hypothesis and $\lambda = \max(\lambda_0, 2\log N + \Lambda_0)$ if all primes of $\mathfrak{G}$ and of the kernel moduli are $\le \mathrm{e}^{\Lambda_0}$. (3)* [PROVED]*, unconditional: if $T \ge W_0$ and every kernel modulus has no prime factor $\le T$ and at most $T^{1/4}$ prime factors, the saving is $\le \log(1 + \varepsilon_T + \varepsilon_T Nh/(W_K - h))$, $\varepsilon_T = C(\log T)^c T^{-1/4}$.*

Proof sketch. $D(\rho) = \sum_\theta \tilde{w}_\theta |\hat{\rho}(\theta)|^2$ with $0 \le \tilde{w}_\theta \le h + (W_K - h)/N$; apply Lemma 14.4 as in Theorem 14.5. (2) Cut $q \ge N$ to a prefix $q^\circ \ge N$ of level $\le \log N + \Lambda_0$: $\mathrm{coll}_q(\rho) \le \mathrm{e}^S/N$. (3) The law of Theorem 14.7 with $W = T$. □

Open: kernels with $Nh/(W_K - h) \ge \exp\{(\log N)^{3/4}\}$ whose moduli have a prime factor $\le (Nh/(W_K - h))^{4+o(1)}$ or more than $T^{1/4}$ prime factors [OPEN]. No failing example is known.

14.4. **What remains of the large sieve.** Of the escapes of [LS, §7] there remain: (E1) frequencies of super-polynomial level against classes with several large primes (by Theorem 14.13 below, only classes with at least two prime factors $> \exp\{(\log N)^{1/4}\}$ remain); (E2) composite kernels as above; (E3) non-CRT information; (E4) the exclusions inherited from Theorem 10.14.

Proposition 14.9 (what (E1) needs; [LS2, Props 5.1, 5.2]; [PROVED]; (1) via Theorem 10.12)**.** *(1) Let all primes of $\mathfrak{G}$ be $\le N^A$. If every frequency has at most r distinct primes $> W$ in its denominator, the saving is $\le \log 2 + C \max(\lambda_0, 2rA \log N + \lambda(Q_0))^{3/4}$. So a saving $\ge (\log N)^{3/4+\varepsilon}$ needs denominators with $\ge (\log N)^{4\varepsilon/3 - o(1)}$ distinct family primes. (2) If some $\rho \in \mathcal{M}(\mathcal{A})$ has $\mathbb{E}_\rho f \le \mathrm{e}^{S_1} \mathbb{E}_U f$ for all $f \ge 0$ of level $\le 2\lambda'$, and $|\hat{\rho}(\theta)|^2 \le \mathrm{e}^{S_2}/N$ for all θ of level $> \lambda'$, then every CRT-admissible large-sieve bound, with any rational frequencies and weights, is $\ge N/(\mathrm{e}^{S_1} + \mathrm{e}^{S_2})$.*

Part (2) has no cross terms, since F_w is diagonal in θ.

Conjecture 14.10 ($(H_{\mathrm{LS}\infty})$; [LS2, §5]; [CONJECTURE])**.** For every family $\mathfrak{G}$ of the four types with primes $\le N^{O(1)}$ there are a fixed $\eta > 0$ and $\rho \in \mathcal{M}(\mathcal{A})$ satisfying

Lemma 14.4 at level $2\lambda' = (2/\eta)\log N$ and $|\hat\rho(\theta)| \le \mathrm{e}^{C\lambda'^{3/4}}\mathrm{e}^{-\eta\lambda(\theta)}$ for every θ of level $> \lambda'$.

By Proposition 14.9(2) this caps every CRT-admissible large sieve at $C_\eta(\log N)^{3/4}$, whatever the frequencies. The sequential law decays only at the top prime of the denominator; a cluster expansion suggests $\eta = 1/2 - o(1)$ (*Assessment*). A proof needs family-specific input, such as the sparsity of forced classes:

Theorem 14.11 (dense band families escape; [LS2, Lemma 8.1, Props 8.2, 8.4, 8.6, Thm 8.5], [LS2r2]; [PROVED])**.** *Let $0 < \eta \le 1/8$, let $\ell_i, \ell_i' \ge 5$ $(i \le K)$ be distinct primes, $D_i = \ell_i\ell_i'$, $\gcd(a_i, D_i) = 1$, and $\mathcal{A} = \bigcap_i \{n : \|na_i/D_i\| \le 1/2 - \eta\}$: an abstract class family with $\asymp \eta D_i$ classes mod D_i. (1) If all $\ell_i, \ell_i' > W$ and $\log D_i > \lambda$ for all i, every majorant of level $\le \lambda$ has $\mathbb{E}\nu \ge 1$. (2) Let $\eta = 1/9$, $R = 324$, $K = \lfloor \log(N/3)/\log(2R+1) \rfloor$ and $D_i \ge 16KRN$. For suitable a_i, a Montgomery–Vaughan large sieve with $\le N/3$ frequencies, each nonzero one with denominator a product of some D_i, gives $|\mathcal{A} \cap I| \le (5N/3)\mathrm{e}^{-K/27}$ for every interval I of length N: a saving $\ge c\log N$, $c = 1/(27\log 649) - o(1)$. (3) If moreover all $\ell_i, \ell_i' \in [N^{1/2+\delta}, 2N^{1/2+\delta}]$ with fixed small $\delta > 0$, then for $N \ge N_0$ every majorant of every level $\lambda' \ge \lambda_0$ has $\log(1/\mathbb{E}\nu) \le C\lambda'^{3/4}$, and (2) still holds.*

Proof sketch. (1) Gale's supply–demand theorem gives a measure on the allowed classes mod D_i with uniform marginals mod ℓ_i and ℓ_i'; the product measure lives on $\mathcal{A}$ and agrees with $\mathbb{E}_U$ on classes of level $< \log D_i$. (2) A Fejér construction gives P of degree R, $P \ge 1$ on $\|t\| \le 1/2 - \eta$, $\int P^2 \le 1 - \eta/3$; with a_i/D_i near $(2R+1)^{-i}$, Cauchy–Schwarz and [MV] applied to $g = \prod_i P(na_i/D_i) \ge 1$ on $\mathcal{A}$ give $|A| \le (5N/3)(1-\eta/3)^K$. (3) Use (1) below level $\min_i \log D_i$; above it Theorem 8.5 gives $O((\lambda'/\log N + 1)\log\log N)$, and the density of $\mathcal{A}$ gives $0.04\log N$. □

So (E1) cannot be closed by comparison measures at any or every level plus the large-sieve axioms. The band family puts mass $\asymp \eta$ on single moduli of high level. Forced families are expected to satisfy (Sp): for D of level $\ge \log N$, the classes with modulus divisible by D carry mass $\le D^{-1+o(1)}$ (*Assessment*, not proved). A proof of Conjecture 14.10 needs something like (Sp) or the moment hypotheses of Section 10. No method in the literature uses such frequencies.

14.5. **Smooth–rough splitting: one large prime per modulus.** The sparsity of forced classes is used in [LS3] (review [LS3r]: no fatal defect, all proofs sound; two overclaims, an "equivalence" and a remark, repaired by restricting the claims as stated below). It closes (E1) for every mixture in which each modulus has at most one prime factor above $z = \exp\{(\log N)^{1/4}\}$. Fix z and write $M_0 = M_sM_r$ with M_s z-smooth and M_r z-rough; by CRT $n \leftrightarrow (c, r)$ with $c = n \bmod M_s$, $r = n \bmod M_r$, and every frequency with denominator dividing M_0 splits uniquely as $\theta = \theta_s + \theta_r$. Let $\mathcal{A}_s \subseteq \mathbb{Z}/M_s$ be the residues avoiding every class of $\mathfrak{G}$ with z-smooth modulus. For $c \in \mathcal{A}_s$ the *fibre family* $\mathfrak{G}_c$ consists of the classes $b_C \bmod G_r(C)$, for the classes $C = (b_C \bmod G_C)$ of $\mathfrak{G}$ with $G_r(C) > 1$ and $c \equiv b_C \pmod{G_s(C)}$, where $G_C = G_s(C)G_r(C)$ is the smooth–rough split of the modulus; $\mathcal{A}_c \subseteq \mathbb{Z}/M_r$ is its avoider set, so that $\mathcal{A} \bmod M_0 = \{(c, r) : c \in \mathcal{A}_s,\ r \in \mathcal{A}_c\}$. For a probability σ on $\mathbb{Z}/m$ and $p' \ge 2$ put $\mathsf{R}_{p'}(\sigma) = \sum_{\theta \in m^{-1}\mathbb{Z}/\mathbb{Z}} |\hat\sigma(\theta)|^{p'}$ (with $\theta = 0$, so $\mathsf{R}_{p'} \ge 1$).

Theorem 14.12 (smooth–rough splitting; [LS3, Thm 1.1]; [PROVED], elementary)**.** *Let $0 < \beta \le 1/2$ and $p' = 2 + 2\beta$. Let π_s be a probability on $\mathcal{A}_s$ with $\pi_s(c) \le \rho/M_s$ for*

all c, and for each $c \in \operatorname{supp}\pi_s$ let π_c be a probability on $\mathcal{A}_c$ (so $\mathcal{A}_c \neq \emptyset$ there). Then every CRT-admissible bound B from any N-large-sieve system, with any rational frequencies and any weights, satisfies

$$\log(N/B) \le \beta \log N + \log\rho + \log \mathbb{E}_{c\sim\pi_s} \mathsf{R}_{p'}(\pi_c).$$

Proof sketch. Let $\pi = \sum_c \pi_s(c)\delta_c \otimes \pi_c$, a measure on $\mathcal{A}$. For fixed θ_r, Hausdorff–Young on $\mathbb{Z}/M_s$ applied to $f(c) = M_s\pi_s(c)\hat{\pi}_c(\theta_r)$, with $p = p'/(p'-1)$, followed by Jensen, gives $\sum_{\theta_s} |\hat{\pi}(\theta_s+\theta_r)|^{p'} \le \rho\, \mathbb{E}_{\pi_s}|\hat{\pi}_c(\theta_r)|^{p'}$, since $(p-1)p'/p = 1$. Summing over θ_r, $\mathsf{R}_{p'}(\pi) \le \rho \mathbb{E}_{\pi_s}\mathsf{R}_{p'}(\pi_c)$. Hölder with $\sum w \le 1$ and $w \le 1/N$ gives $F_w(\pi) \le (\mathsf{R}_{p'}(\pi)/N)^{1/(1+\beta)}$, as in Theorem 14.3; now use $B \ge 1/F_w(\pi)$. □

Nothing is assumed about the smooth frequencies θ_s: their denominators may have any number of prime factors $\le z$, so any level. The smooth coordinates cost only the *density* ρ of a measure on $\mathcal{A}_s$; Fourier control is needed only in the rough coordinates, in the $\ell^{p'}$ sense of Theorem 14.3. The density is affordable up to $z = \exp\{(\log N)^{1/4}\}$: running the sequential law of [KA2] (square base, plain rule at the primes $W_1 < \ell \le z$, W_1 absolute) and conditioning on a good event of probability $\ge 1/2$ (no activated set hit, all activated densities $\le 1/2$, total activated mass $\le 8\mathfrak{M}(z)$) gives π_s on $\mathcal{A}_s$ with

$$\log\rho \le 16\,\mathfrak{M}(z) + 2W_1 + \log 2, \qquad \mathfrak{M}(y) \le K_3(\log y)^3(\log\log y)^3,$$

$\mathfrak{M}(y)$ the first-moment bound of [KA2, §3, Cor. 3.7] [LS3, Lemma 2.1] ([PROVED] given [KA2, §§2–4]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4). Each step of the law has conditional density $\le \ell^{-E_\ell}(1-p_\ell)^{-1}$, and on the good event $\prod(1-p_\ell)^{-1} \le \mathrm{e}^{16\mathfrak{M}(z)}$. (The base modulus of [KA2] need not divide M_s; one takes π_s to be the marginal on $\mathbb{Z}/M_s$, which does not increase the maximal density [LS3, Lemma 2.1].)

Theorem 14.13 (rough-slice mixtures; [LS3, Thm 3.1]; [PROVED] given [KA2, §§2–4]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4)**.** *Let $z = \exp\{(\log N)^{1/4}\}$ and let $\mathfrak{G}$ be any mixture of the four types (any moduli, no B, no bound on their size) in which every modulus has at most one prime factor $> z$, to any power. Then every CRT-admissible N-large-sieve bound, with any rational frequencies, any denominators and any weights, saves*

$$\log(N/B) \le C(\log N)^{3/4}(\log\log N)^3.$$

Proof sketch. Every fibre family $\mathfrak{G}_c$ is a slice system: classes modulo ℓ^v, $\ell > z$ prime, whose union at ℓ is exactly the activated set of the sequential law at ℓ (each class with a prime $> z$ has it as its top prime and a z-smooth cofactor). Take $\beta = (\log N)^{-1/4}$, $\kappa = 3/4$, $\alpha = \beta/2$, and remove from π_s the event that some fibre density exceeds $\ell^{\kappa-1}$ (probability $\le \sum_{\ell>z}\ell^{2-2\kappa}\mathbb{E}p_\ell^2 \le 1/16$, by the second moments $\mathbb{E}p_\ell^2 \ll \ell^{-7/4+o(1)}$ of [KA2, Lemma 4.3]) or that $\sum_{\ell>z} p_\ell(c)\ell^{-\alpha}$ exceeds 16 times its mean J. Let π_c be the product of the uniform laws off the fibre classes, as in Theorem 14.3; then $\log \mathsf{R}_{p'}(\pi_c) \le 64J$. By the chain rule of [KA2, §2] and partial summation, $J \le \int_z^\infty y^{-\alpha}\, d\mathfrak{M}(y) \ll \alpha^{-3}(\log(1/\alpha))^3$. Theorem 14.12 gives $\log(N/B) \le \beta\log N + 16\mathfrak{M}(z) + 64J + O(1)$, and each term is $\ll (\log N)^{3/4}(\log\log N)^3$. □

Compared with Theorem 14.3, Theorem 14.13 drops every condition on the small part of the modulus (any number of small primes, any size) and needs no enlarged threshold ℓ_0': the small coordinates are paid by density, not by a product structure. This extends the *scope* of Theorem 14.3, not its strength: on the families

of Corollary 6.1 Theorem 14.13 is weaker by the factor $(\log\log N)^3$, which is the Rankin loss in $\mathfrak{M}$ [KA2, Rem. 3.8]. The local moment of Lemma 10.9 should remove it; this has not been checked. The same splitting with a level split in place of Hölder is Proposition 14.9(2) with the smooth coordinates removed from both conditions [LS3, Rem. 1.1(c)]. For an *arbitrary* mixture, Theorem 14.12 also caps the frequencies with z-smooth denominators ($\theta_r = 0$), provided $\mathcal{A}_c \neq \emptyset$ for all $c \in \operatorname{supp}\pi_s$; this is not automatic when classes with several primes $> z$ cover whole fibres. Mixed frequencies $\theta_s + \theta_r$ with $\theta_r \neq 0$ in a general mixture are not covered.

So the only remaining source of (E1) over forced families is the set of classes with *two or more* prime factors $> z$, through correlations of the fibre measures between such primes.

Hypothesis 14.14 ((H_{rough}); [LS3, §4]; [CONJECTURE])**.** For every finite family $\mathfrak{G}$ of the four types and every $c \in \mathcal{A}_s$ outside an event of probability $\le 1/8$ under the law of Lemma 2.1 of [LS3], there is a probability π_c on $\mathcal{A}_c$ with $\log \mathsf{R}_{2+2\beta}(\pi_c) \le C(\log N)^{3/4+o(1)}$, $\beta = (\log N)^{-1/4}$.

By Theorem 14.12 together with the density bound for π_s above (which uses the inputs of [KA2]), (H_{rough}) implies a cap $C(\log N)^{3/4+o(1)}$ for every CRT-admissible large sieve over every mixture of the four types, whatever the frequencies; this is a *sufficient* condition only. A cap, or Conjecture 14.10, need not yield fibre measures of this kind (Hölder-type and sup-type criteria are incomparable [LS2, §5]). Theorem 14.13 proves (H_{rough}), up to $(\log\log N)^3$, when every class has one rough prime (product measures); Theorem 14.18 below proves it for residue-sparse mixtures. A level-split form with a cheap comparison at rough level $\le 2\lambda'$ and $|\hat{\pi}_c(\theta_r)|^2 \le \mathrm{e}^{S_r}/N$ above rough level $\lambda' \asymp \log N$ would also suffice, but then for *every* $c \in \operatorname{supp}\pi_s$. In the two-copy form, $\sum_{\operatorname{supp}(\operatorname{den}\theta)=S} |\hat{\sigma}(\theta)|^2 = \mathbb{E}_{x,y\sim\sigma} \prod_{\ell\in S}(\ell^{E_\ell}\mathbf{1}[x \equiv y \ (\ell^{E_\ell})] - 1)$ [LS3, Lemma 4.1] [PROVED]; so (H_{rough}) follows from product decay of these signed two-copy correlations and of the sup over each support class, which is a multi-scale correlation-decay statement.

For a fibre family, a rough prime p and a residue b, the *conditional mass* at b is the deterministic sum $m^*(p,b) = \sum_{C\ni p,\ b_C\equiv b\ (p^v)} p^v/G_r(C)$ over the classes C whose modulus is divisible by p, with $v = v_p(G_r(C))$.

Lemma 14.15 (residue concentration; [LS3, Lemma 4.2]; [PROVED])**.** *The class $-4 \bmod M$ belongs to $\mathscr{R}(M)$ for every $M \equiv 3 \pmod 4$. Hence for the family $\{-4 \bmod pM' :\ M' \le X,\ pM' \equiv 3\ (4),\ M'$ z-rough$\}$ the conditional mass at the residue $-4 \bmod p$ is $m^*(p,-4) = \sum_{M'} 1/M' \asymp \log X/\log z$, which is $\asymp A(\log N)^{3/4}$ for $X = N^A$.*

Proof. Take $D = 1$ in the definition of $\mathscr{R}(M)$; the sum is Mertens over z-rough M' in one class modulo 4. □

So a fibre family can be dense at one residue although sparse on average. This defeats the plain Kotecký–Preiss criterion for a cluster expansion of the uniform measure on $\mathcal{A}_c$ with uniform activities; it does not show that a symmetric route fails (*Assessment*). In the top-prime order the class $-4 \bmod pM'$ is charged at its top prime with activation probability $\asymp 1/(\text{cofactor})$, and the average influence of a coordinate is $\ll (\log N)^{O(1)}/p$ while its worst-case influence is not small; one-step operator bounds in $\ell^{p'}$, $p' > 2$, all lose [LS3, §4.3] (*Assessment*). In fact this family

is covered at every frequency level by Theorem 14.18 below: residue concentration turns out to be the easy case.

14.6. **Damped collisions: residue-sparse mixtures at every level.** (H_{rough}) is reduced further in [LS4] (review [LS4r]: no fatal or major defect, all proofs sound; minor wording and bookkeeping points applied). Let σ be a probability on $\mathbb{Z}/M_r$, $M_r = \prod_{\ell\in\mathcal{P}} \ell^{E_\ell}$, let $\operatorname{supp}\theta$ be the set of primes dividing $\operatorname{den}\theta$, let σ_T be the marginal of σ modulo $M_T = \prod_{\ell\in T} \ell^{E_\ell}$, and let $\mathsf{R}_2(\sigma_T) = M_T\,\sigma\otimes\sigma(x\equiv y\ (M_T))$ be its collision number.

Lemma 14.16 (damped-collision reduction; [LS4, Lemma 1.1]; [PROVED])**.** *If $w_\ell\in[0,1]$ and $|\hat\sigma(\theta)|^{2\beta}\le\prod_{\ell\in\operatorname{supp}\theta} w_\ell$ for every $\theta\neq 0$, then $\mathsf{R}_{2+2\beta}(\sigma)\le\mathbb{E}_T\mathsf{R}_2(\sigma_T)$, where $T\subseteq\mathcal{P}$ contains each ℓ independently with probability w_ℓ.*

Proof. Group the frequencies by $S=\operatorname{supp}\theta$. The two-copy identity of Section 14.5 gives $\sum_{\operatorname{supp}\theta=S}|\hat\sigma|^2 = \mathbb{E}\prod_{\ell\in S}h_\ell \ge 0$, with $h_\ell = \ell^{E_\ell}\mathbf{1}[x\equiv y\ (\ell^{E_\ell})]-1$, so $\mathsf{R}_{2+2\beta}(\sigma)\le\mathbb{E}\prod_\ell(1+w_\ell h_\ell)$. Expand $1+w_\ell h_\ell = (1-w_\ell)+w_\ell\ell^{E_\ell}\mathbf{1}[x_\ell=y_\ell]$, both terms nonnegative. □

The signed correlations of (H_{rough}) are thereby split into a *sup* condition with only weak per-prime decay and a *positive* collision number on a random sparse set of coordinates. For the capped fibre law Q' of [KA2] conditioned on a good event $G_B=\{x\in\mathcal{A}\}\cap\{\sum_\ell w_\ell\tilde p_\ell(x)\le B\}$ ($\tilde p_\ell$ the light activated densities), the collision side is free: $\mathbb{E}_T\mathsf{R}_2((\sigma_B)_T)\le \mathrm{e}^{2B}/Q'(G_B)^2$, by a tilted pair-process identity, and only *first* moments enter [LS4, Lemma 2.1, Prop. 4.1] [PROVED].

Theorem 14.17 ((H_{rough}) from sup decay; [LS4, Thm 4.2]; [PROVED] implication, inputs as Theorem 14.13)**.** *Let $z=\exp\{(\log N)^{1/4}\}$, $\beta=(\log N)^{-1/4}$, $\mathfrak{G}$ any finite mixture of the four types, Q' the sequential law of [KA2] on all coordinates, and for a z-smooth part c let σ_c be its conditional law on the z-rough coordinates, conditioned on G_{B_c} with B_c eight times the conditional mean of $\sum_{\ell>z}w_\ell\tilde p_\ell$. Fix $\gamma\in(0,1]$, $1\le K\le z^{\gamma/2}$ and $w_\ell=(K\ell^{-\gamma})^{2\beta}$. Assume*

- (A^*) *for every c outside an event of Q'-probability $\le 1/32$ and every $\theta\neq0$ with z-rough denominator, $|\hat\sigma_c(\theta)|\le\prod_{\ell\in\operatorname{supp}\theta}K\ell^{-\gamma}$.*

Then for $N\ge N_0(\gamma)=\exp\{C\gamma^{-4}\}$ every CRT-admissible N-large-sieve bound for $\mathcal{A}(\mathfrak{G})$, with any rational frequencies, any denominators and any weights, saves

$$\log(N/B)\le(\log N)^{3/4}+C\gamma^{-3}(\log N)^{3/4}(\log\log N)^3.$$

Proof sketch. Choose the exceptional smooth event of the density lemma to contain the event of (A^*), a large rough leak, and a large conditional mean (Markov). Off it $Q'_c(G_{B_c})\ge1/2$, so Lemma 14.16 and the collision bound give $\log\mathsf{R}_{2+2\beta}(\sigma_c)\le 512J+\log4$, $J=\mathbb{E}_{Q'}\sum_{\ell>z}w_\ell\tilde p_\ell\ll\gamma^{-3}(\log N)^{3/4}(\log\log N)^3$ by partial summation as in Theorem 14.13. Conclude with Theorem 14.12, $\pi_c=\sigma_c$. □

So the only open input is the qualitative-looking sup decay (A^*), at any fixed rate γ and with per-prime losses up to $z^{\gamma/2}$, and no comparison-at-level hypothesis is needed (*Assessment*: in the per-prime sense this is far weaker than Conjecture 14.10). If a sieve uses only frequencies whose rough supports lie in a class $\mathcal{S}$, (A^*) is needed only there.

(A^*) is proved when the residues are fixed in advance. For a prime ℓ let $\mathrm{Res}_\ell(\mathfrak{G})\subseteq\mathbb{Z}/\ell^{E_\ell}$ be the union of the ℓ-parts b_C mod $\ell^{v_\ell(G_C)}$ of the classes $C\in\mathfrak{G}$ with $\ell\mid G_C$.

A pivotal-coin coupling bounds $|\hat\sigma(\theta)| \le Q'(G)^{-1}\prod_{\ell\in\operatorname{supp}\theta} 4U(R_\ell)/(1-\delta_\ell)$ for any fixed sets $R_\ell \supseteq$ the ℓ-parts of the classes through ℓ [LS4, Lemma 5.1] [PROVED]: a coordinate whose two values both avoid R_ℓ cannot change the path.

Theorem 14.18 (residue-sparse mixtures; [LS4, Thm 5.2, Cor. 5.3]; [PROVED], inputs as Theorem 14.13)**.** *Let $z = \exp\{(\log N)^{1/4}\}$, $\gamma \in (0,1]$, and let $\mathfrak{G} = \mathfrak{G}_1 \cup \mathfrak{G}_2$ be any finite mixture of the four types (any moduli, no B), with $\mathfrak{G}_1$ the classes having at most one prime factor $> z$. If*

$$(RS_\gamma) \qquad U(\operatorname{Res}_\ell(\mathfrak{G}_2)) \le \ell^{-\gamma} \quad \text{for every prime } \ell > z,$$

then for $N \ge N_0(\gamma)$ every CRT-admissible N-large-sieve bound for $\mathcal{A}(\mathfrak{G})$, with any rational frequencies, any denominators and any weights, saves $\log(N/B) \le (\log N)^{3/4} + C\gamma^{-3}(\log N)^{3/4}(\log\log N)^3$. In particular this holds with $\gamma = 1/4$ whenever every class with two or more prime factors $> z$ is H-small with $H \le z^{1/4}/2$, i.e. $b \equiv -r/s \pmod G$ with $0 \le r \le H$, $1 \le s \le H$, $\gcd(s,G) = 1$ (for instance the $\mathscr{R}(M)$-classes -4, -1, $-1/4$, and $-4d$, $-1/(4d)$ with $4d \le H$, over all moduli M; (a,D)-classes with $4D + a \le H$; Case-A classes $-m^{-1}$, $m \le H$; selector classes).

Proof sketch. In the fibre at c take $R_\ell(c) = \operatorname{Res}_\ell(\mathfrak{G}_2) \cup F_\ell^r(c)$, the second set being the ℓ-residues of the one-rough-prime classes activated by c; off an event of probability $\le 1/64$, $U(F_\ell^r(c)) \le \ell^{-1/4}$ (as in Theorem 14.13). The pivotal bound gives (A^*) with $K = 32$ and $\gamma' = \min(\gamma, 1/4)$; apply Theorem 14.17. For H-small classes the projection of $\operatorname{Res}_\ell(\mathfrak{G}_2)$ modulo ℓ has at most $(H+1)H \le z^{1/2}$ elements, so $U(\operatorname{Res}_\ell(\mathfrak{G}_2)) \le \ell^{-1/2}$. □

(So H-small families even satisfy $(RS_{1/2})$; the statement uses $\gamma = 1/4$ because the proof works with $\gamma' = \min(\gamma, 1/4)$ anyway.) Theorem 14.18 strictly extends Theorem 14.13 (the case $\mathfrak{G}_2 = \emptyset$) and, for the families it covers, settles the nonempty-fibre proviso of Section 14.5, since $Q'_c(G_{B_c}) \ge 1/2$ forces $\mathcal{A}_c \ne \emptyset$ [LS4r]. It contains the family of Lemma 14.15: residue concentration is the *easy*, deterministic-residue case, not an obstruction. What remains open for the all-level cap over all forced families is (A^*) for *residue-dense* mixtures: those whose classes with two or more primes $> z$ have, jointly, ℓ-residues filling a fraction $> \ell^{-\gamma}$ of $\mathbb{Z}/\ell$ at some $\ell > z$, for every fixed $\gamma > 0$. Residue-sparseness is a property of the union $\operatorname{Res}_\ell(\mathfrak{G}_2)$, not of single classes: many individually H-small classes with different (r,s) can make the union dense (as for generic $\mathscr{R}(M)$ with many moduli through ℓ: the Jacobi obstruction acts modulo the whole modulus, not prime by prime). There the fixed residue set must be replaced by the random set of residues of classes whose other coordinates are matched; the resulting covering-count statement (CC) [LS4, §3.2] is a [CONJECTURE], with no counterexample known.

14.7. **Prime-only majorants for mixtures.** A *prime majorant* of $\mathcal{A}$ is $\nu = \sum_i a_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$ with $\nu(p) \ge 0$ at all primes and $\nu(p) \ge 1$ at all primes $p \in \mathcal{A}$, outside a finite set of primes. Let L be the lcm of the d_i and the moduli of $\mathfrak{G}$, and $\mathbb{E}^*$ the uniform mean on $(\mathbb{Z}/L)^\times$. By Dirichlet's theorem, ν is a prime majorant iff $\nu \ge 0$ on every reduced class mod L and $\nu \ge 1$ on every reduced class contained in $\mathcal{A}$. Under $\mathbb{E}^*$ the prime-power coordinates are independent and uniform on units, and selector classes have mass 0 [PL, Lemmas 1.2, 1.3].

Theorem 14.19 (prime-majorant cap; [PL, Thm 3.1]; [PROVED]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4)**.** *There are absolute W, λ_0, C such that every prime*

majorant ν of level $\lambda \geq \lambda_0$ of $\mathcal{A}(\mathfrak{G})$, $\mathfrak{G}$ any finite mixture of the four types with arbitrary moduli, has $\log(1/\mathbb{E}^*\nu) \leq S^*(\lambda) := C\lambda^{3/4}(\log\lambda)^{3/4}$, *and* $\leq C(B)\lambda^{3/4}$ *if every modulus satisfies* $G \leq P(G)^{1+B}$. *By* [KA3, §4.3], $S^*(\lambda) = C\lambda^{3/4}$ *with Case A as in Theorem 10.12; this is pointer-level and not separately reviewed.*

Proof sketch. Run the proof of [KA2, Thm 5.1] on the unit groups $(\mathbb{Z}/\ell^e)^\times$. The base is the set of unit squares, which avoids every W-smooth class by Lemma 10.2 (as in Lemma 10.3); under the unit measure its R-term is $(\pi(W)+1)\log 2$. The sequential steps are arithmetic-free, non-unit classes have mass 0, and the factor $\ell/(\ell-1)$ on unit classes is absorbed into the inflation weight $\gamma^*(\ell) = (\ell/(\ell-1))(1-\ell^{-1/2})^{-1}$ of the moments and the leak [PL, Lemmas 2.3–2.5]. □

Corollary 14.20 (prime-law methods; [PL, Cor. 4.2]; [PROVED]; Case A as in Theorem 14.19)**.** *Let a method bound* $\#\{p \leq N : p \in \mathcal{A}(\mathfrak{G})\}$ *by* $\pi(N)\mathbb{E}^*\nu + \mathrm{Err}$, *with ν a prime majorant and* $\mathrm{Err} \geq 0$ *(including the correction* $\sum_{p|L,\, p\leq N}(1+|\nu(p)|)$*). Assume either that for every modulus of ν the product of its distinct primes $> W$ is $\leq N^A$, or that every prime of $\mathfrak{G}$ is $\leq N^A$ and $\sum_i |a_i| \leq N^A$. Then the saving against $\pi(N)$ is $\leq C_A(\log N)^{3/4}(\log\log N)^{3/4}$; it is $\leq C_{A,B}(\log N)^{3/4}$ under bounded B, and $\leq C_A(\log N)^{3/4}$ in general at pointer level.*

This is the form of every method that evaluates $\sum_{p\leq N}\nu(p)$ by Dirichlet main terms with errors bounded in absolute value: Siegel–Walfisz, Bombieri–Vinogradov, Bombieri–Davenport–Halberstam, Elliott–Halberstam or GRH, at any polynomial level. Integer majorants that are ≥ 1 only on $\mathcal{A} \cap \{(n, P(z)) = 1\}$, $z \leq N^A$, are majorants of $\mathcal{A}(\mathfrak{G} \cup \{0 \bmod p : p \leq z\})$; so Theorem 10.14 covers the setting of Theorem 13.4(2) for all mixtures [PL, §4.4]. Under GRH signed evaluation does not help either [PL, Prop. 4.3] [CONDITIONAL] on GRH: if the moduli of ν are $\leq N^A$, $\sum_i |a_i| \leq N^{1/2-\varepsilon}$ and $\nu(p) \geq 0$ off a set F of $\leq N^{1/2}$ primes, then $\sum_{p\leq N,\, p\notin F}\nu(p) \geq (1-o(1))\mathrm{li}(N)\mathbb{E}^*\nu$. Unconditional error terms cannot resolve main terms of size $\pi(N)\mathrm{e}^{-(\log N)^{3/4}}$ at all [LS2, §6.3] (*Assessment*).

Proposition 14.21 (finite range; [LS2, Prop. 6.1]; [PROVED])**.** *For a prime $q \in (N, 2N]$, $\nu = \sum_{p\leq N,\, p\in\mathcal{A}} \mathbf{1}[n \equiv p \pmod q]$ is ≥ 0 on $\mathbb{Z}$, ≥ 1 at the primes of $\mathcal{A} \cap [1, N]$, of level $\leq \log 2N$ and coefficient sum $\leq N$, and $\sum_{p\leq N}\nu(p) = \#\{p \leq N : p \in \mathcal{A}\}$.*

So no cap of any kind holds if $\nu \geq 1$ is required only there; what is missing is the location of those primes, which is non-CRT input. The other relaxation, $\nu(p) \geq 0$ only for $p \leq N$ but $\nu \geq 1$ at all primes of $\mathcal{A}$, is [OPEN]; that certifying it needs the location of the primes $\leq N$ is an *Assessment*. Both relaxations reduce to the prime-majorant condition when every reduced class mod L contains a prime $\leq N$, so the gap lives at $L > N^c$ [PL, §6].

14.8. **Interval counts with bounded frequency denominators.** By (10), a direct count of $\sum_{n\leq N}\nu(n)$ may exploit cancellation between frequencies. If all denominators are below N, it cannot.

Theorem 14.22 (Selberg minorant; [IF, Thm 2.2]; [PROVED])**.** *Let $\nu \geq 0$ on $\mathbb{Z}$ be periodic, with every nonzero frequency of denominator $\leq D$. Then $\sum_{n=1}^{N}\nu(n) \geq (N-D)\mathbb{E}\nu$ for every $N \geq 1$.*

Proof. Let $I = [1/2, N+1/2]$ and $\delta = 1/D$. Selberg's minorant [Va] is an integrable $F : \mathbb{R} \to \mathbb{R}$ with $F \le \mathbf{1}_I$, $\hat{F}$ continuous and supported in $[-\delta, \delta]$, and $\hat{F}(0) = |I| - \delta^{-1} = N - D$. F is entire of exponential type $2\pi\delta$ and integrable, so $\sum_n |F(n)| < \infty$, and Poisson summation for $F(x)\mathrm{e}(x\theta)$ gives $\sum_n F(n)\mathrm{e}(n\theta) = \sum_k \hat{F}(k-\theta)$ for every θ. If $\theta \notin \mathbb{Z}$ has denominator $q \le D$, then $\|\theta\| \ge 1/q \ge \delta$, so every term vanishes ($\hat{F}$ is continuous and 0 outside $(-\delta, \delta)$). For $\theta \in \mathbb{Z}$ the sum is $\hat{F}(0)$. Since ν is a finite trigonometric sum, $\sum_n F(n)\nu(n) = \sum_\theta \hat{\nu}(\theta) \sum_n F(n)\mathrm{e}(n\theta) = (N-D)\mathbb{E}\nu$. Finally $F(n) \le \mathbf{1}_{[1,N]}(n)$ and $\nu(n) \ge 0$ for every $n \in \mathbb{Z}$, so $\sum_{n \le N} \nu(n) \ge \sum_n F(n)\nu(n)$. □

Positivity is used at every integer (Remark 3.6). The hypothesis holds iff ν is a combination of classes of moduli $\le D$.

Corollary 14.23 ([IF, Cor. 2.3], [KA3, §4.3]; [PROVED] via Theorem 10.12; Case A as there)**.** *Let ν be a majorant of $\mathcal{A}(\mathfrak{G})$ that is a combination of classes of moduli $\le N/2$, with any coefficients. Then $\sum_{n \le N} \nu(n) \ge (N/2)\exp\{-C(\log N)^{3/4}\}$ for N large, however the interval sum is evaluated. No bound on the family primes is needed.*

Proof. ν has level $\le \log N$; apply Theorem 10.12, then Theorem 14.22 with $D = N/2$. □

This caps Λ^2 and β-sieves of level $< N/2$ with any remainder evaluation (dispersion, Kloosterman sums, Erdős–Turán/Vaaler or floor/ceiling rounding, the exact $|S_N(\theta)|$, smooth windows $\Phi(n/N) \ge \mathbf{1}_{[1,N]}(n)$), with no need for (H_{eq}). Inter-frequency gain must come from classes of combined modulus $> N/2$.

Theorem 14.24 (coefficient budget; [IF, Lemma 2.4, Thm 2.5, Rem. 2.6]; [PROVED] via Theorem 10.12; Case A as there)**.** *Let every prime of $\mathfrak{G}$ be $\le N^A$, let $\nu = \sum_i a_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$ be a majorant of $\mathcal{A}(\mathfrak{G})$, and $T_> = \sum_{d_i > N/2} |a_i|$. If $B = N\mathrm{e}^{-s}$ satisfies $B \ge \sum_{n \le N} \nu(n)$ and $T_> \le cB$, then $s \le \log(2+12c) + C_A(\log N)^{3/4}$ for $N \ge N_0(A)$, with N_0, C_A independent of c. If $T_> \le (N/24)\mathrm{e}^{-C_A(\log N)^{3/4}}$, the saving is $\le C_A(\log N)^{3/4} + \log 4$ whatever B is.*

Proof sketch. For F as above with $D = N/2$, Poisson along a progression gives $|\sum_{n \equiv b\,(d)} F(n) - (N-D)/d| \le 2(N+D)/D = 6$, and 0 if $d \le D$. So $B \ge \sum_n F(n)\nu(n) \ge (N/2)\mathbb{E}\nu - 6T_>$, i.e. $\mathbb{E}\nu \le (2+12c)\mathrm{e}^{-s}$. If $s \le \log(2+12c)$ there is nothing to prove; otherwise $B < N/(2+12c)$, so $T_> \le cB < N/12$. For the mean, follow the proof of Theorem 10.14: project to the family period, then the coarsening of Lemma 3.4 touches only terms of level $> \log(N/2)$, whose coefficient sum is $\le T_> < N/12$. □

$T_>$ may be replaced by its infimum over representations of ν; $c = 1$ contains the class of Theorem 10.14. Not covered: *hybrid* methods that evaluate the classes of modulus $\le N/2$ in any way and charge each larger class its trivial count. Their bound $\sum_{n \le N} \nu + \sum_{d_i > N/2}(|a_i| - a_i r_i)$, with $r_i = \#\{n \le N : n \equiv b_i\ (d_i)\} - N/d_i$, can be far below $T_>$. This is a proof gap [OPEN], not a counterexample [IF, Rem. 2.6]. What is left is to count integers $n \le N$ with prescribed witness sets of combined modulus $> N/2$ better than termwise (*Assessment*).

14.9. **Hybrid methods.** The hybrid gap just stated is narrowed, but not closed, in [IF2] (review [IF2r]: all claims sound; one defect, a Farkas argument on truncated LPs, was repaired by a periodic LP). Fix N, a majorant $\nu = \sum_i a_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$ of $\mathcal{A}(\mathfrak{G})$, and split the terms into *small* ($d_i \le N/2$) and *large* ($d_i > N/2$). Put $c(b,d) = \#\{n \le N : n \equiv b\ (d)\} \in \{\lfloor N/d \rfloor, \lceil N/d \rceil\}$. A *hybrid method* [IF2, Def. 1.2] bounds the small part by any valid upper bound for its interval sum and charges a large term the least position-blind amount $\beta^*(a,d) = a\lceil N/d \rceil$ ($a > 0$), $a\lfloor N/d \rfloor$ ($a < 0$); call the resulting bound $B_{\mathrm{hyb}}(\nu)$. The charge $aN/d + |a|$ of Theorem 14.24 dominates β^*. A large class is *full* if $c(b,d) = \lceil N/d \rceil$ and *sparse* if $c(b,d) = \lfloor N/d \rfloor$.

Lemma 14.25 (sign rule; [IF2, Lemma 3.1, Ex. 3.2]; [PROVED])**.** *For every representation, $B_{\mathrm{hyb}}(\nu) \ge \sum_{n \le N} \nu(n) + \sum_{i\ wrong} |a_i|$, with equality when the small part is counted exactly; a large term is* wrong-signed *if $a_i > 0$ on a sparse, not full class, or $a_i < 0$ on a full, not sparse class. Right-signed large terms are counted exactly. At $N = 20$ the class $\mathbf{1}[0 \bmod 21]$ has a representation (twelve full classes modulo 21 and classes modulo 3 and 7) with hybrid charge 0. Hence no inequality $B_{\mathrm{hyb}}(\nu) \ge cN\mathbb{E}\nu$, $c > 0$, holds for all $\nu \ge 0$.*

Proof. $\beta^*(a,d) - a\,c(b,d)$ equals $a(\lceil N/d \rceil - c)$ for $a > 0$ and $|a|(c - \lfloor N/d \rfloor)$ for $a < 0$, and $c \in \{\lfloor N/d \rfloor, \lceil N/d \rceil\}$ with the two differing by at most 1. The example is checked by direct computation [IF2, §3]. □

So the Selberg accounting of Theorem 14.24 does not extend to hybrids as it stands. It does when the *free* large mass (right-signed terms) is small: if the positive terms on classes of modulus $> N$ meeting $[1,N]$, plus the right-signed mass at moduli in $(N/2, N]$, are $\le cB$, and all family primes are $\le N^A$, the saving is at most $S_A + \log(2 + 12(1+c))$, where S_A is the cap of Theorem 10.14 ([IF2, Cor. 5.1] with Theorem 14.24) [PROVED]. Large free mass is not in itself a counterexample: $(1+K) - K\sum_{b \bmod N} \mathbf{1}[b \bmod N] \equiv 1$ has $B = N$ and right-signed mass KN.

Hypothesis 14.26 (Flat(C, t, s_0, Δ) at N; [IF2, §5]; [OPEN])**.** There is a summable $F : \mathbb{Z} \to \mathbb{R}$ with $F \le \mathbf{1}_{[1,N]}$; $F(s) = tN/d$ for every class s of modulus $d \le N/2$ (where $F(s) = \sum_{n \in s} F(n)$); $F(s) \ge tN/d + s_0$ for every class of modulus $d > CN$ meeting $[1,N]$, and $F(s) \ge tN/d - 1$ for those missing it; and $|F(s) - tN/d| \le \Delta$ for every class of modulus in $(N/2, CN]$.

Unlike Selberg's minorant, which is ≈ 0 near the ends of $[1,N]$, such an F is *flat*: for large moduli it is $\ge s_0$ on $[1,N]$ and ≥ -1 off it.

Theorem 14.27 (conditional hybrid cap; [IF2, Thm 5.2]; [PROVED] implication, [CONDITIONAL] on Flat; Case A as in Theorem 10.14)**.** *Assume Flat(C, t, s_0, Δ) at N with $0 < t \le 1$ and $s_0 \ge N^{-A_1}$. If a hybrid bound B for $\mathcal{A}(\mathfrak{G})$, family primes $\le N^A$, has right-signed mass at moduli in $(N/2, CN]$ at most cB, then for $N \ge N_0(A, A_1, C)$*

$$\log(N/B) \le S' + \log \frac{1 + \Delta(1+c)}{t}, \qquad S' = C_{A,A_1,C} (\log N)^{3/4} (\log\log N)^{3/4}.$$

Terms of modulus $> CN$, wrong-signed terms and free negative terms are unrestricted.

Proof sketch. By $\nu \ge 0$, $F \le \mathbf{1}_{[1,N]}$ and Lemma 14.25, $B \ge \sum_n F(n)\nu(n) + T_{\mathrm{wr}}$, and the small part gives $tN\mathbb{E}\nu_{\mathrm{small}}$. Term by term, the flatness conditions make each large term of modulus $> CN$ contribute ≥ 0 beyond $tN/d_i \cdot a_i$, and each term of

modulus in $(N/2, CN]$ at least $-\Delta|a_i|$ (right-signed) or $(1-\Delta)|a_i|$ (wrong). Hence $(1+\Delta)B \geq tN\mathbb{E}\nu - \Delta T_{\mathrm{mid}} + s_0 W^+_{>CN}$, with $W^+_{>CN}$ the mass of positive terms of modulus $> CN$ meeting $[1, N]$. Unless $(1+\Delta(1+c))B \geq tN$ (then the bound is trivial), this gives $\mathbb{E}\nu < 1$ and total mass $\leq N^{A_1+1} + N$ on terms of modulus $> CN$, which bounds the level after the coarsening of Lemma 3.4; the mean is then capped as in Theorem 10.14 [IF2, §5]. In this non-trivial case $W^+_{>CN} < N/s_0$, so the level does not involve t or Δ, which enter only the trivial case and the final line; hence no lower bound on t or $1/\Delta$ is needed [SPW2, Rem. after Lemma 1.1]. □

[IF2, Thm 5.2] is stated with $t, 1/\Delta \geq \mathrm{e}^{-S_A}$; these only make the conclusion 3/4-shaped. [IF2] uses the cap of [KA2, Thm 5.1] with the case analysis of [KA2, Cor. 6.1], hence the factor $(\log\log N)^{3/4}$. The cap enters only as a black-box bound on $\log(1/\mathbb{E}\nu)$ at a level $\leq (A + A_1 + 2)\log N + S + \lambda_0$, so Theorem 10.12 and the case analysis of Theorem 10.14 give $S' = C_{A,A_1,C}(\log N)^{3/4}$ (observation of this note, pointer-level, not reviewed). A variant [IF2, Thm 5.2′] needs right-signed mass only on $(N, CN]$, under a stronger Flat with sign conditions on $(N/2, N]$ [PROVED] implication.

Flat reduces to a purely combinatorial statement with no arithmetic and no $\mathcal{A}$.

Definition 14.28 (SPW(C, σ, Δ_0) at N, spread pseudo-window; [IF2, §9])**.** A summable $R \geq 0$ on $\mathbb{Z}$ with (P1) $R(s) = c(s)$ for every class s of modulus $\leq N/2$; (P2) $R(s) \leq 1 - \sigma$ for every class of modulus $> CN$; and (P3) $|R(s) - c(s)| \leq \Delta_0$ for every class of modulus in $(N/2, CN]$.

That is, a random integer that looks exactly like a uniform point of $[1, N]$ modulo every $d \leq N/2$, but which no class of modulus $> CN$ sees as a point of $[1, N]$. The window $\mathbf{1}_{[1,N]}$ itself fails only (P2). Version 4 of this note stated SPW with σ fixed as an open hypothesis; that form is false (Theorem 14.30), and the correct target is the weak form of Lemma 14.31.

Proposition 14.29 ([IF2, Prop. 9.1, Lemma 9.3]; [PROVED])**.** *(1) SPW(C, σ, Δ_0) implies Flat(C, t, s_0, Δ) with $s_0 = \sigma/2$, $t = \theta/2$, $\Delta = 6\theta + \Delta_0$, $\theta = \sigma/(2(\sigma + \tau + 1/(2C)))$, τ an absolute constant, via $F = \theta F_S + (1-\theta)(\mathbf{1}_{[1,N]} - R)$ with F_S Selberg's minorant of Theorem 14.22 at $\delta = 2/N$. (2) Every SPW has $\sigma \leq \sigma_C(N) = \min_{q \leq N/2}(1 - \lceil N/q \rceil / k_q)$, $k_q = \lfloor CN/q \rfloor + 1$; this is 2/5 at $C = 2$ in the limit, and SPW is impossible for $C \leq 1$.*

Proof. (2) Take b with $c(b, q) = \lceil N/q \rceil$. The class b mod q is the disjoint union of the k_q classes $b + jq$ mod $k_q q$, of modulus $k_q q > CN$; each has R-mass $\leq 1 - \sigma$, and they sum to $c(b, q)$. (1) is a direct check of the Flat conditions, using the decay of F_S off $[1, N]$ to bound τ [IF2, §9]. □

The bound $\tau = O(1)$ rests on the inequality $0 \leq B(z) - \mathrm{sgn}(z) \leq 2K(z)$ for Beurling's function [Va], quoted in [IF2] from its review with the numbering not re-checked against the source.

Fixed σ is impossible. The next two results are from [SPW] (review [SPWr]: no fatal or major defect; both rational certificates below reproduced independently).

Theorem 14.30 (fixed-σ SPW fails; [SPW, Thm 3.2]; [PROVED])**.** *Let $C > 1$, let $M \geq 2$ with $L_M = \mathrm{lcm}(1, \ldots, M) \leq N$, let e be the least multiple of L_M exceeding CN, and $m_0 = e/\lfloor N/2 \rfloor$ (so $m_0 \leq 2C + 3$ for $N \geq N_0(C)$). If R satisfies (P1) and*

(P2) of SPW$(C,\sigma,\cdot)$ at N, then for every integer $1 \le r < \min(N-1,(C-1)N-1)/2$

$$\sigma \le \frac{e}{(M+1)r} + \frac{2\pi m_0(2r+1)}{e-\pi m_0},$$

and hence $\sigma \le 4\sqrt{\pi m_0/(M+1)}\,(1+o(1))$ as $M\to\infty$. With $M \asymp \log N$: $\sigma \ll_C (\log N)^{-1/2}$. So for every fixed C and $\sigma>0$, SPW(C,σ,Δ_0) fails for all large N, whatever Δ_0.

Proof sketch. Project R to ρ on $\mathbb{Z}/e$; points of $\mathbb{Z}/e$ are classes of modulus $e>CN$, so $0\le\rho\le 1-\sigma$, and (P1) pins $\hat\rho(k)=\hat{\mathbf{1}}_W(k)$, $W=\{1,\dots,N\}$, for $m_0\le|k|\le M$ (such k divide e). Convolve $\rho-\mathbf{1}_W$ with the Fejér kernel of degree M on $\mathbb{Z}/e$: the result T is a trigonometric polynomial of degree $<m_0$ with $|T|\le 1$ on $\mathbb{Z}/e$, so $\|T\|_\infty \le (1-\pi m_0/e)^{-1}$ by sampling. At $x_{\rm in}=1+r$ and $x_{\rm out}=-r$ the kernel tails give $T(x_{\rm out})-T(x_{\rm in})\ge\sigma-e/((M+1)r)$, while Bernstein's inequality bounds the difference by $2\pi m_0\|T\|_\infty(2r+1)/e$. Optimise r. □

The mechanism is the jump of the window: once all frequencies $m_0\le|k|\le M$ are pinned, a smoothing at scale e/M is a polynomial of bounded degree, which cannot jump by σ across the edge of W. The bound is numerically weak (it beats σ only for $\log N \gtrsim 200/\sigma^2$ at $C=2$), but exact single-modulus certificates already give $\sigma\le 72/185<2/5$ at $N=300$ and $\sigma\le 0.381133$ at $N=1150$ [SPW, §3] [PROVED]. The LP pattern "$\sigma^*=\sigma_C(N)=2/5$" seen at $N\le 60$ [IF2, §9] is a small-N artefact. The same argument bounds the margin of Flat: Flat(C,t,s_0,Δ) with $t\ge 0$ forces $s_0\le 6\sqrt{\pi m_0/(M+1)}(1+o(1))$, for any Δ [SPW, Cor. 3.3] [PROVED]. Theorem 14.27 needs only $s_0\ge N^{-A_1}$, so this does not touch it. A pointwise bounded-density strengthening of SPW looks perfect in LPs up to $N=24$ but provably needs density $\gg\sqrt{N}$ [SPW, §2] [PROVED].

What the hybrid route needs. The following lemma, from [SPW2] (review [SPW2r]: no fatal defect; one major label defect, an overclaimed necessity, repaired as stated here), makes the requirement exact for this route.

Lemma 14.31 (weak SPW suffices; [SPW, Lemma 1.4], [SPW2, Lemma 1.1]; [PROVED] implication; Case A as in Theorem 10.14)**.** *Suppose that at every large N there is $R\ge 0$ on $\mathbb{Z}$ with (P1), with $R(s)\le 1-\eta_N$ on every class of modulus $>CN$ meeting $[1,N]$, $R(s)\le K_N$ on every class of modulus $>CN$ missing $[1,N]$, and $|R(s)-c(s)|\le\Delta'_N$ on every class of modulus in $(N/2,CN]$, where C is fixed and $\eta_N\le 1/2\le K_N$. Put $\ell_N=\log(K_N/\eta_N)+\log(1+\Delta'_N/K_N)$. If $\eta_N/K_N\ge 4N^{-A_1}$, then every hybrid bound B for $\mathcal{A}(\mathfrak{G})$, family primes $\le N^A$, with right-signed mass at moduli in $(N/2,CN]$ at most cB, satisfies for $N\ge N_0$*

$$\log(N/B)\le S'+\ell_N+\log(1+c)+O(1),$$

with S' as in Theorem 14.27.

Proof. $R_1=(1-1/(2K))\mathbf{1}_{[1,N]}+R/(2K)$ satisfies SPW$(C,\eta/(2K),\Delta'/(2K))$: full classes get mass $\le 1-\eta/(2K)$, sparse ones $\le 1/2$. Proposition 14.29(1) gives Flat with $t,s_0\asymp\eta/K$ and $\Delta=6\theta+\Delta'/(2K)$, and Theorem 14.27 gives the bound with $\log(1/t)+\log(1+\Delta(1+c))\le\ell_N+\log(1+c)+O(1)$; here $s_0=\eta/(4K)\ge N^{-A_1}$. □

So the 3/4 cap for hybrids by this route needs exactly $\log(K_N/\eta_N)+\log(1+\Delta'_N/K_N)=O((\log N)^{3/4})$; a bound $(\log N)^\theta$, $\theta<1$, gives a θ-cap; and polynomially small margins $\eta_N/K_N=N^{-c}$ give only $B\ge N^{1-c-o(1)}$, nothing of the form

$(\log N)^\theta$. This is a limitation of the route (Lemma 14.31 via Proposition 14.29 and Theorem 14.27), not a proved necessity. The exponent $3/4$ in S' is pointer-level only, as stated after Theorem 14.27; otherwise S' keeps the factor $(\log\log N)^{3/4}$.

Hypothesis 14.32 (weak SPW; [SPW, Lemma 1.4], [SPW2, §1]; [OPEN])**.** For a fixed $C > 1$ and all large N there is R as in Lemma 14.31 with $\log(K_N/\eta_N) + \log(1 + \Delta'_N/K_N) \le C'(\log N)^{3/4}$.

Theorem 14.30 forces only $\eta_N/K_N \ll (\log N)^{-1/2}$, far above this threshold. If the sparse classes are left free, the margin still decays, but only logarithmically:

Theorem 14.33 (K-free edge bound; [SPW2, Thm 3.1]; [PROVED])**.** *Let $C > 1$ and M, e, m_0 be as in Theorem 14.30. If $R \ge 0$ has (P1) and $R(s) \le 1 - \eta$ on every class of modulus e meeting $[1, N]$, with no condition on any other class, then for every integer $1 \le r < \min(N - 1, (C - 1)N - 1)/2$*

$$\eta \le \frac{eN}{4(M+1)r^2} + \frac{e}{(M+1)r} + \frac{4\pi m_0(m_0+1)N(2r+1)}{e^2}.$$

With $r = \lceil NM^{-1/3} \rceil$: $\eta \ll_C M^{-1/3} \asymp_C (\log N)^{-1/3}$, uniformly in K.

Proof sketch. As for Theorem 14.30, but $|T| \le 1$ is replaced by $|\hat{T}(k)| \le 2N$, which gives a Lipschitz bound for T independent of K, and the mass of ρ off W enters only through the kernel tail at distance $> r$ from x_{in}. □

So the relaxation with fixed η and $K = \infty$ is also false for large N, but weak forms ($\eta_N \ge \mathrm{e}^{-(\log N)^{3/4}}$) are untouched. Finite-support LPs at $C = 2$, $N \le 80$, show no $N^{-1/2}$-type decay of the optimal η, but they depend strongly on the support length [SPW2, §2] [EVIDENCE] (floating-point LPs). LPs find SPW at $C = 2$ for $N \le 60$ and Flat at sampled $N \le 100$ [IF2, §§5, 9] [EVIDENCE]; by Theorem 14.30 such small-N values are not indicative of large N. Open: weak SPW (Hypothesis 14.32), and right-signed mass at moduli in $(N, CN]$; periodic certificates show $(N/2, N]$ cannot be freed at $C = 1.5$ [IF2, §10] [EVIDENCE]. No hybrid beating $3/4$ is known.

15. Tuple counts and truncated weights

Among the non-CRT inputs of Section 16, item 6, the most natural is tuple counting: aggregate control of witness correlations of growing order, with combined moduli far above N. The sources are [TU] and [TU2], whose reviews [TUr, TU2r] found them sound after minor repairs (incorporated), and [KA3, §§5, 7]. The review [KA3r] covers [KA3, §5]; [KA3, §7] (Lemma 15.15 and the Assessment closing Section 15.3) was added later and checked by the coordinator only. Nothing here gives $\theta > 3/4$; the section locates what a tuple-count proof of $\theta > 3/4$ would have to contain.

15.1. **Witness correlations and the tuple-count hypothesis.** Let $\mathcal{P}_y$ be the primes $\ell \le y$ with $\ell \equiv 3 \pmod 4$, $A_\ell = (\ell + 1)/4$, $F(\ell) = |\mathscr{R}(\ell)|$ and $p_\ell = F(\ell)/\ell$. The *prime family* consists of the classes of $\mathscr{R}(\ell)$, $\ell \in \mathcal{P}_y$; its *witness count* is $f_y(n) = \sum_{\ell \in \mathcal{P}_y} \mathbf{1}[n \bmod \ell \in \mathscr{R}(\ell)]$. Under the CRT law the summands are independent $\mathrm{Bern}(p_\ell)$, and $c(\log y)^2 \le \mu_y := \sum_{\ell \in \mathcal{P}_y} p_\ell \le C(\log y)^2$ for $y \ge 3$ [TU, (1.3)]. Put

$$S_j(N) = \sum_{n \le N} \binom{f_y(n)}{j} = \sum_{|T| = j} C_T(N), \qquad e_j = e_j(p) = \sum_{|T| = j} \delta_T,$$

where T runs over j-sets of classes with distinct primes, $C_T(N)$ counts the $n \le N$ in all classes of T, and $\delta_T = \prod_{\ell\in T} \ell^{-1}$. Trivially $|S_j(N) - Ne_j| \le \#\{T\}$ [TU, Lemma 1.2]. A tuple with $\prod_T \ell > N$ has $C_T(N) \in \{0,1\}$ while $N\delta_T < 1$; every tuple-count hypothesis beyond the trivial range is about such tuples.

Lemma 15.1 (shift form; [TU, Lemma 1.3]; [PROVED]). *For $\ell \in \mathcal{P}_y$ let $\mathcal{D}_\ell$ be a set of divisors of A_ℓ^2 whose residues $-4D \bmod \ell$ are the classes of $\mathscr{R}(\ell)$, each once, and put $\omega_{y,D}(m) = \#\{\ell \in \mathcal{P}_y : D \in \mathcal{D}_\ell,\ \ell \mid m\}$. Then $f_y(n) = \sum_{D\ge 1} \omega_{y,D}(n+4D)$ for all $n \in \mathbb{Z}$, and $S_k(N) = \sum \#\{n \le N :\ \ell_i \mid n + 4D_i,\ i \le k\}$, summed over k-sets $\{(\ell_i, D_i)\}$ with distinct $\ell_i \in \mathcal{P}_y$ and $D_i \in \mathcal{D}_{\ell_i}$. All shifts satisfy $4D \le (y+1)^2/4$.*

Proof. The classes $-4D$, $D \in \mathcal{D}_\ell$, are distinct modulo ℓ, so $\mathbf{1}[n \bmod \ell \in \mathscr{R}(\ell)] = \sum_{D\in\mathcal{D}_\ell} \mathbf{1}[\ell \mid n + 4D]$; regroup by D. Expanding $\binom{f_y(n)}{k}$ counts k-sets of hit primes, each with its unique D; and $D \le A_\ell^2$. □

So S_k is the distinct-prime part of k-point correlations of ω-type functions along up to k shifts $4D$, with $\prod \ell_i \gg N$.

For even K let y_K be the largest y with $\mu_y \le K/\mathrm{e}^2$, so $\log y_K \asymp K^{1/2}$, and put $\eta_K = \mathrm{e}^{-K/\mathrm{e}^2}/K$. Write $\mathrm{TC}(N; K, y, \eta)$ for: $|S_j(N) - Ne_j| \le \eta N$ for all $1 \le j \le K$.

Hypothesis 15.2 (TC_θ; [TU, §2]; [OPEN]). Let $\theta \in (0,1)$ and $K_N = 2\lceil (\log N)^\theta \rceil$. For all large N, $\mathrm{TC}(N; K_N, y_{K_N}, \eta_{K_N})$ holds.

It is proved for $\theta < 2/3$ (Proposition 15.4); for $\theta \in (2/3, 1)$ it needs a compensating excess (Corollary 15.18) and is expected false (Assessment there).

Theorem 15.3 ([TU, Thm 2.1, Cors 2.2, 2.3]; [PROVED]; the bound on $E(N)$ [CONDITIONAL] on Hypothesis 15.2). *If K is even and $\mathrm{TC}(N; K, y, \eta)$ holds, then $\#\{n \le N : f_y(n) = 0\} \le N(\prod_{\ell\in\mathcal{P}_y}(1-p_\ell) + e_K + K\eta)$. For $y = y_K$, $\eta = \eta_K$ this is $\le (\mathrm{e}+2)N\mathrm{e}^{-K/\mathrm{e}^2}$. Hence Hypothesis 15.2 implies $E(N) \le (\mathrm{e}+2)N\exp\{-(2/\mathrm{e}^2)(\log N)^\theta\}$ for large N.*

Proof. For integers $f \ge 1$, $\sum_{j=0}^{K}(-1)^j\binom{f}{j} = (-1)^K\binom{f-1}{K}$; for $f = 0$ the sum is 1. For even K it is $\ge \mathbf{1}[f = 0]$. Summing over $n \le N$, $\#\{f_y = 0\} \le \sum_j (-1)^j S_j \le N\sum_j(-1)^j e_j + K\eta N$, and under the CRT law the identity gives $\sum_j(-1)^j e_j \le \prod(1-p_\ell) + \mathbb{E}\binom{f}{K} = \prod(1-p_\ell) + e_K$. For $y = y_K$: each $p_\ell < 1$, so $\mu_{y_K} > K/\mathrm{e}^2 - 1$ by maximality, $\prod(1-p_\ell) < \mathrm{e}^{1-K/\mathrm{e}^2}$, $e_K \le (\mathrm{e}\mu/K)^K \le \mathrm{e}^{-K}$ and $K\eta_K = \mathrm{e}^{-K/\mathrm{e}^2}$. The classes are forced. □

Proposition 15.4 (trivial range and squares; [TU, Props 2.4, 4.2]; [PROVED]). *(1) $\mathrm{TC}(N; K, y_K, \eta_K)$ holds for $K \le c_0(\log N)^{2/3}$; so Hypothesis 15.2 holds for $\theta < 2/3$. (2) It is false for even $K \ge (\mathrm{e}^2/2 + \varepsilon)\log N$, $N \ge N_0(\varepsilon)$.*

Proof. (1) The termwise bound gives $|S_j - Ne_j| \le (\sum_\ell F(\ell))^K \le (Cy_K(\log y_K)^2)^K \le N\eta_K$ once $K^{3/2} \ll \log N$; this is Brun's pure sieve. (2) By Lemma 2.6 every class of $\mathscr{R}(\ell)$ is a non-residue, and $0 \notin \mathscr{R}(\ell)$. So squares avoid the family, $\#\{f_y = 0\} \ge \lfloor\sqrt{N}\rfloor$, against the bound $(\mathrm{e}+2)N\mathrm{e}^{-K/\mathrm{e}^2} \le (\mathrm{e}+2)N^{1/2-\varepsilon'}$ of Theorem 15.3. □

For $\theta > 2/3$ the content of Hypothesis 15.2 is aggregate cancellation among tuples of combined modulus $\exp\{(\log N)^{3\theta/2+o(1)}\}$, each prime $\le y_K = N^{o(1)}$; no total-variation (Kubilius-type) model exists there [TU, Prop. 4.1] [PROVED].

15.2. **Bounded order cannot pass 3/4.** Take a prime-slice system (Definition 3.1) with $p_\ell(c) \le 1/4$ and $|F_\ell(c)| < \ell$ for all $\ell \in \mathcal{P}$, $c \in R$, and for a term of ν put $T_i = \{\ell \in \mathcal{P} : \ell \mid d_i\}$.

Definition 15.5 (mixed majorants; [TU, Def. 3.0])**.** ν is (λ_0, k)-*mixed* if every term has level $\sum_{T_i} \log \ell \le \lambda_0$ or *(prime) order* $|T_i| \le k$.

A k-point correlation of slice classes has order $\le k$ whatever its modulus; so $P \circ f$ with $\deg P \le k$ and the majorant of Theorem 15.3 with $K = k$, plus any terms of level $\le \lambda_0$, are mixed.

Theorem 15.6 (truncated weights on prime slices; [TU, Thm 3.1]; [PROVED])**.** *Let ν be (λ_0, k)-mixed, $L_0 > 0$, $s_\ell = \min(\log \ell, L_0)$ and $\lambda = \max(\lambda_0, kL_0)$. Then (4) holds with this λ, with $s_* = \min_{\mathcal{P}} s_\ell$, with $\bar\mu = \sum_{\ell \in \mathcal{P}} \bar p_\ell$, and with the mass sum replaced by $\sum_{\ell \in \mathcal{P}} \bar p_\ell \mathrm{e}^{-\alpha s_\ell}$, i.e. by $\sum_{\log \ell \le L_0} \bar p_\ell \ell^{-\alpha} + \mathrm{e}^{-\alpha L_0} \sum_{\log \ell > L_0} \bar p_\ell$.*

Proof sketch. Prove Theorem 4.5 with Proposition 4.4 applied to the weights s_ℓ, which it allows. A term has weighted level $\sum_{T_i} s_\ell \le \min(\sum_{T_i} \log \ell, |T_i| L_0) \le \lambda$. As $s_\ell \le \lambda$ for all ℓ, no coordinate is discarded, so $p_\ell(c) \le 1/4$ is needed for every slice prime; it is assumed. The weights do not depend on c, so the Jensen step is unchanged. □

Corollary 15.7 (order needed; [TU, Cors 3.2, 3.3]; [PROVED]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4)**.** *For the families of Corollary 6.1 with slice primes $\le N^A$, every $(A \log N, k)$-mixed majorant has $\log(1/\mathbb{E}\nu) \le C_9(A, C)[(\log N)^{3/4} + k \log\log N] + \log(P/\varphi(P))$.*

Proof. Take $\lambda_0 = A \log N = \lambda$, $L_0 = \lambda_0/k$, $\alpha = \max(\lambda_0^{-1/4}, 3k \log \lambda_0/\lambda_0)$. As in Corollary 6.1, the profile is $\ll \min(\alpha, 1)^{-3}$ and $\bar\mu \ll \lambda_0^3$. So $19\alpha\lambda \le 19\lambda_0^{3/4} + 57k \log \lambda_0$, the first mass sum is $\ll \lambda_0^{3/4}$, the second is $O(1)$ as $\alpha L_0 \ge 3 \log \lambda_0$, and the G-terms are $O(\log^2(k + \log N))$. □

Call a method *CRT-main-term* if its bound for $\#(\mathcal{A} \cap [1, N])$ is $\ge \frac{1}{2} N \mathbb{E}\nu$, i.e. its evaluation of $\sum_{n \le N} \nu(n)$ asserts the CRT main term up to half of it. Hypotheses "order-j correlation sums equal their CRT values up to a small error", $j \le k$, are of this kind; a low-level part evaluated exactly with a large favourable deficit is not. For such methods (and $\log(P/\varphi(P)) = o((\log N)^\theta)$) bounded order gives no $\theta > 3/4$, and for $\theta > 3/4$ a saving $(\log N)^\theta$ needs order $k \ge c(\log N)^\theta / \log\log N$ [PROVED]. Order $2\lceil (\log N)^\theta \rceil$ suffices under Hypothesis 15.2 (Theorem 15.3) [CONDITIONAL].

Corollary 15.8 (class order, all four types; [TU, Cor. 3.4], [KA3, §4.3]; [PROVED] (proof given here; [KA3, §4.3] only points to it), given [KA]/[KA2] as reviewed; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4, Thm 7.1, Cor. 7.4, (7.10))**.** *Let $\mathfrak{G}$ be as in Definition 10.1, and let every term of a majorant ν of $\mathcal{A}(\mathfrak{G})$ be an intersection of at most $k \ge 1$ residue classes of modulus $\le N^A$, or have level $\le A \log N$. Then $\log(1/\mathbb{E}\nu) \le C(kA \log N)^{3/4}$ for $N \ge N_0(A)$. So for CRT-main-term methods a saving $(\log N)^\theta$ needs $k \ge c(\log N)^{4\theta/3 - 1}$.*

Proof. Such an intersection is empty or one class whose modulus divides the product of the k moduli; its level is $\le kA \log N$. Apply Theorem 10.12. □

([TU] had the factor $(\log(kA \log N))^{3/4}$ of the previous cap.) The window of k between $(\log N)^{4\theta/3-1}$ and $(\log N)^\theta / \log\log N$ is treated next.

Proposition 15.9 (fixed shifts; [TU, Prop. 4.3 and extension]; [PROVED]). *Let $\mathcal{S}$ be a set of r positive integers. Let $\nu \geq 0$ be a finite combination of intersections of the events $M \mid n+4D$ with $D \in \mathcal{S}$, $M \equiv 3 \pmod 4$, $D \mid A_M^2$, $M \leq N^A$, and let $\nu \geq 1$ on the avoider set of a forced-class family containing these classes. Then $\mathbb{E}\nu \geq \exp\{-2r(\log\log N^A + 3)\}$.*

Proof. $\nu = G(\text{indicator vector})$. The avoider set is nonempty ($0 \notin \mathscr{R}(M)$ as $(D, M) = 1$) and has vector 0, so $G(0) \geq 1$. A hit $M \mid n+4D$ gives a prime $q \mid n+4D$, $q \equiv 3 \pmod 4$, $q \leq N^A$, $q \nmid D$. So $\mathbb{E}\nu \geq \prod_q (1 - r_q/q)$, $r_q \leq \min(r, q-1)$; use $1 - x \geq \mathrm{e}^{-2x}$ for $q > 2r$ and $1 - r_q/q \geq 1/q$ for $q \leq 2r$. □

So witness information of any order and precision along r fixed shifts saves $\leq 2r(\log\log N + O(1))$ for CRT-main-term methods. Known divisor-correlation theorems (Heath-Brown, Deshouillers–Iwaniec, Matomäki–Radziwiłł–Tao, Tao–Teräväinen) have fixed or averaged shifts; Hypothesis 15.2 needs $\asymp K$ shifts jointly. That no known theorem or standard conjecture supplies it for any $\theta > 2/3$ is an *Assessment* [TU, Ass. 4.4]. Not covered: non-witness information along the shifts ($q \equiv 1 \pmod 4$, $\tau(n+4D)$, primes $> N^A$), and combinations over all shift tuples.

15.3. **Truncated weights.** Theorem 15.11 and Corollaries 15.12 and 15.13 carry the proviso of Theorem 10.12: proved given [KA, KA2] as reviewed; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4, Thm 7.1, Cor. 7.4, (7.10).

Definition 15.10 ([KA3, Def. 5.0]). Fix $L_0 > 0$ and put $s_\ell = \min(\log \ell, L_0)$ for primes $\ell > W$. A term $a\mathbf{1}[n \equiv b \pmod d]$ has *truncated level* $\sum_{\ell \mid d,\, \ell > W} s_\ell$ and *prime order* $\omega_{>W}(d)$; ν is (λ_0, k)*-mixed over* W if every term has level $\leq \lambda_0$ or prime order $\leq k$ (Definition 15.5 with all primes $> W$ in place of the slice primes). With $L_0 = \lambda_0/k$ such a ν has truncated level $\leq \lambda_0$.

Theorem 15.11 (truncated-weight cap; [KA3, Thm 5.1]; [PROVED]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4, Thm 7.1, Cor. 7.4, (7.10)). *Let W, λ_0, C be as in Theorem 10.12, K_3' as in Lemma 10.9, and C_0 as in Corollary 8.6. Let $\mathfrak{G}$ be as in Definition 10.1 with all family primes $\leq \mathrm{e}^\Lambda$, and let $\nu \geq 0$ on $\mathbb{Z}$, $\nu \geq 1$ on $\mathcal{A}(\mathfrak{G})$, have truncated level $\leq \lambda$, with $\lambda_0 \leq \lambda \leq \Lambda$ and $L_0 \leq \lambda$. With $d_0 = \lfloor \lambda/L_0 \rfloor$,*

$$\log \frac{1}{\mathbb{E}\nu} \leq C\lambda^{3/4} + 2d_0 \log \frac{C_0(K_3'\Lambda^3 + 4d_0)}{d_0} + \frac{8}{3}d_0 + \log(22d_0 + 22) + 6.$$

Proof sketch. Theorem 9.1 holds for any class $\mathcal{F}$ of functions closed under conditioning on earlier coordinates, if (S_w) holds on $\mathcal{F}$. Take for $\mathcal{F}$ the sums of terms each depending on a prime set T with $\sum_T s_\ell \leq \lambda$; conditioning only shrinks T. The cost ledger, in increasing order of primes: (i) singletons $W < \ell \leq \mathrm{e}^{\min(s_1, L_0)}$, $s_1 = \lambda^{1/4}$, and blocks $\{2^i s_1 < \log \ell \leq \min(2^{i+1}s_1, L_0, \lambda/2)\}$, where $s_\ell = \log \ell$ and the depths are those of Theorem 10.12: total $\leq C\lambda^{3/4} + O(\log^2 \lambda)$; (ii) if $L_0 > \lambda/2$, the linear block $(\mathrm{e}^{\lambda/2}, \mathrm{e}^{L_0}]$: $\leq 2\log(1 + 3\mathrm{e}^{-\lambda/4})$; (iii) one *top block* $V_{\mathrm{top}} = \{\log \ell > L_0\}$ of any width, where $s_\ell = L_0$, so every $f \in \mathcal{F}$ is d_0-local. Theorem 8.5 is arithmetic-free, and $\mathbb{E}m_{V_{\mathrm{top}}} \leq \mathfrak{M}(\mathrm{e}^\Lambda) \leq K_3'\Lambda^3$ (Lemma 10.9). Corollary 8.6 with t chosen per history, and Jensen, give $\mathbb{E}\Phi_{\mathrm{top}} \leq d_0 \log(C_0(K_3'\Lambda^3 + 4d_0)/d_0) + \frac{4}{3}d_0 + \frac{1}{2}\log(22d_0 + 22) + 3$, doubled by Theorem 9.1; (iv) base and Jensen: $2W + \log 2$. The leak bound of Lemma 10.11 is per prime, so $\mathfrak{L} \leq 1/2$. □

Corollary 15.12 (prime order; [KA3, Cor. 5.2]; [PROVED]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4, Thm 7.1, Cor. 7.4, (7.10)). *Let $\mathfrak{G}$ be as in Definition 10.1*

with family primes $\le N^A$, and let every term of a majorant ν of $\mathcal{A}(\mathfrak{G})$ have level $\le A\log N$ or prime order $\le k$, $1 \le k \le (A\log N)^3$. Then $\log(1/\mathbb{E}\nu) \le C_A[(\log N)^{3/4} + k\log\log N]$. So a CRT-main-term saving $(\log N)^\theta$, $\theta > 3/4$, needs prime order $k \ge c_A(\log N)^\theta/\log\log N$.

Proof. Project onto the lcm of the family moduli, as in the proof of Theorem 10.14; prime sets shrink, so both term types are kept, and all primes are $\le \mathrm{e}^\Lambda$, $\Lambda = A\log N$. For $k = 1$ use Theorem 10.12; for $k \ge 2$, Theorem 15.11 with $\lambda = \Lambda$, $L_0 = \lambda/k$, $d_0 = k$. □

Corollary 15.13 (boundedly many large primes; [KA3, Cor. 5.3]; [PROVED]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4, Thm 7.1, Cor. 7.4, (7.10))**.** *Let all family primes of $\mathfrak{G}$ be $\le N^A$, and let every modulus of $\mathfrak{G}$ have at most r prime factors $> W$ (W the absolute constant of Theorem 10.12). An intersection of k classes of $\mathfrak{G}$ (or of any residue classes whose moduli have $\le r$ primes $> W$) has prime order $\le kr$. Hence every majorant ν of $\mathcal{A}(\mathfrak{G})$ each of whose terms has level $\le A\log N$ or is such an intersection has $\log(1/\mathbb{E}\nu) \le C_A[(\log N)^{3/4} + kr\log\log N]$.*

Scope. W is fixed, not a moving cutoff: the twin moduli of [TW2, TW4] and the atoms $\mathscr{R}(k\ell)$ of [TQ] may have cofactors with many primes above W, and are then covered by Theorem 10.12 but not by Corollary 15.13.

Proposition 15.14 (block-sparse families; [TU2, Prop. 7.1]; [PROVED] relative to [KA2, Thm 5.1]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4)**.** *Let all moduli of $\mathfrak{G}$ be $\le N^A$, each with at most r prime factors in every interval $(x, x^2]$, $x \ge 2$. If every term of ν has level $\le \lambda_0 = A\log N$ or is an intersection of at most k classes of $\mathfrak{G}$, then $\log(1/\mathbb{E}\nu) \le C\lambda_0^{3/4}(\log\lambda_0)^{3/4} + Ckr(\log\log N)^2$.*

Proof sketch. The level enters [KA2, Thm 5.1] only through d-locality, and now $d \le \lambda_0/s + kr$ in $(\mathrm{e}^s, \mathrm{e}^{2s}]$. The step cost $d\log(C_0(M+4d)/d)$ is subadditive, so the kr part costs $\ll kr\log\log N$ in each of $O(\log\log N)$ dyadic blocks up to N^A. □

Whether Lemma 10.9 removes $(\log\lambda_0)^{3/4}$ here is unchecked. The next lemma (coordinator-checked only) concerns the literal hypothesis of Corollary 15.8.

Lemma 15.15 (packing; [KA3, Lemma 7.2]; [PROVED])**.** *Let $Z \ge 2$ and $k \ge 1$. Every residue class $b \bmod d$ such that every prime power $p^e \parallel d$ is $\le Z$ and $\log d \le (k-1)\log Z/2$ is an intersection of at most k residue classes of moduli $\le Z$.*

Proof. Pack the prime powers $p^e \parallel d$, in any order, greedily (next-fit) into bins whose products stay $\le Z$; each fits into an empty bin. Let there be $m \ge 2$ bins ($m \le 1$ is trivial), with products $b_1, \dots, b_m$. The first element of bin $i+1$ did not fit into bin i, so $b_i b_{i+1} > Z$ for $i < m$. Hence $d \ge \prod_{j<\lfloor m/2\rfloor} b_{2j+1}b_{2j+2} > Z^{\lfloor m/2\rfloor}$, so $\lfloor m/2\rfloor < \log d/\log Z \le (k-1)/2$ and $m \le 2\lfloor m/2\rfloor + 1 \le k$. The bins are pairwise coprime with product d, and by CRT $\{n \equiv b\ (d)\} = \bigcap_i \{n \equiv b\ (b_i)\}$. □

With $Z = N^A$, $L = \log N$: majorants with prime powers $\le N^A$ and $\log d_i \le (k-1)AL/2$ satisfy the hypothesis of Corollary 15.8, which in turn forces level $\le kAL$. So, up to a factor 2 in k, it is a level hypothesis at $\lambda \asymp kL$. Closing the window from it alone would require $C[L^{3/4} + (\lambda/L)\,\mathrm{polylog}\,L]$ in place of $C\lambda^{3/4}$ for all such majorants. By inspection the k-ary ledger cannot give this for $k \le L^3$ (it is optimal at $s_1 = \lambda^{1/4} \le AL$, where truncation at e^{AL} is invisible; *Assessment*), and whether $\lambda^{3/4}$ is attained there is [OPEN] ([EVIDENCE] only, [ET, §2.5]). The window is meaningful

for structured orders: prime order (closed, Corollary 15.12) and intersections of k *family* classes (closed for bounded $r = \max_G \omega_{>W}(G)$, Corollary 15.13). For the latter the open window is

$$\max\bigl(L^{4\theta/3-1},\, L^{\theta}/(r\log L)\bigr) \lesssim k \lesssim L^{\theta}/\log L, \tag{11}$$

for family moduli $\le N^A$ (needed for the lower end, which comes from Corollary 15.8), and except as narrowed by Proposition 15.14 for block-sparse families; it is nonempty only when $r \to \infty$ [KA3, §5] [OPEN].

Assessment [KA3, §7]. In the pattern model of Section 8, class degree k over m-prime patterns is degree km, and binomial extrapolation (Lemma 8.3) is tight as for prime order km. If this persists for forced classes, the window (11) is genuine; deciding it needs ES-specific input.

15.4. **Forced zeros and the repaired hypothesis.**

Lemma 15.16 (forms; [TU2, Lemma 1.1, Cor. 1.2]; [PROVED])**.** *Let $\ell \equiv 3 \pmod 4$ be prime and $A = A_\ell$. The map $(r,s,m) \mapsto D = r^2 m$ is a bijection from the triples with $rsm = A$, $(r,s) = 1$, onto the divisors of A^2, and $n \equiv -4D \pmod \ell$ if and only if $\ell \mid ns + r$. In particular $(1,1,A)$ gives the class $-1 \bmod \ell$, which lies in $\mathscr{R}(\ell)$ for every ℓ.*

Proof. Write $D/A = r/s$ in lowest terms; then $A = sa'$, $D = ra'$, and $D \mid A^2$ gives $r \mid a'$, so $a' = rm$. The converse is clear. As $4rsm = \ell + 1$, $4r^2 m \equiv r/s \pmod \ell$. □

Call (r,s) the *form* of the class; fix one form per residue, $(1,1)$ for -1. For a tuple T let T_φ be its pairs of form $\varphi = (r,s)$. If $\prod_{T_\varphi} \ell > Ns + r$ for some φ, then $C_T(N) = 0$ although $\delta_T > 0$, since $\prod_{T_\varphi} \ell$ divides $ns + r \le Ns + r$. Call T *admissible* if this happens for no φ; let $e_j^{\mathfrak{A}}$ be the CRT mass of admissible j-tuples and $Z_j = e_j - e_j^{\mathfrak{A}}$ that of the *forced zeros*. Then $S_j(N) - Ne_j = \sum_{T\ \mathrm{adm.},\,|T|=j} (C_T - N\delta_T) - NZ_j$.

Theorem 15.17 (forced-zero mass; [TU2, Thms 2.1, 2.2]; [PROVED])**.** *Fix $\theta \in (2/3, 1)$, take $K = K_N$, $y = y_K$ as in Hypothesis 15.2, and $u_0 = \lceil \log(N+2)/\log(y/2) \rceil$. For $N \ge N_0(\theta)$, $u_0 \le K$ and $Z_{u_0} \ge \eta_K \exp\{K/(2\mathrm{e}^2)\}$.*

Proof sketch. Let $\mathcal{Q}$ be the primes $\ell \equiv 3 \pmod 4$ in $(y/2, y]$, $\sigma_y = \sum_{\mathcal{Q}} 1/\ell \ge c/\log y$. A u_0-set $U \subseteq \mathcal{Q}$ with class -1 at each prime has $\prod_U \ell > N + 1$, so is a forced zero, and $Z_{u_0} \ge e_{u_0}((1/\ell)_{\mathcal{Q}}) \ge (\sigma_y/2)^{u_0}/u_0!$. Here $u_0 \asymp (\log N)^{1-\theta/2}$, so $\log Z_{u_0} \ge -C(\log N)^{1-\theta/2} \log\log N$, while $\log(1/\eta_K) \asymp (\log N)^{\theta}$. And $1 - \theta/2 < \theta$ for $\theta > 2/3$. □

Corollary 15.18 (a necessary condition; [TU2, Cor. 2.3]; [PROVED])**.** *If $\theta \in (2/3, 1)$, $N \ge N_0(\theta)$ and $\mathrm{TC}(N; K_N, y_K, \eta_K)$ holds, then $\sum_{T\ \mathrm{adm.},\,|T|=u_0} (C_T(N) - N\delta_T) \ge N(Z_{u_0} - \eta_K) > 0$: the admissible u_0-tuples carry an aggregate excess over their CRT values.*

Assessment [TU2, Ass. 3.2]. For the class -1 the count $\lfloor (N+1)/q \rfloor$ is never above $(N+1)/q$ [TU2, Lemma 3.1], so the excess must come from tuples that are not pure class -1 (heuristically, multi-form tuples). It is $N^{1/2-o(1)}$ times a square-root noise scale, and no mechanism for it is known. We expect Hypothesis 15.2 to be false for every $\theta \in (2/3, 1)$; this is not proved. Toy moments on $[1, N]$ fall below CRT but not on far translates [TU2, §5] [EVIDENCE].

Theorem 15.3 uses only the alternating sum, i.e. the Brun pure-sieve majorant $\nu_K(n) = \sum_{j\le K}(-1)^j\binom{f_y(n)}{j}$ (K even).

Hypothesis 15.19 ($\mathrm{TC}_\theta^{\mathrm{alt}}$; [TU2, §4]; [OPEN])**.** With $K = K_N$, $y = y_K$, $\eta = \eta_K$ as in Hypothesis 15.2, for all large N,

$$\sum_{n\le N}\nu_K(n) \le N\Big(2\prod_{\ell\in\mathcal{P}_y}(1-p_\ell) + 2\mathrm{e}^{-K}\Big) + K\eta N.$$

The CRT mean of ν_K lies in $[\prod(1-p_\ell), \prod(1-p_\ell) + \mathrm{e}^{-K}]$, so this is a one-sided statement with a factor 2, not an accuracy statement. The intermediate hypothesis $\mathrm{TC}_\theta^{\mathfrak{A}}$ asks $|S_j(N) - Ne_j^{\mathfrak{A}}| \le \eta_K N$ for $j \le K$: accuracy on admissible tuples only.

Theorem 15.20 (Euler-characteristic cancellation; [TU2, Thm 4.1, Cor. 4.2]; [PROVED])**.** *Let $K \ge \mathrm{e}^2\mu_y$ be even, $u_1 = \lfloor \log N/\log y\rfloor + 1 \ge 4(1+\log y)$ and $\varepsilon = 4(2\mathrm{e}(1+\log y)/u_1)^{u_1} \le 1$. Then $|\sum_{j\le K}(-1)^j e_j^{\mathfrak{A}} - \prod(1-p_\ell)| \le 2\varepsilon\prod(1-p_\ell) + \mathrm{e}^{-K}$, and $|\sum_{j\le K}(-1)^j Z_j| \le 2\varepsilon\prod(1-p_\ell) + 2\mathrm{e}^{-K}$. Consequently, for $\theta \in (0,1)$ and $N \ge N_0(\theta)$, $\mathrm{TC}_\theta^{\mathfrak{A}}$ implies $E(N) \le (\mathrm{e}+3)N\exp\{-(2/\mathrm{e}^2)(\log N)^\theta\}$, and Hypothesis 15.19 implies $E(N) \le (2\mathrm{e}+2)N\exp\{-(2/\mathrm{e}^2)(\log N)^\theta\}$.*

Proof sketch. For the CRT hit set H, $\sum_j(-1)^j e_j^{\mathfrak{A}} = \mathbb{E}\sum_{T\subseteq H\text{ adm.},\,|T|\le K}(-1)^{|T|}$, and dropping $|T| \le K$ costs $\le \sum_{j>K} e_j \le \mathrm{e}^{-K}$. Admissibility is a condition on each form group, so the full sum is $\prod_\varphi \chi_\varphi(H_\varphi)$ with $\chi_\varphi(G) = \sum_{U\subseteq G\text{ adm.}}(-1)^{|U|}$. Subsets of admissible sets are admissible, so $\chi_\varphi(G) = \mathbf{1}[G = \emptyset]$ for admissible G, and always $|\chi_\varphi(G)| \le 2^{|G|}$. So the sum differs from $\mathbf{1}[H = \emptyset]$ only if every nonempty group is inadmissible, hence has $\ge u_1$ primes. With $p_\ell < 1/2$ and $\sum_{\ell\equiv -1\,(4rs)} 4/\ell \le 2(1+\log y)/(rs)$, the tails summed over forms give 2ε. In the calibration $u_1/\log y_K \to \infty$, so $\varepsilon \to 0$; conclude as in Theorem 15.3. □

The second implication is just $\nu_K \ge \mathbf{1}[f_y = 0]$: Hypothesis 15.19 is essentially the conclusion restricted to one majorant (its factor 2 matters: at $N = 10^7$, $y = 1000$ the avoiders on $[1, N]$ are 1.22 times CRT, an excess of about the $\sqrt{N}$ squares, [EVIDENCE]). It fails for even $K \ge (\mathrm{e}^2/2+\varepsilon)\log N$ as in Proposition 15.4(2). That it is the correct form of the tuple-count door is an *Assessment* [TU2, §4]: forced zeros provably cancel in it, single-form floor deficits heuristically do; $\mathrm{TC}_\theta^{\mathfrak{A}}$ is expected false above 2/3.

15.5. **The prime family at 2/3.**

Corollary 15.21 ([TU2, Cor. 6.1]; [PROVED])**.** *Fix $A \ge 1$ and $y \le N^A$. Let ν be a CRT majorant of $\{f_y = 0\}$ whose terms have $\sum_{\ell\in\mathcal{P}_y,\,\ell\mid d_i,\,\ell\ge\ell_0}\log\ell \le \lambda$, $\lambda \ge \log N$, where ℓ_0 is an absolute constant such that $\tau(A_\ell^2)/\ell \le 1/4$ for all primes $\ell \ge \ell_0$. Then $\log(1/\mathbb{E}\nu) \le C_A\lambda^{2/3}$. So CRT-main-term methods at level $\le A'\log N$ save at most $C(\log N)^{2/3}$, the order of Brun's pure sieve (Proposition 15.4(1)).*

Proof sketch. Put the primes $\ell < \ell_0$ into Q_0; then $|R|/Q_0 \gg 1$ and every slice prime has $p_\ell \le \tau(A_\ell^2)/\ell \le 1/4$. Apply Theorem 15.6 with $k = 1$, $L_0 = \lambda_0 = \lambda$. By $\mu_x \le C(\log x)^2$ and partial summation, $\sum_\ell p_\ell\ell^{-\alpha} \le 2C\alpha^{-2}$: the profile is quadratic, not cubic. Take $\alpha = \lambda^{-1/3}$. □

So the prime family's CRT exponent is exactly 2/3, where literal TC_θ starts to see forced zeros. Hence Hypothesis 15.19 for $\theta > 2/3$ is non-CRT: a saving

$(\log N)^\theta$ needs level $\geq c(\log N)^{3\theta/2}$, so ν_K must be CRT-accurate on $[1, N]$ at moduli $\exp\{c(\log N)^{3\theta/2}\}$ (Brun's sieve far beyond its level of distribution; it improves $E(N)$ only for $\theta > 3/4$, by [TQ]). That no known theorem reaches this level is an *Assessment* [TU2, Ass. 6.2].

15.6. **What is open.**

- Hypothesis 15.19 for some $\theta > 3/4$ (it would give $\theta > 3/4$); no structured obstruction below $\theta = 1$ is known.
- The window (11): a k-ary comparison in class coordinates, for the dependent indicators $\mathbf{1}[n \in C]$.
- Deciding the literal Hypothesis 15.2 and $\mathrm{TC}_\theta^{\mathfrak{A}}$ on $(2/3, 1)$ (only an *Assessment* so far), and at $\theta = 1$.
- Removing the factor $(\log \lambda_0)^{3/4}$ from Proposition 15.14, e.g. via Lemma 10.9 (unchecked).
- Interval deficits (the second branch of the dichotomy of Section 13); fixed-shift statements combined over all shift tuples; non-witness information along shifts; bounded-order input with weights below 1.

16. What the theorems exclude

The following list is checked against the proofs of Theorems 4.5, 4.9, 7.6, 9.1, 9.3, 10.12, 10.14 and Lemma 3.4 [ET, §6.1], [KA, §4.3], [NC, §6], [KA2, §6], and updated with Sections 14 and 15 [LS, §7], [LS2, §0], [LS3, §4], [LS4, §5], [IF2, §0], [SPW, §4], [SPW2, §1], [TU, §6], [TU2, §0], [KA3, §5]. Anything below lies outside the cap of Theorem 10.14; where another theorem covers part of an item, this is said.

(1) *The large sieve at super-polynomial frequency levels.* Covered: every CRT-admissible large sieve with rational frequencies of polynomial level, fibrewise or not, with any weights and any moduli (Theorem 14.2); multiplicative, Gauss-sum, twisted and hybrid Bessel forms with periodic rows (Theorem 14.5); the large sieve applied to the primes (Theorem 14.6); prime slices with any rational frequencies (Theorem 14.3); and every mixture in which each modulus has at most one prime factor $> z = \exp\{(\log N)^{1/4}\}$, with any rational frequencies, at the cost of a factor $(\log\log N)^3$ (Theorem 14.13); more generally every mixture whose classes with at least two prime factors $> z$ are residue-sparse, (RS_γ) (for instance all H-small), with a factor $\gamma^{-3}(\log\log N)^3$ (Theorem 14.18). Outside: frequencies of super-polynomial level against mixtures whose classes with at least two prime factors $> z$ violate (RS_γ) for every fixed $\gamma > 0$ (escape (E1); residue-sparseness is a property of the union Res_ℓ over the family, not of single classes; Conjecture 14.10; the sup-decay hypothesis (A^*) of Theorem 14.17, or (H_{rough}) of Hypothesis 14.14, would suffice). For such mixtures the frequencies with z-smooth denominators are covered when no fibre $\mathcal{A}_c$, $c \in \operatorname{supp} \pi_s$, is empty, which is not automatic (Section 14.5). By Theorem 14.11 this cannot be closed by comparison measures and large-sieve axioms alone: a proof needs family-specific input such as the sparsity of forced classes. Non-periodic rows (such as n^{it}) are non-CRT input.

(2) *The larger sieve with bad composite kernels.* Gallagher's sieve with prime-power kernels saves at most $26 \log\log N + C$ over any mixture, unconditionally (Theorem 14.7); composite kernels are capped up to the factor $1 + Nh/(W_K - h)$ (Theorem 14.8). Outside: kernels for which that factor exceeds $e^{(\log N)^{3/4}}$ and whose moduli have small prime factors.

(3) *Cancellation between frequencies above modulus $N/2$.* Covered: majorants built from classes of modulus $\le N/2$, with any coefficients and any evaluation of the interval sum (dispersion, Kloosterman, Erdős–Turán/Vaaler, floor/ceiling, smooth windows) (Corollary 14.23); bounds that dominate the large coefficient mass up to $e^{O((\log N)^{3/4})}$ (Theorem 14.24); hybrid methods whose free large mass is $\le cB$ (Section 14.9). Outside: hybrids with large right-signed mass at moduli in $(N/2, CN]$, and all hybrids unless Flat holds with margins $\log(1/t), \log\Delta = O((\log N)^{3/4})$ (Theorem 14.27); such Flat follows from weak SPW (Hypothesis 14.32, [OPEN], Lemma 14.31), while SPW with a fixed margin is false (Theorem 14.30); direct counts of multi-witness patterns of combined modulus $> N/2$. For finite prime-slice families every per-frequency rounding with weight ≥ 1 is covered for any moduli (Theorem 13.3).

(4) *Per-frequency weights below one* with classes of modulus $> N/2$: covered only under Theorem 14.24's hypothesis; for smooth windows on prime slices, [CONDITIONAL] on (H_{eq}) (Section 13).

(5) *Majorants ≥ 1 only on part of the avoider set.* Theorem 10.14 needs $\nu \ge 0$ on $\mathbb{Z}$ and $\nu \ge 1$ on all of $\mathcal{A}(\mathfrak{G}) \subseteq \mathbb{Z}$. Covered: majorants ≥ 0 at all primes and ≥ 1 at the primes of $\mathcal{A}$, for every mixture, hence prime-law methods with BV/BDH/EH/GRH-type input at polynomial level (Theorem 14.19, Corollary 14.20). The relaxation "$\nu \ge 1$ only at the primes of $\mathcal{A} \cap [1, N]$" admits no cap of any kind (Proposition 14.21): the gap there is certification, not majorant design. Outside: $\nu \ge 0$ only on $[1, N]$ (Remark 3.6); $\nu \ge 1$ only on $\mathcal{A} \cap [1, N]$ or on the exceptional set; $\nu \ge 0$ only at primes $\le N$ (with $\nu \ge 1$ at all primes of $\mathcal{A}$), which is open (*Assessment*); unconditional signed error terms for prime laws (*Assessment*).

(6) *Non-CRT input.* The theorems see only the CRT law of residues. Covered: CRT-evaluated moment methods (Section 13), correlation input of bounded order or of prime order $o((\log N)^{\theta}/\log\log N)$ under CRT-main-term evaluation (Corollaries 15.8 and 15.12), and the large sieve for primes (Theorem 14.6). Outside: Type I/II sums, the arithmetic of $(p + a)/4$, Halász-type arguments, non-witness divisor information along fixed shifts, tuple counts whose main term is not the CRT density (any $\theta > 3/4$ input of this kind for the prime family needs accuracy at moduli $\exp\{c(\log N)^{3\theta/2}\}$, see the discussion after Corollary 15.21), and family-class order k in the window (11) for families whose moduli have unboundedly many primes above W (Corollary 15.13, Lemma 15.15).

(7) *Other class types*, including non-selector admissible sets. The proof of Theorem 10.12 uses only three properties of the classes: (a) no W-smooth class contains a unit square modulo its modulus (for the base); (b) first moments $\ll (\log y)^3$ over $P(G) \le y$; (c) cofactor second moments $\ll \ell^{o(1)}$. The same proof covers any family with (a)–(c) [KA2, §6, item 2]; families without them are outside. Any admissible set that cannot be expressed

by selector classes (or by classes of the four types) has an uncontrolled term $\log(Q_0/|R|)$ (Remark 4.7; [ET, §6]); this includes admissible sets that exclude classes failing (a), such as classes of unit squares. The one exception in this note is the quadratic-residue set R_W for fixed-B families of $\mathscr{R}(M)$-classes (Theorem 9.3). Schinzel's theorem, as quoted in [ElT], may make (a) automatic for polynomial-identity families; this has not been checked.

(8) *Primes beyond $N^{O(1)}$.* Families with primes beyond $N^{O(1)}$ are outside Theorem 10.14, since Lemma 3.4 needs Λ_0. Covered without such a bound: finite prime-slice families with per-frequency rounding (Theorem 13.3), majorants from classes of modulus $\le N/2$ (Corollary 14.23), and large sieves on prime slices with any rational frequencies (Theorem 14.3), and large sieves over mixtures with one prime factor $> \exp\{(\log N)^{1/4}\}$ per modulus (Theorem 14.13), and over residue-sparse mixtures (Theorem 14.18).

The exact exponent is now settled for the class of Theorem 10.14: the factor $(\log\log N)^{3/4}$ of the previous version is gone (Lemma 10.9), so no gain $(\log N)^{3/4}\omega(N)$ with $\omega \to \infty$ is possible in that class, with or without B. The same holds for the results of Section 14 that use Theorem 10.12 as a black box; for the prime-measure results (Theorem 14.19 and what rests on it) the improvement is pointer-level only [KA3, §4.3]. All constants are absolute but astronomically large (Section 10).

No longer excluded (compared with version 2 of this note). The following are inside Theorem 10.14: $\mathscr{R}(M)$-moduli with $\log M/\log P(M)$ unbounded; (a, D)- and Case-A moduli without a dominant prime; families mixing all four types; twin moduli (Lemma 16.1); any number of prime factors per modulus; selector admissible sets (as selector classes, with selector primes $\le N^A$); and multiplier conditions with a small part larger than the top prime, used without a separate level budget, provided the conditions are themselves classes of the four types and all their primes, those of the multiplier included, are $\le N^A$. For these, Theorem 10.14 charges every prime above W and bounds the level through $\sum_i |a_i| < N$ together with the bound N^A on the family primes. *New in version 5*: every large sieve, at any frequency level, over mixtures with at most one prime factor $> \exp\{(\log N)^{1/4}\}$ per modulus (Theorem 14.13, with $(\log\log N)^3$), and over mixtures whose classes with several such primes are residue-sparse (Theorem 14.18), which narrows (E1) to mixtures whose multi-rough part violates (RS_γ) for every fixed $\gamma > 0$, and there to the sup-decay hypothesis (A^*) (Theorem 14.17); and, for hybrids, the refutation of fixed-margin SPW with the weak requirement that replaces it as the target of this route (Theorem 14.30, Lemma 14.31). This version does not change the cap of Theorem 10.14. *New in version 4*: the removal of the factor $(\log\log N)^{3/4}$ without B (Theorem 10.12); the large sieve beyond prime slices at polynomial level, its twisted, hybrid and prime forms, and Gallagher's larger sieve (Section 14); prime-only majorants for all mixtures; interval counts with all moduli $\le N/2$; bounded-order and prime-order tuple input (the prime forms and prime-only majorants with the factor $(\log\log N)^{3/4}$, prime order k with a term $k\log\log N$); and the Λ^2 middle range $r \asymp \log L$ as a cap (Remark 12.8). For the external input, the Case-A part uses Elsholtz–Tao Prop. 1.4 and Thm 7.1, Cor. 7.4 and (7.10) of their §7 (published, not re-proved; Remark 10.7), the latter as Lemma 10.8 does; families without Case-A classes use only Shiu's theorem.

Lemma 16.1 (balanced moduli carry cubic supply; [ET, Lemma 3.8]; [PROVED])**.** *There are $c > 0$ and x_0 such that for $x \geq x_0$,*

$$\sum_{M \leq x,\ M \equiv 3\ (4),\ P(M) \leq M^{1/2}} \frac{|\mathscr{R}(M)|}{M} \geq c(\log x)^3.$$

More precisely, for each fixed $\eta \in (0, 1/480]$, the moduli $M = k\ell_1\ell_2$ with primes $\ell_1 \in (Y, Y^{1+\eta/2}]$, $\ell_2 \in (Y^{1+\eta/2}, Y^{1+\eta}]$, odd $k \in (Y^{2\eta}, Y^{3\eta}]$ and $Y = x^{1/(2+4.5\eta)}$ already contribute $\geq c(\eta)(\log x)^3$; these are same-scale pairs, $\log \ell_2 / \log \ell_1 \leq 1 + \eta$.

Proof sketch. Each such M is $\leq x$ and balanced, and determines (k, ℓ_1, ℓ_2). For coprime $u, v \leq z = Y^{1/8}$ with $(uv, k) = 1$ and $4uv \mid M+1$, the class $-uv^{-1} \pmod M$ lies in $\mathscr{R}(M)$, and distinct (u, v) give distinct classes modulo ℓ_2. The ℓ_2-sum over the progression $-(k\ell_1)^{-1} \pmod{4uv}$ is evaluated by partial summation, with the error terms controlled on average by Cauchy–Schwarz against Brun–Titchmarsh and the Bombieri–Vinogradov theorem [D, Ch. 28]. The (u, v)-sum is bounded below by the lattice lemma of [LL, Lemma 3.1], which needs $\eta \leq 1/480$. The k- and ℓ_1-sums give $\gg \eta \log Y$ and $\gg \eta$. Full details are in [ET, §3.8], checked in [ETr, item 4]. □

The share of the weighted Case-B mass carried by balanced moduli is 0.098, 0.121, 0.141, 0.157 and 0.171 at $x = 10^3, \ldots, 10^7$, still growing [EVIDENCE]. Lemma 16.1 does not by itself show that every window partition loses: singleton windows have no internal pairs. So the twin moduli had to be covered, not discarded. For fixed B they cannot beat the cap (Theorem 9.3) [PROVED]; without the B-hypothesis they are inside Theorem 10.12, with the same cap $C\lambda^{3/4}$ [PROVED].

17. Open problems

A proof of $\theta > 3/4$ by a congruence sieve must therefore contain at least one of the following [ET, §6], [KA, §4.3], [NC, §6], [KA2, §6], [LS, §7], [LS2, §0], [IF2, §0], [TU2, §0], [KA3, §5], [LS3, §4], [LS4, §5], [SPW, §4], [SPW2, §1]:

(i) classes of a type outside Definition 10.1 that fail one of the properties (a)–(c) of Section 16, item 7, or a non-selector admissible set with a large saving of its own;

(ii) a majorant that is ≥ 1 only on part of the avoider set (on $[1, N]$ or on the exceptional set), or a prime majorant that is ≥ 0 only at primes $\leq N$;

(iii) large-sieve frequencies of super-polynomial level used against classes with at least two prime factors $> \exp\{(\log N)^{1/4}\}$ (Conjecture 14.10, or (H_{rough}) of Hypothesis 14.14, would exclude this, and so would (A^*) of Theorem 14.17; Theorems 14.13 and 14.18 exclude it for one such prime per modulus and for mixtures satisfying (RS_γ), so only mixtures whose multi-rough part violates (RS_γ) for every fixed $\gamma > 0$ remain), or larger-sieve kernels with an uncontrolled factor $1 + Nh/(W_K - h)$;

(iv) a direct count above modulus $N/2$: a hybrid method that counts classes of modulus in $(N/2, CN]$ exactly with large right-signed mass, or any hybrid at those N for which Flat fails (Theorem 14.27; weak SPW, Hypothesis 14.32, would exclude this; SPW with fixed margin is false, Theorem 14.30), or cancellation between frequencies of combined modulus $> N/2$;

(v) a non-CRT tuple count: Hypothesis 15.19 for some $\theta > 3/4$, i.e. CRT-accuracy on $[1, N]$ at moduli $\exp\{c(\log N)^{3\theta/2}\}$; or family-class order in the

window (11) for moduli with many primes above W; or other non-CRT input (Type I/II sums, Halász-type arguments);

(vi) family primes beyond $N^{O(1)}$, outside the settings of Theorem 13.3, Corollary 14.23, Theorem 14.3, Theorem 14.13 and Theorem 14.18.

Gone since version 2: moduli with $\log M/\log P(M)$ unbounded, (a, D)- and Case-A moduli without a dominant prime, mixed families, and multipliers larger than the slice primes (provided the multiplier conditions are classes of the four types with all primes $\le N^A$); unresolved η-twin moduli, item (i) of the list in [ET, §6]. Gone since version 3: the large sieve beyond prime slices at polynomial level, in twisted, hybrid and prime forms (the prime form with the factor $(\log\log N)^{3/4}$, removed at pointer level only); Gallagher's larger sieve; prime-only majorants of mixtures (same proviso); per-frequency rounding and inter-frequency cancellation when all moduli are $\le N/2$; tuple input of bounded order, and of prime order k up to $k\log\log N$; and the factor $(\log\log N)^{3/4}$ in Theorem 10.14. Gone since version 4: super-polynomial large-sieve frequencies against mixtures with one prime factor $> \exp\{(\log N)^{1/4}\}$ per modulus (Theorem 14.13), and against mixtures satisfying (RS_γ) (Theorem 14.18). Changed since version 4: the target in (iv) is weak SPW; the fixed-margin SPW of version 4 is refuted (Theorem 14.30).

The exact exponent without B is settled for the class of Theorem 10.14: the factor $(\log\lambda)^{3/4}$ of the previous version is removed by the local-weight moment (Lemma 10.9). The heuristic bound $\sum_{M\le x,\, P(M)\le y}\tau(A_M^2) \ll \Psi(x,y)(\log x)^2$ for shifted smooth numbers [KA2, Remark 3.8] was not needed and remains [OPEN]; it would replace the crude decay u^{-4} of Lemma 10.6 by a $\rho(u)$-type decay, which does not affect the cap.

Constants. All constants of Section 10 are absolute but astronomically large; the saving of [TQ] has an ineffective constant. Comparing them is [OPEN] and not attempted.

The following statement would close (i) for forced classes with a bounded-saving admissible set.

Hypothesis 17.1 (H_{MS}; [ET, §5.6]; [OPEN])**.** For every CRT system of forced classes with moduli $\le \mathrm{e}^\lambda$, every majorant of level λ satisfies $\log(1/\mathbb{E}\nu) \le C\inf_\alpha\big[\alpha\lambda + \sum_{\mathrm{cond}}\mathbb{P}(\mathrm{cond})(\mathrm{mod\ cond})^{-\alpha}\big] + O(\log^2\lambda)$.

Under H_{MS} every CRT-majorant architecture built from Case-B and Case-A forced classes, with a bounded-saving admissible set and rounding treated in absolute value, is capped at $(\log N)^{3/4}$ [CONDITIONAL]. H_{MS} is open even for one scale of primes with random pair conditions.

For the twin residual, the reduction of [TW, §6] isolated an arithmetic-free statement. Let V be a block of primes with costs $s_\ell \in (s, (1+\eta)s]$ and $d = \lfloor\lambda/s\rfloor$.

Conjecture 17.2 (k-ary comparison inequality; [TW, Conj. 6.4]; [OPEN])**.** Let $\nu = \bigotimes\nu_\ell$ be a product law on V, and let $\mathcal{F}$ be a family of hard k-ary constraints $(2 \le k \le r)$ on V with all incident weights $\le \theta \le \theta_0(r)$ and total ν-mass $\mu_{\ge 2}$. Let σ be the sequential law: in increasing order of ℓ, ν_ℓ conditioned off the residues activated by earlier coordinates, or ν_ℓ itself if that set has ν_ℓ-mass $> \delta_\ell$. Then for every λ-level $f \ge 0$,

$$\mathbb{E}_\sigma f \le \exp\big(C_r[d\log(2+\mu_{\ge 2}) + 1]\big)\,\mathbb{E}_\nu f.$$

Here the *incident weight* of a residue a at ℓ is the ν-probability that some activated constraint through (ℓ, a) is completed by the other coordinates.

The statement of Conjecture 17.2 is the *unweighted* form. Its weighted form, for the same law σ, is Theorem 8.5 [PROVED], and without the incident-weight hypothesis the unweighted form is false (Remark 8.7). With it, the unweighted form remains [OPEN], but it is no longer needed for the 3/4 question [KA, §5].

Proposition 17.3 ([TW, Prop. 6.5]; [CONDITIONAL] on Conjecture 17.2 and Hypothesis K2)**.** *Assume Conjecture 17.2 for $r = \lfloor (1+B)(1+\eta) \rfloor$ and the k-ary second-moment hypothesis K2 of* [TW, §6.3]. *Then for every fixed B, every family of $\mathscr{R}(M)$-classes with $M \le P(M)^{1+B}$ (twin, prime-power top and all other shapes included), plus W_0-smooth classes, has saving $\ll_B \eta^{-1}\lambda^{3/4}\log\lambda$.*

Its conclusion, without the factor $\log\lambda$ and without either hypothesis, is Theorem 9.3. Proposition 17.3 is kept only to record the route.

For Λ^2-majorants the core is a sparse noise-stability statement.

Conjecture 17.4 (sparse noise stability; [TW, Conj. 6.8]; [OPEN])**.** Let X_i be independent and $\mathcal{E}$ a finite family of events, each depending on coordinates $S(E)$, with incident weights $\sum_{E \ni i} \mathbb{P}(E) \mathrm{e}^{a|S(E)|} \le \theta \le \theta_0(a)$ and a codegree condition for $|S(E)| \ge 3$. Let $\mathcal{A}$ be "no event" and $\mathcal{A}'$ its ρ-correlated copy. Then

$$\log \frac{\mathbb{P}(\mathcal{A} \cap \mathcal{A}')}{\mathbb{P}(\mathcal{A})^2} \le (1+O(\theta)) \sum_E \mathbb{P}(E) \Big(\prod_{i \in S(E)} (\rho_i + (1-\rho_i) P_i(E)) - \mathbb{P}(E) \Big),$$

where $P_i(E)$ is the marginal factor of E at i.

Theorem 12.4 proves a bound of this type for binary systems ($|S(E)| \le 2$) with the diagonal and one-site terms on the right; the any-arity bound of [TW3, Lemmas 6.1, 6.2], with codegree terms and a free codegree quarantine, is what Theorem 12.11 uses. The conjecture itself is [OPEN].

Many large primes in the Λ^2 route [OPEN]. The Λ^2 caps of Section 12 are uniform for $r \le \epsilon \log L / \log\log L$ prime factors above $(\log X)^8$, and the moduli with $r \ge 330 \log L$ are free. The middle range, which by the *Assessment* of [TW4, §9.3] carries most of the α^{-3}-weighted mass, needs per-prime-efficient bounds for the number of vertex hubs and for an off-diagonal pair sum [TW4, §12]. This now concerns only the Λ^2 method: Theorem 9.3 under the B-hypothesis, and Theorem 10.12 without it for majorants ≥ 1 on the whole avoider set, cover every r (Remark 12.8).

Non-CRT counts [OPEN]. Conjecture (H_{eq}) for smooth windows and the local-limit hypothesis H_{node} for exact interval counts (Section 13) are open, as is any treatment of non-symmetric hit-pattern majorants on $[1, N]$ [NC, §8.6].

18. One sieve limit behind both ceilings

This closing section places the cap in a wider picture, following [CU] (review [CUr]: no fatal or major defect; ten minor label and wording points, applied). Nothing in it improves the exponent 3/4, or the pointwise exponent 1/4 below.

The pointwise problem. For a prime p let $W(p)$ be the least $M \equiv 3 \pmod 4$ with $p \bmod M \in \mathscr{R}(M)$, the least witness modulus, and call p *Mordell-hard* if it is a square modulo 840. Write $\mathcal{L} = \log T$ for a cutoff T, and F_T for the indicator of "no class of $\mathscr{R}(M)$, $M \le T$". Internal results of the programme give

$W(p) \geq \exp\{c(\log p)^{1/4}(\log\log p)^{-1/4}\}$ for infinitely many Mordell-hard primes [O13, Thm 5.1], and show that 1/4 is the ceiling of the Haar-minorant architecture used for it [O14, Thm 4.5, Cor. 4.6]; both are [PROVED MOD] the published and internal inputs listed there: Gallagher's theorem on primes in progressions [G2, Thm 7] (not to be confused with his larger sieve [G]), the Nair–Tenenbaum bound [NT], the effective Page bound [D, Ch. 20], the fundamental lemma of sieve theory [FI], and, for [O13, Thm 5.1], the internal [O10, Thm 3.4]; all with internal hostile reviews. The unit-Haar avoider exponent satisfies $\mathcal{L}^3 \ll \log(1/\delta^*(T)) \ll \mathcal{L}^3(\log\mathcal{L})^5$: the lower bound is the fibre product of [TQ] read under Haar measure [CU, Prop. 1.1] ([PROVED] given [TQ]; ineffective), the upper bound is [O13, Thm 3.4] ([PROVED MOD] Nair–Tenenbaum [NT]).

The abstract limit. Take independent coordinates, one small coordinate X_s and big coordinates X_b of cost $\geq L$ each, and events each depending on X_s and on exactly one big coordinate. Given x_s, let p_b be the probability that X_b hits an event, with $p_b \leq 1/4$, let $P = \sum_b p_b$, let F be the indicator of "no event", and let $\mathcal{V}_k$ be the span of the functions of X_s and at most k big coordinates.

Theorem 18.1 (two-sided order-k sieve limit; [CU, Thm 4.1]; [PROVED]; no novelty claimed)**.** *Fix x_s. (U−) Every $G \in \mathcal{V}_k$ with $G \geq F$ has $\log(1/\mathbb{E}_s G) \leq k\log(C_0(P+4k)/k) + \frac{4}{3}k + \frac{1}{2}\log(22k+22) + 3$; so a saving $\geq P/2$ needs $k \geq c_0 P$. (U+) For even $k \geq \mathrm{e}^2 P$ the Bonferroni majorant $Q_k(H) \in \mathcal{V}_k$ has $\mathbb{E}_s Q_k(H) \leq 2\mathrm{e}^{-P}$. (L−) Every $B \in \mathcal{V}_k$ with $B \leq F$ has $\mathbb{E}_s B \leq 0$ whenever $k \leq 0.6P - 1$. (L+) For odd $k \geq \mathrm{e}^2 P$, $Q_k(H) \leq F$ and $\mathbb{E}_s Q_k(H) > 0$.*

Here H counts the hit big coordinates and $Q_k(h) = \sum_{j\leq k}(-1)^j\binom{h}{j}$. (U−) is the binomial extrapolation behind Theorem 8.5, (L−) is the planting lemma of [O14, Lemma 1.1, Thm 1.3] (the LP dual of lower-bound sieves), and (U+), (L+) are Bonferroni. So the critical order is $k \asymp P$ for majorants and for minorants alike; with comparable big costs the critical level is $\asymp L \cdot P$, dimension times log-size: the large-dimension sieve limit $\beta_\kappa \asymp \kappa$.

The two instances. Both ceilings meet this setting through a subfamily with one big prime per event, big cost $\asymp \mathcal{L}$ and big mass $P \asymp \mathcal{L}^3$: on the majorant side the atoms of [TQ], of mass $\asymp t^3$ on every unit fibre; on the minorant side the same atoms, which block every low-level minorant. Precisely [CU, Prop. 4.2] ([PROVED] given [TQ] and [O14, Thm 1.3]; ineffective): for $T \geq T_0$, every modulus Q with $\log Q \leq T^{0.05}$ and every unit class $r \bmod Q$, every function $B \leq F_T$ on the fibre $n \equiv r \pmod Q$ of level $\log D \leq c\mathcal{L}^4$ has $\mathbb{E}B \leq 0$. This removes a factor $\log\mathcal{L}$ from [O14, Thm 4.5].

Theorem 18.2 (the Erdős–Straus sieve limit; [CU, Thm 4.3]; [PROVED] as a conjunction of the cited results, each with its own label and scope)**.** *Up to constants and, where marked, factors $(\log\lambda)^{O(1)}$: (1) the largest saving $S(\lambda)$ of a level-λ CRT majorant over all forced-class mixtures satisfies $c\lambda^{3/4} \leq S(\lambda) \leq C\lambda^{3/4}$: the upper bound for every mixture is Theorem 10.12, the lower bound for one mixture is the majorant of [TQ] (Section 11); (2) on every fibre with $\log Q \leq T^{0.05}$ no positive minorant of F_T has level $\leq c\mathcal{L}^4$, and on the good fibre of [O13, §5] a positive minorant of level $\leq C\mathcal{L}^4\log\mathcal{L}$ exists, so the least positive-minorant level is $\asymp \mathcal{L}^4$ up to one factor $\log\mathcal{L}$; (3) with budget $\lambda \asymp \log N$ (coefficient-sum rounding), (1) is the exceptional ceiling $S \asymp \lambda^{3/4}$; with budget $\lambda \asymp \log x$ (transfer to the primes $\leq x$),*

(2) is the pointwise ceiling $\mathcal{L} \lesssim \lambda^{1/4}$*, attained up to* $(\log\lambda)^{1/4}$*. Both are the single relation* $\lambda \asymp \mathcal{L}\cdot\mathcal{L}^3$ *between level and usable cutoff.*

Item (3) is a [PROVED] consequence of (1) and (2) within the scopes of this note (congruence sieves of the class of Theorem 10.14, and the large sieves of Section 14) and of [O14] and its sequel [O15] (Haar-minorant transfers with $\log x \gg \log Z$, orbit-uniform linear certificates); outside these scopes it is an *Assessment*. With Haar exponent a the two exponents are $a/(a+1)$ and $1/(a+1)$; this extrapolation is [CONDITIONAL] on the same one-big-prime mechanism, uniform in the small coordinates, and does not follow from a alone. Two cautions (*Assessment*). The full system is not a one-big-coordinate system: a majorant of the full avoider set need not majorise the avoider set of the subfamily, so (U−) for the subfamily does not by itself bound majorants of the full avoider set; that bound is Theorem 10.12. On the minorant side the subfamily is used only for the obstruction, and the positive minorant of [O13] is built differently. The two sides are also not the same inequality: (U−) yields a measure on the avoider set within $\mathrm{e}^{O(k\log(P/k))}$ of the true law on nonnegative order-k tests (the comparison measure of Lemma 14.4), while (L−) yields a measure on the hit set that agrees with the true law on all order-k tests. Below the threshold majorants still save an amount growing with k, while minorants are useless; exact toy LPs show both thresholds at $k \approx 2P$ [CU, §4.4] [EVIDENCE]. On the pointwise side the heuristic truth is $1/3$ (*Assessment*, from a log-scale random-set model). Exponent $1/3$ (up to a power of $\log\log p$) follows from a "Linnik for sifted sets" hypothesis LS (every unit-class sieve system with moduli $\le T$ and avoider density $\delta > 0$ contains a prime $p > T$ with $\log p \le C(\log T + \log(1/\delta))$) [O16, Thm 1.2] ([CONDITIONAL] on LS, and [PROVED MOD] Nair–Tenenbaum). By contrast, Haar-centred progression input used through linear certificates (BV, EH(θ), the progression part of GEH, GRH truncated to moduli $\le x$; not full GRH or GEH) stays capped at $1/4$ [O16, Prop. 3.1] ([PROVED] as scoped there).

The $3/4$ *bound for primes.* The same atoms give the prime form with the correct normalisation [CU, Thm 2.1, Prop. 3.1] ([PROVED] given [TQ] and the prime number theorem in progressions with the Landau–Page term; constants ineffective): uniformly for $3 \le T \le \exp\{c_1(\log x)^{1/4}\}$,

$$\#\{p \le x:\ W(p) > T\} \ll \pi(x)\exp\{-c(\log T)^3\},$$

and in particular the number of non-representable primes $p \le N$ is $\ll \pi(N)\exp\{-c(\log N)^{3/4}\}$. At this scale this is the prime part of [TQ] with $\pi(N)$ in place of N: a reproof, not an improvement. The "Haar-side route" to $3/4$ (a Haar void at cutoff $\exp\{c(\log N)^{1/4}\}$ plus a majorant certifying it at level $\le \log N$) is the route of [TQ], and by Theorem 10.12 it cannot be improved within the class of Theorem 10.14: an integer-CRT saving s against the classes of modulus $\le T$ with selector classes needs level $\ge (s/C)^{4/3}$ [CU, Cor. 3.3] [PROVED].

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